Forgetting under noise is undecidable: ergodicity of positive-rate cellular automata at a fixed noise rate

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Abstract

Can noise make the long-run behaviour of a cellular automaton predictable? A probabilistic cellular automaton with positive rates, in which every site can take every value with positive probability at every step, is ergodic when it forgets its initial configuration. Deciding ergodicity is known to be impossible when some transitions may have probability zero. We show that positive noise does not restore decidability. There are a dimension d, a rate $\varepsilon_0 > 0$ and a computable family of positive-rate cellular automata on $\mathbb{Z}^d$, each a deterministic local rule followed by uniform random replacement, such that for every noise rate $\varepsilon \in (0, \varepsilon_0]$ the automaton built from a Turing machine M is ergodic if and only if M halts.

The proof links a logical quantity to a physical one. A tile set that tiles the plane makes a noisy quorum rule remember. A tile set whose contradictions can always be cornered by a fixed number of local tokens, a refutation of bounded width in the sense of existential pebble games, makes the same rule forget. We build a computable family of tile sets in which every contradiction that exists has a refutation of one fixed width, by reading a recognizable substitution with certificates of bounded treewidth and running the machine in a layer closed under intersection. Two consequences: no computable function bounds how long the ergodic members take to forget, and uniqueness of the translation-invariant Gibbs state of the associated space–time interactions is undecidable.

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1 Introduction

1.1 Does noise make forgetting decidable?

A probabilistic cellular automaton updates every site of a lattice at once, each site reading a finite neighbourhood through one fixed table of probabilities. When every entry of that table is positive, the automaton has *positive rates*: no value is ever forbidden at any site. Run such an automaton from two different configurations. Does it eventually forget where it started? If it does, from every start, it is *ergodic*: it has one invariant measure, and every initial law converges to it.

For deterministic rules and for rules with forbidden transitions, the answer can encode any computation, and it is undecidable. Kurdyumov [17] and Toom [30] proved that ergodicity of probabilistic cellular automata cannot be decided by any algorithm, using rules in which some transitions have probability zero; see also [4] for deterministic rules on $\mathbb{Z}$. Positive rates were the natural escape hatch. Noise blurs every configuration, and one may hope that it also blurs away the fine structure that lets an automaton simulate a Turing machine. Toom [29] and Gács [9] showed that noise does not always erase memory: there are positive-rate automata that are not ergodic. Marsan, Sablik and Törmä [21] gave a perturbed automaton with two phase transitions for ergodicity. Marcovici, Sablik and Taati [19] studied which noisy automata are ergodic, and Marcovici and Taati [20] proved that, at one fixed positive rate, automata with a stationary Bernoulli measure cannot be computably separated from those without one. Whether ergodicity itself remains undecidable at a fixed positive noise rate is the question of this paper.

The answer is that it does. Noise does not make forgetting decidable.

Theorem A (Theorem 8.2). There are integers j, k, a rate $\varepsilon_0 > 0$ and a total computable map $M \mapsto R_M$ from Turing machines to finite Wang sets with the following property. For every $\varepsilon \in (0, \varepsilon_0]$, the quorum rule of R_M on $\mathbb{Z}^{2+k}$, followed by uniform random replacement at rate ε, is a positive-rate probabilistic cellular automaton that is ergodic if and only if M halts.

The rule, the noise and the rate are given by explicit formulas in k (Section 2 and Remark 3.2); the integer k comes from the construction, which shows that it exists without computing it. Every member of the family has positive rates, and the noise is the same uniform replacement at the same rate for every member.

1.2 How it works: a team of inspectors

The automaton is built from a Wang tile set R [31]. Each site holds a legal 3×3 block of tiles and a check bit, or a blank symbol $\perp$. A checker erases every block that disagrees with a neighbour, and a quorum over k extra copies of the plane restores what most copies agree on (Definition 2.7). Two theorems tie the behaviour of this noisy rule to a property of R.

Picture j inspectors walking on the plane. Each one sees a 3×3 patch and may step to a neighbouring site or jump onto another inspector; together they try to corner a contradiction in the tile rules. The bookkeeping of what they can still believe is a finite deletion procedure, which we call the j-token tables (Section 2). When the tables empty, the team has refuted R, and we say R is *j-token refutable*. In the language of constraint satisfaction, the team wins

whenever Spoiler wins the existential j-pebble game [16, 1], and a win of the team is a win of Spoiler with $j+1$ pebbles in the game where Duplicator answers with 3×3 blocks (the live target of Theorem 2.25): a refutation of bounded width.

Theorem B (Theorems 3.1 and 4.2). Fix $j \geq 1$, $k \geq \max(2, 2j)$ and $0 < \varepsilon \leq \varepsilon_k$, where ε_k is the explicit rate of Remark 3.2. For every finite Wang set R:

(i) if R tiles the plane, the noisy quorum rule of R remembers a bit forever and is not ergodic;

(ii) if R is j-token refutable, the noisy quorum rule of R forgets its initial configuration exponentially fast after a delay set by the refutation, and is ergodic.

Part (i) is a Toom-type stability argument [29, 28]. Part (ii) is the new mechanism: every failed relation among j inspectors is covered by a field that depends on the noise alone, and contours of that field force linearly many distinct noise events along any path that carries a discrepancy. The bounded width of the refutation is what lets a quorum transport it through noise. A refutation needing an unbounded number of inspectors would give no such cover.

The two parts leave a gap: a tile set that does not tile but needs ever more inspectors to refute. If every undecidable family of tile sets fell into that gap, Theorem B would decide nothing. The third theorem closes the gap.

Theorem C (Theorem 7.8). There are a fixed integer j and a total computable map $M \mapsto R_M$ to finite Wang sets such that R_M tiles the plane if and only if M does not halt, and R_M is j-token refutable whenever M halts.

So a fixed team of inspectors can always find the flaw, yet nobody can tell in advance whether there is one. The construction has two halves, and each is chosen so that bounded width suffices for it. The geometry is a fixed hierarchical tiling, Ollinger's 2×2 substitution [25], in the transcription of [18], decorated with ruler bits. Recognizability lets a bounded number of inspectors read the type of a supertile of any size, through certificates of bounded treewidth [23, 27, 6]. The computation runs in a layer of Horn clauses. Constraints closed under intersection are decided by local consistency alone [14, 8], so once the geometry is pinned, the inspectors can reconstruct a whole computation square from pairwise checks. Theorem A follows by applying Theorem B to the family of Theorem C with $k = \max(2, 2j)$ (Section 8).

1.3 Consequences

Five corollaries come with the proof (Section 8).

- *One family for every small rate.* The conditions on ε in Theorem B are monotone, so the same family works at every rate in $(0, \varepsilon_0]$ (Corollary 8.4). Other replacement laws work as well (Corollary 8.6).
- *Tiling reduces to forgetting.* A computable map sends every finite Wang set to a noisy automaton, in one fixed dimension and at every rate in $(0, \varepsilon_0]$, that is ergodic exactly when the set does not tile the plane (Corollary 8.5).
- *Forgetting cannot be timed.* Every halting member forgets, but at every rational rate no computable function of M bounds the time it takes (Corollary 8.7). Otherwise a finite simulation would decide halting.
- *Phase coexistence is undecidable.* A positive-rate automaton defines a translation-invariant, finite-range interaction one dimension up, whose Gibbs states contain the trajectories of

the dynamics [12, 7]. For our family, this interaction has a unique translation-invariant Gibbs state exactly when M halts (Corollary 8.8).

1.4 Objections answered

The dimension is enormous. It is. The parser of the hierarchical tiling uses $\binom{276}{2} + 1$ observation points, so j and the dimension $2 + k$ are fixed but very large. This construction gives at least 761,042 dimensions. The size comes entirely from reading the scaffold; the computation layer adds only a handful of inspectors to those observation points, and part (ii) of Theorem B is proved for every j. How small the dimension can be made is the first question of Section 9.

The alphabet grows with the machine. Each member of the family has its own finite alphabet, so the smallest transition probability, ε divided by the alphabet size, is not bounded below over the family. "A fixed noise rate" means the same replacement rate ε, with uniform replacement, for every member. This is the setting of the question above and of [20].

Is this the classical reduction with noise added? No. The classical reductions [17, 30] keep a computation exact because some transitions never happen. Here every transition happens. The undecidability survives because the dichotomy of Theorem B is itself stable under noise: memory from tilings, forgetting from bounded-width refutations, at one rate chosen before the tile set.

1.5 Reading guide

Section 2 defines the automaton, the noise and the token tables, and relates refutations by tokens to refutations in the existential pebble game. Sections 3 and 4 prove the two halves of Theorem B; a reader interested only in the noise can stop there. Sections 5 to 7 prove Theorem C: the hierarchical scaffold and its parser, the routing of a machine onto the scaffold, and the bounded-width refutation of every halting member. Section 8 assembles Theorem A and proves the corollaries, and Section 9 lists open problems. The explicit tile tables are in Section A.

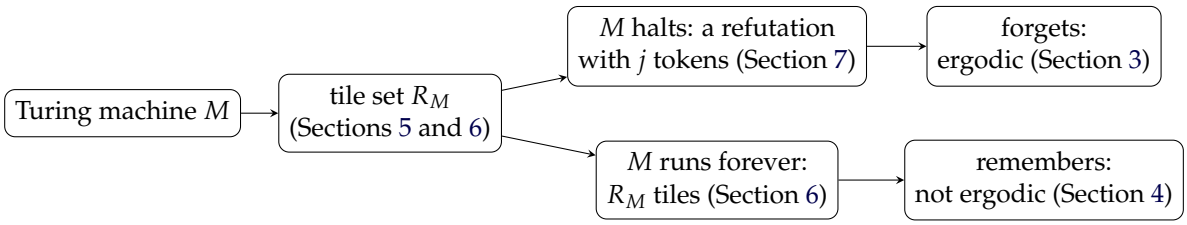


Figure 1: The proof of Theorem A. The same noisy quorum rule, at the same rate, is built from R_M in both cases.

2 The model and the token game

This section fixes the cellular automaton, its noise, and the notions of ergodicity, forgetting and memory used in the paper. It then defines token tables, a finite procedure that may refute a Wang set. Theorem 2.25 shows two things. The procedure never refutes a Wang set that

tiles. When it fails to refute a set, it leaves behind an existential pebble family, and the proof of Theorem 7.8 works with that family.

For $r \geq 0$ write $V_r = \{-r, \dots, r\}^2 \subseteq \mathbb{Z}^2$, and put $\Lambda = V_1 = \{-1, 0, 1\}^2$. For $u \in \mathbb{Z}^2$, $|u|_\infty$ is the largest absolute value of a coordinate of u. A *window* is a finite set of sites.

2.1 Wang sets, blocks and the checker

Definition 2.1 (Wang sets)**.** A *Wang set* is a finite set R of tiles. Each tile has a north, east, south and west edge colour, written with the subscripts $\mathrm{N}, \mathrm{E}, \mathrm{S}, \mathrm{W}$. A *pattern* on a set $U \subseteq \mathbb{Z}^2$ is a map $p \colon U \to R$. It is *legal* if $p(u)_\mathrm{E} = p(u + (1,0))_\mathrm{W}$ whenever $u, u + (1,0) \in U$, and $p(u)_\mathrm{N} = p(u + (0,1))_\mathrm{S}$ whenever $u, u + (0,1) \in U$. A *tiling* is a legal pattern on $\mathbb{Z}^2$, and R *tiles* if it has a tiling.

Restrictions and translates of legal patterns are legal.

Lemma 2.2 (Compactness)**.** *If R does not tile, then there is $n \geq 1$ such that no legal pattern on V_n exists. For such n, no square of side at least $2n + 1$ carries a legal pattern.*

Proof. Take the legal patterns on the boxes $V_0, V_1, \dots$ as the nodes of a tree. The parent of a pattern on V_{n+1} is its restriction to V_n, and a root is joined to every pattern on V_0. Each box carries finitely many patterns, so the tree is finitely branching. If every box carried a legal pattern, the tree would be infinite, and König's lemma would give a branch $p_0, p_1, \dots$ in which each p_n is the restriction of p_{n+1}. The union of the branch would be a tiling. So some box V_{n_0} carries no legal pattern, and neither does any larger box; take $n = \max(n_0, 1)$. A legal pattern on a square of side at least $2n + 1$ would restrict and translate to a legal pattern on V_n. □

Definition 2.3 (Blocks, live symbols and compatibility)**.** Let R be a Wang set. The set T_R of *legal* 3×3 *blocks* is the set of legal patterns on Λ. A *live symbol* is a pair $s = (\lambda, b)$ of a block $\lambda \in T_R$ and a *checker bit* $b \in \{0, 1\}$, and $L_R = T_R \times \{0, 1\}$ is the set of live symbols. The alphabet is $\mathcal{A}_R = L_R \cup \{\bot\}$, where $\bot$ (bottom) is one further symbol. The *Moore directions* are $D = \Lambda \setminus \{(0,0)\}$. For live symbols $s = (\lambda, b)$ and $s' = (\lambda', b')$ and for $d \in D$, the symbol s' is *compatible with s at displacement d* if $b = b'$ and $\lambda(w) = \lambda'(w - d)$ for every $w \in \Lambda \cap (d + \Lambda)$. The *centre* of $s = (\lambda, b)$ is the tile $\mathrm{ctr}(s) = \lambda(0,0)$.

Place s at a site u and s' at $u + d$. Their blocks cover $u + \Lambda$ and $u + d + \Lambda$. Compatibility says that the two bits are equal and that the two blocks agree on every common cell. The relation is symmetric: s' is compatible with s at displacement d exactly when s is compatible with s' at displacement $-d$, as the substitution $w \mapsto w - d$ shows. The alphabet has $2|T_R| + 1$ symbols; if T_R is empty, $\bot$ is the only symbol.

Lemma 2.4 (Centres match)**.** *Let s, s' be live. If s' is compatible with s at displacement $(1,0)$, then $\mathrm{ctr}(s)_\mathrm{E} = \mathrm{ctr}(s')_\mathrm{W}$. If s' is compatible with s at displacement $(0,1)$, then $\mathrm{ctr}(s)_\mathrm{N} = \mathrm{ctr}(s')_\mathrm{S}$.*

Proof. Write $s = (\lambda, b)$ and $s' = (\lambda', b)$; the bits agree by compatibility. Let $d = (1,0)$. The cell $w = (1,0)$ lies in $\Lambda \cap (d + \Lambda)$, so compatibility gives $\lambda(1,0) = \lambda'(0,0) = \mathrm{ctr}(s')$. The block λ is legal, so $\lambda(0,0)_\mathrm{E} = \lambda(1,0)_\mathrm{W}$. Hence $\mathrm{ctr}(s)_\mathrm{E} = \mathrm{ctr}(s')_\mathrm{W}$. The vertical case is the same, with $d = (0,1)$ and the colours N and S. □

For a tiling z and a site u, the *block of z at u* is $z^{[u]} \in T_R$, given by $z^{[u]}(w) = z(u + w)$ for $w \in \Lambda$. The *block configuration* of z is $\hat{z} \in L_R^{\mathbb{Z}^2}$, $\hat{z}(u) = (z^{[u]}, 0)$. For every u and every $d \in D$, the symbol $\hat{z}(u + d)$ is compatible with $\hat{z}(u)$ at displacement d: both bits are 0, and $z^{[u]}(w) = z(u + w) = z^{[u+d]}(w - d)$ for $w \in \Lambda \cap (d + \Lambda)$.

Definition 2.5 (Planar checker)**.** The *checker* $C\colon \mathcal{A}_R^{\mathbb{Z}^2} \to \mathcal{A}_R^{\mathbb{Z}^2}$ is given by $C(x)(u) = x(u)$ if $x(u)$ is live and, for every $d \in D$, $x(u+d)$ is live and compatible with $x(u)$ at displacement d. Otherwise $C(x)(u) = \perp$.

Thus $C(x)(u) \in \{x(u), \perp\}$ for every x and u, and $C(\hat{z}) = \hat{z}$ for every tiling z.

Remark 2.6 (The checker reduction)*.* The checker is the planar form of the classical reduction from tilings to the nilpotency of cellular automata [5, 15]. If R tiles, C has the live fixed point $\hat{z}$. If R does not tile, take $n \geq 1$ from Lemma 2.2; then $C^n(x) = \perp$ for every $x \in \mathcal{A}_R^{\mathbb{Z}^2}$.

To see this, let $m \geq 1$, $x \in \mathcal{A}_R^{\mathbb{Z}^2}$ and $u \in \mathbb{Z}^2$ with $C^m(x)(u)$ live. We show that R has a legal pattern on $u + V_m$. A live output of C at a site needs live inputs at the nine sites of its window, so induction on m shows that $C(x)(v)$ is live for every $v \in u + V_{m-1}$. Hence, for $v \in u + V_{m-1}$, the symbol $x(v)$ is live, say with block λ_v, and $x(v+d)$ is compatible with $x(v)$ at displacement d for every $d \in D$. For $w \in u + V_m$ choose $v \in u + V_{m-1}$ with $|w - v|_\infty \leq 1$ and put $p(w) = \lambda_v(w - v)$. The value does not depend on the choice. For two choices v, v' with $|v - v'|_\infty \leq 1$ this is compatibility. Otherwise define v'' coordinatewise by $v''_c = v_c$ if $|v_c - v'_c| \leq 1$ and $v''_c = w_c$ if $|v_c - v'_c| = 2$. Then $v'' \in u + V_{m-1}$, and v'' lies within distance 1 of v, of v' and of w, so two applications of compatibility link the values at v and v'. Finally, let $\{w, w + \Delta\}$ with $\Delta \in \{(1,0),(0,1)\}$ be an edge inside $u + V_m$, and let v be a point of $u + V_{m-1}$ nearest to $w + \Delta/2$ in each coordinate. Both w and $w + \Delta$ lie in $v + \Lambda$, and λ_v is legal, so the edge matches. So p is legal. With $m = n$ this contradicts the choice of n.

2.2 The quorum rule and replacement noise

Definition 2.7 (Quorum rule)**.** Let $k \geq 2$. A *site* is a pair $(u, P) \in \mathbb{Z}^2 \times \mathbb{Z}^k$, and the sites with a fixed transverse coordinate P form the *plane* P. The configuration space is $\Omega = \mathcal{A}_R^{\mathbb{Z}^2 \times \mathbb{Z}^k}$. For $x \in \Omega$, $x(\cdot, P) \in \mathcal{A}_R^{\mathbb{Z}^2}$ is its restriction to the plane P. The *providers* are $c_0 = 0$ and $c_i = e_i$ for $i = 1, \ldots, k$, where e_i is the i-th unit vector of $\mathbb{Z}^k$. Put $y(u, P) = C(x(\cdot, P))(u)$. The *quorum rule* is

$$F_R^{[k]}(x)(u,P) = \begin{cases} s & \text{if } s \text{ is live and } \#\{i \in \{0,\ldots,k\} : y(u, P + c_i) = s\} \geq k, \\ \perp & \text{otherwise.} \end{cases}$$

Each plane runs the checker. The rule then reads the checked symbol at the site itself and at its k transverse neighbours $(u, P + e_i)$, and keeps a live symbol only when at least k of these $k + 1$ readings agree on it. Two different live symbols cannot both be read k times, since $2k > k + 1$; so $F_R^{[k]}$ is well defined. The value $F_R^{[k]}(x)(v)$ depends only on x restricted to $v + \mathcal{N}$, where $\mathcal{N} = \Lambda \times \{c_0, \ldots, c_k\}$. So $F_R^{[k]}$ is a cellular automaton on $\mathbb{Z}^2 \times \mathbb{Z}^k = \mathbb{Z}^{2+k}$, and its local rule is a finite table computable from R and k. If z is a tiling, the *cloned tiling configuration* $\xi_z \in \Omega$, $\xi_z(u, P) = \hat{z}(u)$, satisfies $F_R^{[k]}(\xi_z) = \xi_z$, because all $k + 1$ providers read $\hat{z}(u)$.

Definition 2.8 (Replacement noise)**.** Fix $\varepsilon \in (0,1)$. Let $\eta_t(v) \in \{0,1\}$ and $\zeta_t(v) \in \mathcal{A}_R$, for $t \geq 1$ and $v \in \mathbb{Z}^2 \times \mathbb{Z}^k$, be independent random variables with $\mathbb{P}(\eta_t(v) = 1) = \varepsilon$ and $\zeta_t(v)$ uniform on $\mathcal{A}_R$. The *process from* $x \in \Omega$ is given by $X_0 = x$ and, for $t \geq 0$,

$$X_{t+1}(v) = \begin{cases} \zeta_{t+1}(v) & \text{if } \eta_{t+1}(v) = 1, \\ F_R^{[k]}(X_t)(v) & \text{if } \eta_{t+1}(v) = 0. \end{cases}$$

A site v at a time $t \geq 1$ with $\eta_t(v) = 1$ is a *replacement event*, whatever value $\zeta_t(v)$ takes.

So after each application of $F_R^{[k]}$, each site independently is replaced, with probability ε, by a uniformly random symbol of $\mathcal{A}_R$. Given X_t, the values $X_{t+1}(v)$ are independent, and for every symbol s

$$\mathbb{P}\big(X_{t+1}(v) = s \mid X_t\big) = (1-\varepsilon)\,\mathbf{1}\big[F_R^{[k]}(X_t)(v) = s\big] + \frac{\varepsilon}{|\mathcal{A}_R|} \;\geq\; \frac{\varepsilon}{|\mathcal{A}_R|} > 0.$$

The process is therefore a probabilistic cellular automaton with positive rates. We write Law_x for the law of $(X_t)_{t\geq 0}$ from x, and $\mathbb{P}_x$, $\mathbb{E}_x$ for probabilities and expectations under it. The symbol $\perp$ also denotes the all-bottom configuration. For a probability measure μ on Ω, Law_μ is the law of the process when X_0 has law μ and is independent of the variables $\eta_t(v)$, $\zeta_t(v)$.

By induction on t, $X_t(v)$ is a function of X_0 restricted to the finite set $v + \mathcal{N}_t$, where $\mathcal{N}_0 = \{0\}$ and $\mathcal{N}_{t+1} = \mathcal{N}_t + \mathcal{N}$, and of finitely many noise variables. Hence, for a window O, the law of $X_t|_O$ under Law_x depends on only finitely many coordinates of x, and

$$\mathbb{P}_\mu(X_t|_O \in \mathcal{E}) = \int \mathbb{P}_x(X_t|_O \in \mathcal{E})\,\mu(dx) \qquad \text{for every } \mathcal{E} \subseteq \mathcal{A}_R^O.$$

Running all starts with the same variables $\eta_t(v), \zeta_t(v)$ couples them. This is the common-noise coupling used in Theorem 3.1.

Write $\mu\Phi^t$ for the law of X_t under Law_μ, and $\Phi f(x) = \mathbb{E}_x f(X_1)$ for bounded measurable $f\colon \Omega \to \mathbb{R}$, so that $(\mu\Phi)(f) = \mu(\Phi f)$. The noise after time t is independent of X_t, so given X_t the process $(X_{t+t'})_{t'\geq 0}$ is the process from X_t. In particular $\mu\Phi^{t+1} = (\mu\Phi^t)\Phi$. A probability measure μ on Ω is *invariant* if $\mu\Phi = \mu$.

2.3 Ergodicity, forgetting and memory

The space Ω carries the product topology; it is compact and metrizable.

Definition 2.9 (Ergodicity)**.** The process is *ergodic* if it has exactly one invariant probability measure π and $\mu\Phi^t \to \pi$ weakly as $t \to \infty$ for every probability measure μ on Ω.

For a window O and $\mathcal{E} \subseteq \mathcal{A}_R^O$, the cylinder $\{\omega \in \Omega : \omega|_O \in \mathcal{E}\}$ is open and closed. Functions that depend on finitely many coordinates are finite combinations of cylinder indicators, and they are dense in the continuous functions on Ω. Hence $\mu_t \to \mu$ weakly if and only if $d_{\mathrm{TV}}(\mu_t|_O, \mu|_O) \to 0$ for every window O. Here $\mu|_O$ is the marginal law on $\mathcal{A}_R^O$, and $d_{\mathrm{TV}}(\nu,\nu') = \max_{\mathcal{E}\subseteq\mathcal{A}_R^O} |\nu(\mathcal{E}) - \nu'(\mathcal{E})|$.

Lemma 2.10 (Invariant measures exist)**.** *For every continuous $f\colon \Omega \to \mathbb{R}$, the function Φf is continuous. For every probability measure μ on Ω and every sequence $n_m \to \infty$, the averages $\bar\mu_n = \frac{1}{n}\sum_{t=0}^{n-1} \mu\Phi^t$ have a weak limit along a subsequence of (n_m), and every such limit is invariant.*

Proof. If f depends only on the coordinates in a window O, then $\Phi f(x)$ depends only on x restricted to the window $O + \mathcal{N}$, so Φf is continuous. A general continuous f is uniformly continuous on the compact space Ω, so it is a uniform limit of functions f_ℓ that depend on finitely many coordinates. Since $\|\Phi g\|_\infty \leq \|g\|_\infty$, the function Φf is the uniform limit of the continuous functions Φf_ℓ, hence continuous. The probability measures on the compact metrizable space Ω form a weakly sequentially compact set, which gives the convergent subsequence. Let $\bar\mu_{n_m} \to \pi$ weakly. For continuous f, using that Φf is continuous,

$$|\pi(\Phi f) - \pi(f)| = \lim_{m\to\infty} \big|\bar\mu_{n_m}(\Phi f) - \bar\mu_{n_m}(f)\big| = \lim_{m\to\infty} \frac{|\mu\Phi^{n_m}(f) - \mu(f)|}{n_m} \leq \lim_{m\to\infty} \frac{2\|f\|_\infty}{n_m} = 0.$$

So $\pi\Phi = \pi$. □

Definition 2.11 (Forgetting and memory)**.** For a window O and $t \geq 0$ put

$$\delta_O(t) = \sup_{x \in \Omega} d_{\mathrm{TV}}\big(\mathrm{Law}_x(X_t|_O),\, \mathrm{Law}_\perp(X_t|_O)\big).$$

The process *forgets* if $\delta_O(t) \to 0$ as $t \to \infty$ for every window O. It has *memory* if there are configurations $x, x' \in \Omega$, a window O, a set $\mathcal{E} \subseteq \mathcal{A}_R^O$ and $\alpha > 0$ such that

$$\mathbb{P}_x(X_t|_O \in \mathcal{E}) \;\geq\; \mathbb{P}_{x'}(X_t|_O \in \mathcal{E}) + \alpha \qquad \text{for every } t \geq 0.$$

Under memory the laws of $X_t|_O$ from x and from x' stay at total variation distance at least α for all t, and one fixed event sees the distance. In Theorem 4.2 the two starts are a cloned tiling configuration and the all-bottom configuration, and the event is that one site carries its tiling symbol. For forgetting, any coupling of Law_x and $\mathrm{Law}_\perp$ bounds the distance on O by the probability that the two processes differ somewhere on O; Theorem 3.1 bounds $\delta_O(t)$ in this way.

Lemma 2.12 (Forgetting implies ergodicity)**.** *If the process forgets, it is ergodic. Its invariant measure π satisfies $d_{\mathrm{TV}}(\mu\Phi^t|_O, \pi|_O) \leq 2\delta_O(t)$ for every probability measure μ, every window O and every $t \geq 0$.*

Proof. Let μ, ν be probability measures on Ω, let O be a window, $t \geq 0$ and $\mathcal{E} \subseteq \mathcal{A}_R^O$. By the mixture formula,

$$\mathbb{P}_\mu(X_t|_O \in \mathcal{E}) - \mathbb{P}_\nu(X_t|_O \in \mathcal{E}) = \iint \big(\mathbb{P}_x(X_t|_O \in \mathcal{E}) - \mathbb{P}_{x'}(X_t|_O \in \mathcal{E})\big)\, \mu(dx)\, \nu(dx').$$

By the triangle inequality through $\mathrm{Law}_\perp$, the integrand has absolute value at most $2\delta_O(t)$. Hence $d_{\mathrm{TV}}(\mu\Phi^t|_O, \nu\Phi^t|_O) \leq 2\delta_O(t)$. By Lemma 2.10 there is an invariant measure π. Taking $\nu = \pi$, for which $\pi\Phi^t = \pi$, gives the stated bound, and so $\mu\Phi^t \to \pi$ weakly for every μ. If π' is invariant, the bound with $\mu = \pi'$ gives $d_{\mathrm{TV}}(\pi'|_O, \pi|_O) \leq 2\delta_O(t)$ for every t. So π' and π agree on every cylinder, and $\pi' = \pi$. □

Lemma 2.13 (Memory excludes ergodicity)**.** *If the process has memory, it has two different invariant measures. In particular it is not ergodic.*

Proof. Let $\bar{\mu}_n$ and $\bar{\mu}'_n$ be the averages of Lemma 2.10 for the point masses at x and at x'. Choose $n_m \to \infty$ with $\bar{\mu}_{n_m} \to \pi$ weakly, and then pass to a subsequence along which also $\bar{\mu}'_{n_m} \to \pi'$. Both limits are invariant by Lemma 2.10. The indicator of the cylinder $\Gamma = \{\omega : \omega|_O \in \mathcal{E}\}$ is continuous, so

$$\pi(\Gamma) - \pi'(\Gamma) = \lim_{m \to \infty} \frac{1}{n_m} \sum_{t=0}^{n_m - 1} \big(\mathbb{P}_x(X_t|_O \in \mathcal{E}) - \mathbb{P}_{x'}(X_t|_O \in \mathcal{E})\big) \;\geq\; \alpha.$$

Hence $\pi \neq \pi'$. □

A weaker hypothesis already excludes ergodicity: if $d_{\mathrm{TV}}(\mathrm{Law}_x(X_t|_O), \mathrm{Law}_{x'}(X_t|_O)) \geq \alpha$ for every t, then the process is not ergodic, since under ergodicity both marginals converge to $\pi|_O$. That hypothesis alone does not produce two invariant measures through averages, because the separating event may change with t; memory fixes the event for this reason.

2.4 Token tables

Fix a Wang set R, a number $j \geq 1$ of tokens and a radius $r \geq 0$, and put $V = V_r$. A position tuple $\mathbf{u} = (u_1, \ldots, u_j) \in V^j$ and a symbol tuple $\mathbf{s} = (s_1, \ldots, s_j) \in L_R^j$ describe j tokens: token i sits at u_i and carries s_i. Write $\mathbf{s}[i \leftarrow s']$ for $\mathbf{s}$ with entry i replaced by s', and $\mathbf{u}[i \leftarrow w]$ likewise. A *table* S assigns to each $\mathbf{u} \in V^j$ a set $S(\mathbf{u}) \subseteq L_R^j$. Tables are compared by inclusion at every $\mathbf{u}$.

Definition 2.14 (Token tables)**.** The deletion rules act on a table S as follows.

(D1) Coincidence rule. Delete $\mathbf{s}$ from $S(\mathbf{u})$ if $u_i = u_{i'}$ and $s_i \neq s_{i'}$ for some i, i'.

(D2) Move rule. Delete $\mathbf{s}$ from $S(\mathbf{u})$ if there are an index i and a direction $d \in D$ with $u_i + d \in V$ such that no live symbol s' compatible with s_i at displacement d has $\mathbf{s}[i \leftarrow s'] \in S(\mathbf{u}[i \leftarrow u_i + d])$.

(D3) Clone rule. Delete $\mathbf{s}$ from $S(\mathbf{u})$ if $\mathbf{s}[i' \leftarrow s_i] \notin S(\mathbf{u}[i' \leftarrow u_i])$ for some $i \neq i'$.

$\mathrm{Inst}(S)$ is obtained from S by applying (D1) and (D3) until neither applies. $\mathrm{Timed}(S)$ is obtained from S by applying (D2) once, simultaneously, to every tuple of S, with the condition evaluated in S. The *token tables* are

$$S_0 = \mathrm{Inst}(\text{all tuples}), \qquad S_{n+1} = \mathrm{Inst}\big(\mathrm{Timed}(S_n)\big) \quad (n \geq 0),$$

where the table of all tuples has $S(\mathbf{u}) = L_R^j$ for every $\mathbf{u} \in V^j$. We write $S_n^{(r)}$ when the radius matters.

The clone rule makes token i' jump onto token i and take its symbol. The move rule moves token i by one Moore step inside V and asks for a compatible new symbol at the new site. The rules commute with translations of $\mathbb{Z}^2$, so tables on a translate $u + V_r$ are defined by translating positions.

Lemma 2.15 (Tables are well defined)**.** *Fix j and r. Call a table* closed *if neither (D1) nor (D3) deletes anything from it.*

(i) $\mathrm{Inst}(S)$ *is the largest closed table contained in S. In particular it does not depend on the order of the deletions.*

(ii) If $S \subseteq S'$, then $\mathrm{Inst}(S) \subseteq \mathrm{Inst}(S')$ *and* $\mathrm{Timed}(S) \subseteq \mathrm{Timed}(S')$*. Moreover $S_{n+1} \subseteq S_n$ for every $n \geq 0$.*

(iii) There is n_0 with $S_n = S_{n_0}$ for every $n \geq n_0$. We write $S_\infty = S_\infty^{(r)}$ for this final table.

Proof. (i) Closedness asks, of each tuple separately, that coincident tokens carry equal symbols and that its clones lie in the table. So a union of closed tables is closed, and S contains a largest closed table S^*. Run the deletions in any order. By induction on the number of deletions, the current table contains S^*. A tuple deleted by (D1) violates the coincidence condition, so it is not in S^*. A tuple deleted by (D3) has a clone missing from the current table, hence missing from S^*; since S^* is closed, the tuple is not in S^*. The run stops, because the table is finite, and it stops at a closed table contained in S, hence contained in S^*. So it stops at S^*.

(ii) If $S \subseteq S'$, then $\mathrm{Inst}(S)$ is a closed table contained in S', so it is contained in $\mathrm{Inst}(S')$ by (i). A tuple of S that survives (D2) evaluated in S has, for each allowed move, a witness in $S \subseteq S'$, so it survives (D2) evaluated in S'. Hence $\mathrm{Timed}(S) \subseteq \mathrm{Timed}(S')$. Now $S_1 \subseteq \mathrm{Timed}(S_0) \subseteq S_0$, and if $S_n \subseteq S_{n-1}$, then $S_{n+1} = \mathrm{Inst}(\mathrm{Timed}(S_n)) \subseteq \mathrm{Inst}(\mathrm{Timed}(S_{n-1})) = S_n$.

(iii) The tables are subsets of the finite set $V^j \times L_R^j$, and they decrease by (ii). □

Let $\mathbf{0} = (0, \dots, 0) \in V^j$. By (D1), every tuple in $S_n(\mathbf{0})$ is constant, $(s, \dots, s)$ with s live.

Definition 2.16 (Refutability)**.** The Wang set R is *j-token refutable* if $S_\tau^{(r)}(\mathbf{0}) = \emptyset$ for some radius $r \geq 0$ and some time $\tau \geq 0$. For a j-token refutable R, the *refutation time* $\tau(R)$ is the least τ for which some r works. The *token width* of R is the least j for which R is j-token refutable, and ∞ if there is none.

So a refutation deletes every live symbol at the origin. Each table $S_n^{(r)}$ is a finite computation from R, j, r and n. Hence, for each j, the j-token refutable Wang sets form a computably enumerable set.

2.5 The token game

The tables have a game reading. Two players act on j tokens. Spoiler either makes one token jump onto another, so that it takes over that token's position and symbol, or moves one token by a Moore step. After a move, Duplicator chooses the new symbol of the moved token, which must be compatible with its old symbol at that displacement. Duplicator loses when two tokens on one site carry different symbols, or when she has no reply. In a box, Spoiler may move only inside the box. We work with the following sets of game states, from which Duplicator can play forever by keeping the play inside the set.

Definition 2.17 (States and safe sets)**.** A *j-token state* is a pair $(\mathbf{u}, \mathbf{s}) \in (\mathbb{Z}^2)^j \times L_R^j$. For $i \neq i'$, the *clone* of token i onto token i' is the state $(\mathbf{u}[i' \leftarrow u_i], \mathbf{s}[i' \leftarrow s_i])$. For an index i and $d \in D$, a *move* of token i by d is a state $(\mathbf{u}[i \leftarrow u_i + d], \mathbf{s}[i \leftarrow s'])$ with s' live and compatible with s_i at displacement d. Let V be a box V_r or the plane $\mathbb{Z}^2$. A set $\mathcal{W}$ of states with positions in V^j is *V-safe* if every $(\mathbf{u}, \mathbf{s}) \in \mathcal{W}$ satisfies:

(W1) if $u_i = u_{i'}$, then $s_i = s_{i'}$;

(W2) for all $i \neq i'$, the clone of token i onto token i' lies in $\mathcal{W}$;

(W3) for every i and every $d \in D$ with $u_i + d \in V$, some move of token i by d lies in $\mathcal{W}$.

A *plane-safe* set is a $\mathbb{Z}^2$-safe set. For a state satisfying (W1), $h_{\mathbf{u},\mathbf{s}}$ denotes the map $\{u_1, \dots, u_j\} \to L_R$ with $h_{\mathbf{u},\mathbf{s}}(u_i) = s_i$.

We identify a table S on V_r with the set of states $(\mathbf{u}, \mathbf{s})$ with $\mathbf{s} \in S(\mathbf{u})$. Then (W1) says that (D1) deletes nothing, (W2) says that (D3) deletes nothing, and (W3) with $V = V_r$ says that (D2) deletes nothing.

Lemma 2.18 (Walking to the origin)**.** *Let V be a box V_r or the plane $\mathbb{Z}^2$. Every nonempty V-safe set contains a state $(\mathbf{0}, \mathbf{s})$.*

Proof. Take a state $(\mathbf{u}, \mathbf{s})$ in the set. Cloning token 1 onto each other token gives, by (W2), a state of the set with all tokens at u_1. While token 1 is not at the origin, move it by the step $d \in D$ whose coordinates are the signs of the coordinates of its negated position, with the sign of 0 equal to 0. The new position lies in V, since no coordinate grows in absolute value, so (W3) gives a move in the set. After finitely many steps token 1 is at the origin. Cloning it onto the other tokens gives a state $(\mathbf{0}, \mathbf{s}')$ of the set. □

Lemma 2.19 (Finite time and finite boxes)**.** *Fix $j \geq 1$.*

(i) For every r, $S_\infty^{(r)}$ is the largest V_r-safe set.

(ii) If $r \leq r'$, the states of $S_\infty^{(r')}$ with positions in V_r^j lie in $S_\infty^{(r)}$.

(iii) R is j-token refutable if and only if $S_\infty^{(r)} = \emptyset$ for some r. In that case $S_\infty^{(r')} = \emptyset$ for every $r' \geq r$.

Proof. (i) By Lemma 2.15(iii), $S_\infty = S_{n_0} = S_{n_0+1} = \mathrm{Inst}(\mathrm{Timed}(S_\infty)) \subseteq \mathrm{Timed}(S_\infty) \subseteq S_\infty$. So $\mathrm{Timed}(S_\infty) = S_\infty$, which is (W3), and $\mathrm{Inst}(S_\infty) = S_\infty$, which is (W1) and (W2). Conversely, let $\mathcal{W}$ be V_r-safe. It is a closed table, so $\mathcal{W} \subseteq S_0$ by Lemma 2.15(i). If $\mathcal{W} \subseteq S_n$, then every state of $\mathcal{W}$ survives (D2) evaluated in S_n, because (W3) supplies its witnesses in $\mathcal{W} \subseteq S_n$. So $\mathcal{W} \subseteq \mathrm{Timed}(S_n)$, and since $\mathcal{W}$ is closed, $\mathcal{W} \subseteq \mathrm{Inst}(\mathrm{Timed}(S_n)) = S_{n+1}$. Hence $\mathcal{W} \subseteq S_\infty$.

(ii) The states of $S_\infty^{(r')}$ with positions in V_r^j form a V_r-safe set. Indeed (W1) is inherited, clones do not create new positions, and a move with $u_i + d \in V_r$ is allowed in $V_{r'}$ and its witness has positions in V_r^j. Apply (i).

(iii) If $S_\tau^{(r)}(\mathbf{0}) = \emptyset$, then $S_\infty^{(r)}(\mathbf{0}) = \emptyset$ by Lemma 2.15(ii). By (i) and Lemma 2.18, a nonempty $S_\infty^{(r)}$ would contain a state at $\mathbf{0}$, so $S_\infty^{(r)} = \emptyset$. Conversely, if $S_\infty^{(r)} = \emptyset$, then $S_{n_0}^{(r)}(\mathbf{0}) = \emptyset$. If $r' \geq r$, then (ii) gives $S_\infty^{(r')}(\mathbf{0}) \subseteq S_\infty^{(r)}(\mathbf{0}) = \emptyset$, and the same argument gives $S_\infty^{(r')} = \emptyset$. □

Lemma 2.20 (Boxes and the plane)**.** *R is j-token refutable if and only if there is no nonempty plane-safe set of j-token states.*

Proof. Let $\mathcal{W}$ be a nonempty plane-safe set. By Lemma 2.18 it contains a state $(\mathbf{0}, \mathbf{s})$. For every r, the states of $\mathcal{W}$ with positions in V_r^j form a V_r-safe set: (W1) is inherited, clones keep positions in V_r^j, and for a move with $u_i + d \in V_r$ the move supplied by (W3) in the plane has positions in V_r^j. By Lemma 2.19(i) this set lies in $S_\infty^{(r)}$, so $(\mathbf{0}, \mathbf{s}) \in S_\infty^{(r)} \subseteq S_n^{(r)}$ for every n. Hence $S_n^{(r)}(\mathbf{0}) \neq \emptyset$ for all r and n, and R is not j-token refutable.

Conversely, suppose that R is not j-token refutable. Then $S_\infty^{(r)}(\mathbf{0}) \neq \emptyset$ for every r. Let $\mathcal{W}$ be the set of states that lie in $S_\infty^{(r)}$ for infinitely many r. Since L_R^j is finite, some $\mathbf{s}$ lies in $S_\infty^{(r)}(\mathbf{0})$ for infinitely many r, so $\mathcal{W}$ is nonempty. Each $S_\infty^{(r)}$ is V_r-safe, and we check (W1)–(W3) for $\mathcal{W}$. Property (W1) holds because it holds in each $S_\infty^{(r)}$. If $(\mathbf{u}, \mathbf{s}) \in \mathcal{W}$, each of its clones lies in $S_\infty^{(r)}$ whenever $(\mathbf{u}, \mathbf{s})$ does, by (W2) in $S_\infty^{(r)}$, so the clone lies in $\mathcal{W}$. For (W3), fix i and $d \in D$. For each of the infinitely many $r \geq |u_i|_\infty + 1$ with $(\mathbf{u}, \mathbf{s}) \in S_\infty^{(r)}$, the site $u_i + d$ lies in V_r, so (W3) in $S_\infty^{(r)}$ gives a live s'_r with $(\mathbf{u}[i \leftarrow u_i + d], \mathbf{s}[i \leftarrow s'_r]) \in S_\infty^{(r)}$ and s'_r compatible with s_i at displacement d. There are finitely many live symbols, so one value s' occurs for infinitely many r, and the corresponding move lies in $\mathcal{W}$. So $\mathcal{W}$ is a nonempty plane-safe set. □

The next lemma is where the move rule and the coincidence rule combine.

Lemma 2.21 (Neighbouring tokens are compatible)**.** *Let $\mathcal{W}$ be plane-safe, $(\mathbf{u}, \mathbf{s}) \in \mathcal{W}$, and $u_{i'} = u_i + d$ with $d \in D$. Then $s_{i'}$ is compatible with s_i at displacement d.*

Proof. By (W3) some move $(\mathbf{u}[i \leftarrow u_i + d], \mathbf{s}[i \leftarrow s'])$ lies in $\mathcal{W}$, with s' compatible with s_i at displacement d. In this state tokens i and i' both sit at $u_{i'}$; note $i \neq i'$ because $d \neq 0$. By (W1), $s' = s_{i'}$. □

Definition 2.22 (Existential pebble families)**.** A *partial homomorphism into the live target* is a map $h\colon U \to L_R$ on a finite set $U \subseteq \mathbb{Z}^2$ such that $h(u + d)$ is compatible with $h(u)$ at displacement d whenever $d \in D$ and $u, u + d \in U$. A *partial homomorphism into R* is a legal pattern on a finite set $U \subseteq \mathbb{Z}^2$. Fix one of the two targets. An *existential j-pebble family*, or *E_j family*, is a nonempty set H of partial homomorphisms into the target such that:

(Size) every $h \in H$ has at most j sites in its domain;

(Restriction) if $h \in H$ and $U \subseteq \mathrm{dom}\, h$, then $h|_U \in H$;

(Extension) if $h \in H$, $|\operatorname{dom} h| < j$ and $w \in \mathbb{Z}^2$, then some $h' \in H$ extends h and has $w \in \operatorname{dom} h'$.

These families are the winning strategies of Duplicator in the existential j-pebble game of Kolaitis and Vardi [16], played from the grid $\mathbb{Z}^2$ to the target; see also [1]. In that game Spoiler places and removes up to j pebbles on sites, Duplicator answers each placement with a target symbol, and Duplicator must keep the pebbled correspondence a partial homomorphism. For the live target the grid carries the eight Moore relations; for R it carries the horizontal and vertical edge relations.

Remark 2.23. Let H be an E_j family. By (Restriction) it contains the empty map. For every set $U \subseteq \mathbb{Z}^2$ with $|U| \leq j$, some $h \in H$ has domain exactly U: add the sites of U one at a time with (Extension), and after each step restrict with (Restriction) to the sites added so far. For $1 \leq K \leq j$, the members of H with at most K sites form an E_K family, since an extension supplied by (Extension) can be restricted to the old domain plus the new site.

Lemma 2.24 (Pebble families from safe sets)**.** *Let $\mathcal{W}$ be a nonempty plane-safe set of j-token states. Let $H(\mathcal{W})$ be the set of restrictions $h_{\mathbf{u},\mathbf{s}}|_U$ with $(\mathbf{u},\mathbf{s}) \in \mathcal{W}$ and $U \subseteq \{u_1,\dots,u_j\}$. Then $H(\mathcal{W})$ is an E_j family for the live target, and $\operatorname{ctr} \circ H(\mathcal{W}) = \{\operatorname{ctr} \circ h : h \in H(\mathcal{W})\}$ is an E_j family for R.*

Proof. Each $h_{\mathbf{u},\mathbf{s}}$ is well defined by (W1). It is a partial homomorphism into the live target: if u and $u+d$ are sites of $\mathbf{u}$ with $d \in D$, pick i, i' with $u_i = u$ and $u_{i'} = u+d$ and apply Lemma 2.21. Restrictions of partial homomorphisms are partial homomorphisms. The family is nonempty, has at most j sites per map, and is closed under restriction by construction.

For (Extension), let $h = h_{\mathbf{u},\mathbf{s}}|_U$ with $(\mathbf{u},\mathbf{s}) \in \mathcal{W}$ and $|U| < j$, and let $w \in \mathbb{Z}^2$. If $w \in U$ there is nothing to do. Otherwise, for each $u \in U$ choose one index $i(u)$ with $u_{i(u)} = u$. These are fewer than j indices, so some index i_0 is not among them. Choose finitely many Moore steps leading from u_{i_0} to w, and apply (W3) to token i_0 once for each step. This gives states of $\mathcal{W}$ in which every token other than i_0 keeps its position and symbol, ending with a state $(\mathbf{u}^*,\mathbf{s}^*) \in \mathcal{W}$ with $u^*_{i_0} = w$. The map $h' = h_{\mathbf{u}^*,\mathbf{s}^*}|_{U\cup\{w\}}$ lies in $H(\mathcal{W})$, contains w in its domain, and extends h, since $h'(u) = s^*_{i(u)} = s_{i(u)} = h(u)$ for $u \in U$.

For the second family, Lemma 2.4 shows that $\operatorname{ctr} \circ h$ is a legal pattern whenever h is a partial homomorphism into the live target, because $(1,0)$ and $(0,1)$ are Moore directions. Restriction commutes with composition by ctr. For (Extension), write a member as $\operatorname{ctr} \circ h$ with $h \in H(\mathcal{W})$, extend h in $H(\mathcal{W})$, and compose the extension with ctr. □

Theorem 2.25 (Tokens and pebble families)**.** *Let R be a finite Wang set and $j \geq 1$.*

(i) R is j-token refutable if and only if there is no nonempty plane-safe set of j-token states.

(ii) For every nonempty plane-safe set $\mathcal{W}$ of j-token states, the restrictions of the maps $h_{\mathbf{u},\mathbf{s}}$, $(\mathbf{u},\mathbf{s}) \in \mathcal{W}$, form an E_j family for the live target, and their compositions with ctr *form an E_j family for R.*

(iii) If R is not j-token refutable, there is an E_j family for R. Equivalently, if there is no E_j family for R, then R is j-token refutable.

(iv) If R tiles the plane, then R is not j-token refutable.

(v) If there is an E_{j+1} family for the live target, then R is not j-token refutable.

Proof. Part (i) is Lemma 2.20, and part (ii) is Lemma 2.24. Part (iii) follows: if R is not j-token refutable, (i) gives a nonempty plane-safe set, and (ii) turns it into an E_j family for R.

For (iv), let z be a tiling and let $\mathcal{W}_z$ be the set of states $(\mathbf{u}, (\hat{z}(u_1),\dots,\hat{z}(u_j)))$ with $\mathbf{u} \in (\mathbb{Z}^2)^j$. It is nonempty. Property (W1) holds because equal positions carry equal symbols. The clone

of token i onto token i' is the state of $\mathcal{W}_z$ at the positions $\mathbf{u}[i' \leftarrow u_i]$, so (W2) holds. For (W3), the symbol $\hat{z}(u_i + d)$ is compatible with $\hat{z}(u_i)$ at displacement d, and the resulting move is the state of $\mathcal{W}_z$ at the positions $\mathbf{u}[i \leftarrow u_i + d]$. So $\mathcal{W}_z$ is plane-safe, and (i) shows that R is not j-token refutable. Concretely, for every r and n the tuple $(\hat{z}(0), \dots, \hat{z}(0))$ lies in $S_n^{(r)}(\mathbf{0})$.

For (v), let H be an E_{j+1} family for the live target, and let $\mathcal{W}$ be the set of j-token states $(\mathbf{u}, \mathbf{s})$ that satisfy (W1) and have $h_{\mathbf{u},\mathbf{s}} \in H$. Let $g \in H$, and let $(\mathbf{u}', \mathbf{s}')$ be a j-token state with $u_i' \in \operatorname{dom} g$ and $s_i' = g(u_i')$ for every i. Then $(\mathbf{u}', \mathbf{s}')$ satisfies (W1), and $h_{\mathbf{u}',\mathbf{s}'}$ is the restriction of g to $\{u_1', \dots, u_j'\}$, so it lies in H by (Restriction); hence $(\mathbf{u}', \mathbf{s}') \in \mathcal{W}$. By Remark 2.23, H contains the empty map and, for every $w \in \mathbb{Z}^2$, a map with domain $\{w\}$. The state with all j tokens at w, each carrying the value of that map, therefore lies in $\mathcal{W}$, and $\mathcal{W}$ is nonempty. Now let $(\mathbf{u}, \mathbf{s}) \in \mathcal{W}$ and $h = h_{\mathbf{u},\mathbf{s}}$. Every token of a clone of $(\mathbf{u}, \mathbf{s})$ sits at a site of $\operatorname{dom} h$ and carries the value of h there. So the map of the clone is a restriction of h, the clone lies in $\mathcal{W}$, and (W2) holds. For (W3), let i be an index and $d \in D$. The map h has at most j sites, fewer than $j + 1$, so (Extension) gives $h' \in H$ that extends h and has $u_i + d \in \operatorname{dom} h'$. Put $s' = h'(u_i + d)$. Since h' is a partial homomorphism into the live target and $h'(u_i) = s_i$, the live symbol s' is compatible with s_i at displacement d, so $(\mathbf{u}[i \leftarrow u_i + d], \mathbf{s}[i \leftarrow s'])$ is a move of token i by d. Every token of this move sits at a site of $\operatorname{dom} h'$ and carries the value of h' there. So the map of the moved state is a restriction of h', and the move lies in $\mathcal{W}$. Hence $\mathcal{W}$ is a nonempty plane-safe set, and (i) shows that R is not j-token refutable. □

Part (ii) applies to every nonempty plane-safe set. An argument may therefore first pass to a smaller nonempty plane-safe set with extra properties, and then take its family.

2.6 One token, and finite token width

Proposition 2.26 (One token is arc consistency)**.** *Let $L^{(0)} = L_R$, and for $n \geq 0$ let $L^{(n+1)}$ be the set of $s \in L^{(n)}$ such that for every $d \in D$ some $s' \in L^{(n)}$ is compatible with s at displacement d. The sets $L^{(n)}$ decrease, and $L^{(n)} = L^{(|L_R|)}$ for every $n \geq |L_R|$; call this set $L^{(\infty)}$. Then R is 1-token refutable if and only if $L^{(\infty)} = \emptyset$. In particular, 1-token refutability is decidable.*

Proof. By definition $L^{(n+1)} \subseteq L^{(n)}$. Once $L^{(n+1)} = L^{(n)}$, the sequence is constant from n on, and each strict decrease removes at least one of the $|L_R|$ symbols; this gives the second claim. For $j = 1$ there is no pair $i \neq i'$, so (W1) and (W2) hold for every set of 1-token states, and such a set is plane-safe exactly when (W3) holds.

If $L^{(\infty)} \neq \emptyset$, the set $\mathbb{Z}^2 \times L^{(\infty)}$ of 1-token states is plane-safe. Indeed $L^{(\infty)} = L^{(|L_R|+1)}$, so every $s \in L^{(\infty)}$ has, for every $d \in D$, a partner $s' \in L^{(\infty)}$ compatible with s at displacement d. By Theorem 2.25(i), R is not 1-token refutable.

Conversely, let $\mathcal{W}$ be a nonempty plane-safe set of 1-token states, and let Q be the set of symbols carried by its states. Then $Q \neq \emptyset$ and $Q \subseteq L^{(0)}$. If $Q \subseteq L^{(n)}$ and $(u, s) \in \mathcal{W}$, then for each $d \in D$, (W3) gives $(u + d, s') \in \mathcal{W}$ with s' compatible with s at displacement d, and $s' \in Q \subseteq L^{(n)}$; so $s \in L^{(n+1)}$. Hence $\emptyset \neq Q \subseteq L^{(\infty)}$. By Theorem 2.25(i), if R is not 1-token refutable then $L^{(\infty)} \neq \emptyset$.

The set $L^{(\infty)}$ is computed from R by a finite procedure, which gives decidability. □

The proof also shows that $L^{(\infty)}$ is the largest set of live symbols in which every symbol has a compatible partner in every Moore direction. This is arc consistency for the network on $\mathbb{Z}^2$ in which every site has domain L_R and Moore neighbours must carry compatible symbols.

Corollary 2.27 (One token is not enough)**.** *Let $\ell \mapsto R_\ell$ be a total computable map from $\mathbb{N}$ to finite Wang sets such that every R_ℓ that does not tile is 1-token refutable. Then $\{\ell : R_\ell \text{ tiles}\}$ is decidable.*

Proof. Given ℓ, compute $L^{(\infty)}$ for R_ℓ. If $L^{(\infty)} = \emptyset$, then R_ℓ is 1-token refutable by Proposition 2.26, so it does not tile by Theorem 2.25(iv). If $L^{(\infty)} \neq \emptyset$, then R_ℓ is not 1-token refutable, so it tiles by the hypothesis on the family. □

So the fixed number of tokens in Theorem 7.8 is at least 2. For $j \geq 2$ the tokens of a state can be arbitrarily far apart, the states up to translation form an infinite set, and the argument of Proposition 2.26 does not apply.

Proposition 2.28 (Finite token width)**.** *If R is j-token refutable, then it is $(j+1)$-token refutable. If no legal pattern exists on a square of side m, then R is m^2-token refutable. Hence the token width of R is finite if and only if R does not tile, and R is j-token refutable exactly when j is at least its token width.*

Proof. Let $\mathcal{W}$ be a nonempty plane-safe set of $(j+1)$-token states. Deleting the last token from every state gives a nonempty set of j-token states, and this set is plane-safe. Indeed (W1) is inherited, a clone among the first j tokens is the image of a clone in $\mathcal{W}$, and a move of one of the first j tokens is the image of a move in $\mathcal{W}$. By Theorem 2.25(i), if R is not $(j+1)$-token refutable, it is not j-token refutable. This is the first claim.

Suppose that no legal pattern exists on the square $U = \{0, \dots, m-1\}^2$, and that R is not m^2-token refutable. By Theorem 2.25(iii) there is an E_{m^2} family H for R. Since $|U| = m^2$, Remark 2.23 gives a member of H with domain U. It is a legal pattern on U, a contradiction. This is the second claim.

If R does not tile, Lemma 2.2 gives such a square, so the token width is finite. If R tiles, Theorem 2.25(iv) shows that it is not j-token refutable for any j. The last claim follows from the first. □

3 Bounded-width refutations give forgetting

This section proves Theorem 3.1. If R is j-token refutable and $k \geq \max(2, 2j)$, then the noisy rule $F_R^{[k]}$ forgets its initial configuration at every noise rate below a threshold that depends only on k. The refutation time enters the bound only through a constant factor.

The proof has four steps. First, a violated tuple of tokens that carries no replacement has violated parent tuples in at least $k+1-j$ provider planes, all on one common set of at most j planar sites (Lemma 3.4). Second, a pigeonhole step turns this into a binary field E. The field is computed from the replacement indicators alone, and after the refutation time it covers every live site of every run (Lemma 3.6 and Corollary 3.7). Third, Toom's contour argument bounds the size of an explanation of a marked point by the number of replacements in it (Lemma 3.8). Fourth, contours placed along a discrepancy path in disjoint time slabs force linearly many distinct replacements (Lemma 3.10), and a Peierls count ends the proof (Section 3.6). The field E (Definition 3.5) and the contour lemma (Lemma 3.8) are stated in the generality in which the proof of Theorem 4.2 uses them again.

3.1 Setting and statement

Throughout this section $j \geq 1$ and $k \geq \max(2, 2j)$ are integers, $q = k+1$, and R is a Wang set with live symbols L_R. We write $F = F_R^{[k]}$. A site of $\mathbb{Z}^{2+k} = \mathbb{Z}^2 \times \mathbb{Z}^k$ is written $v = (u, P)$, with planar part $u \in \mathbb{Z}^2$ and transverse part $P = (P_1, \dots, P_k) \in \mathbb{Z}^k$. The providers are $c_0 = 0$ and

$c_i = e_i$ for $1 \le i \le k$, and $\bar{D} = D \cup \{0\} = \{-1,0,1\}^2$. The value $C(x(\cdot,P))(u)$ depends only on the symbols $x(u+d,P)$ with $d \in \bar{D}$. Hence $F(x)(v)$ depends only on the restriction of x to

$$\mathcal{N}(v) = \big\{(u+d,\, P+c_i) :\ d \in \bar{D},\ 0 \le i \le k\big\}. \tag{1}$$

Noise and coupling. Let $\eta_t(v) \in \{0,1\}$ and $\zeta_t(v) \in \mathcal{A}_R$, for $t \ge 1$ and $v \in \mathbb{Z}^{2+k}$, be independent random variables with $\mathbb{P}(\eta_t(v) = 1) = \varepsilon$ and $\zeta_t(v)$ uniform on $\mathcal{A}_R$. The run from a configuration x is given by $X_0 = x$ and, for $t \ge 1$,

$$X_t(v) = \begin{cases} \zeta_t(v) & \text{if } \eta_t(v) = 1, \\ F(X_{t-1})(v) & \text{if } \eta_t(v) = 0. \end{cases} \tag{2}$$

Its law is Law_x. A space-time point (v,t) with $t \ge 1$ and $\eta_t(v) = 1$ is an *event*. Being an event does not depend on the value $\zeta_t(v)$: a replacement whose value equals $F(X_{t-1})(v)$ is still an event. The *common-noise coupling* runs X from x and Y from $\perp$ with the same variables η and ζ. Then Y has law $\mathrm{Law}_\perp$.

Token tables. We recall the token tables. Fix $r \ge 0$ and the box $V = \{-r,\dots,r\}^2$. A *table* S assigns to each position tuple $\mathbf{z} = (z_1,\dots,z_j) \in V^j$ a set $S(\mathbf{z})$ of symbol tuples $\mathbf{b} = (b_1,\dots,b_j) \in L_R^j$. Token l sits at z_l and carries b_l. We write $\mathbf{b}[l \leftarrow b]$ and $\mathbf{z}[l \leftarrow z]$ for the tuples with entry l replaced. We write $S \subseteq S'$ if $S(\mathbf{z}) \subseteq S'(\mathbf{z})$ for every $\mathbf{z}$. Three rules delete $\mathbf{b}$ from $S(\mathbf{z})$, with S itself as the reference table:

- the *coincidence rule*, if $z_l = z_{l'}$ and $b_l \ne b_{l'}$ for some l, l';
- the *clone rule*, if $\mathbf{b}[l' \leftarrow b_l] \notin S(\mathbf{z}[l' \leftarrow z_l])$ for some $l \ne l'$;
- the *move rule*, if for some l and some $d \in D$ with $z_l + d \in V$ there is no live symbol b such that b_l at z_l and b at $z_l + d$ are compatible and $\mathbf{b}[l \leftarrow b] \in S(\mathbf{z}[l \leftarrow z_l + d])$.

$\mathrm{Inst}(S)$ is obtained from S by applying the coincidence and clone rules, one deletion at a time, until neither applies. $\mathrm{Timed}(S)$ is obtained from S by deleting at once every tuple that the move rule deletes with reference S. Let $S_{\mathrm{all}}(\mathbf{z}) = L_R^j$ for every $\mathbf{z}$, and put

$$S_0 = \mathrm{Inst}(S_{\mathrm{all}}), \qquad S_{n+1} = \mathrm{Inst}(\mathrm{Timed}(S_n)) \quad (n \ge 0).$$

The set R is *j-token refutable with time τ* if $S_\tau(0,\dots,0) = \emptyset$ for some r.

For $w \in \mathbb{Z}^2$ and $\mathbf{z} \in (w+V)^j$ put $S_n^w(\mathbf{z}) = S_n(\mathbf{z} - w)$, where $\mathbf{z} - w = (z_1 - w, \dots, z_j - w)$. Fix a run X. A pair $(\mathbf{z}, \mathbf{b})$ with $\mathbf{z} \in (w+V)^j$ and $\mathbf{b} \in L_R^j$ is *present at* (P,t) if $X_t(z_l, P) = b_l$ for every l. It is *clean* if none of the points $((z_l,P),t)$ is an event. It *violates* S_n^w if $\mathbf{b} \notin S_n^w(\mathbf{z})$.

Theorem 3.1 (forgetting). *Let $j \ge 1$ and $k \ge \max(2, 2j)$, and put $q = k+1$, $a = 2q^2+1$ and $D_0 = 50q$. Let R be a Wang set that is j-token refutable with time τ. Let $0 < \varepsilon < 1$ be such that $\beta = 2D_0^2 \varepsilon^{1/a} < 1$. Under the common-noise coupling, for every configuration x, every site v and every time $t \ge 0$,*

$$\mathbb{P}\big(X_t(v) \ne Y_t(v)\big) \le 2\varepsilon^{-\tau/a}\, \frac{\beta^t}{1-\beta}. \tag{3}$$

Consequently, for every finite set $O \subset \mathbb{Z}^{2+k}$ and every $t \ge 0$,

$$\sup_x\ d_{\mathrm{TV}}\big(\mathrm{Law}_x(X_t|_O),\, \mathrm{Law}_\perp(X_t|_O)\big) \le 2|O|\, \varepsilon^{-\tau/a}\, \frac{\beta^t}{1-\beta}, \tag{4}$$

which tends to 0 as $t \to \infty$. So the noisy rule forgets its initial configuration, and it is ergodic.

Remark 3.2. The rate β depends only on k and ε. The set R, the box and the time τ enter only through the factor $\varepsilon^{-\tau/a}$. If $0 < \varepsilon \le \varepsilon_k = (4D_0^2)^{-a}$, then $\beta \le 2D_0^2(4D_0^2)^{-1} = 1/2$. The hypothesis $k \ge 2j$ is used once, in the pigeonhole step of Lemma 3.6.

3.2 Instant rules and transport

A table S' is *closed* if neither the coincidence rule nor the clone rule, with reference S', deletes a tuple of S', as in Lemma 2.15.

Lemma 3.3. *Let S be a table.*

(a) $\mathrm{Inst}(S)$ *is the largest closed table contained in S. In particular it does not depend on the order of the deletions.*

(b) Let $\mathcal{Q}$ be a set of pairs $(\mathbf{z}, \mathbf{b}) \in V^j \times L_R^j$ with three properties: if $z_l = z_{l'}$ then $b_l = b_{l'}$; if $(\mathbf{z}, \mathbf{b}) \in \mathcal{Q}$ and $l \ne l'$, then $(\mathbf{z}[l' \leftarrow z_l], \mathbf{b}[l' \leftarrow b_l]) \in \mathcal{Q}$; and $\mathbf{b} \in S(\mathbf{z})$ for every $(\mathbf{z}, \mathbf{b}) \in \mathcal{Q}$. Then $\mathbf{b} \in \mathrm{Inst}(S)(\mathbf{z})$ for every $(\mathbf{z}, \mathbf{b}) \in \mathcal{Q}$.

Proof. (a) This is Lemma 2.15(i).

(b) Let $S^* = \mathrm{Inst}(S)$, which is closed by (a), and let $S'(\mathbf{z}) = S^*(\mathbf{z}) \cup \{\mathbf{b} : (\mathbf{z}, \mathbf{b}) \in \mathcal{Q}\}$. Then $S' \subseteq S$. The table S' is closed. For a pair of $\mathcal{Q}$, the coincidence rule does not apply by the first property, and the clone rule consults a member of $\mathcal{Q}$ by the second. A tuple of S^* is kept because S^* is closed and $S^* \subseteq S'$. By (a), $S' \subseteq S^* = \mathrm{Inst}(S)$. □

Present pairs give such families. Fix a run, w, P, t and a set $Z \subseteq w + V$. Let $\mathcal{Q}$ consist of the pairs $(\mathbf{z} - w, \mathbf{b})$ such that $(\mathbf{z}, \mathbf{b})$ is present at (P, t) and all positions of $\mathbf{z}$ lie in Z. Two tokens on one site read the one symbol there, and cloning a token of a present pair gives a present pair. So $\mathcal{Q}$ has the first two properties of Lemma 3.3(b).

The rule F commutes with planar translations. Replacing $X_t(u, P)$ by $X_t(u + w, P)$, and the noise variables likewise, therefore turns a run into a run, present pairs on $w + V$ into present pairs on V, and S_n^w into S_n. So it suffices to state the next lemma for every w and to prove it for $w = 0$.

Lemma 3.4 (transport)**.** *Fix a configuration x, a realisation of the noise, and the run X from x. Let $w \in \mathbb{Z}^2$, $P \in \mathbb{Z}^k$, $t \ge 1$ and $n \ge 1$. Let $(\mathbf{z}, \mathbf{b})$, with $\mathbf{z} \in (w + V)^j$, be a clean pair present at (P, t) that violates S_n^w. Then there exist*

- *a set $\Gamma \subseteq w + V$ with $|\Gamma| \le j$,*
- *a map $\pi : \Gamma \to \{z_1, \dots, z_j\}$ with $z - \pi(z) \in \bar{D}$ for every $z \in \Gamma$, and*
- *a set $I \subseteq \{0, \dots, k\}$ with $|I| \ge k + 1 - j$,*

such that the following holds for every $i \in I$: there is a pair $(\mathbf{z}', \mathbf{b}')$ with $\mathbf{z}' \in \Gamma^j$ that is present at $(P + c_i, t - 1)$ and violates S_{n-1}^w.

In the statement, Γ and π are chosen before the index i. The same set Γ and the same map π serve every $i \in I$; only the pair $(\mathbf{z}', \mathbf{b}')$ depends on i.

Proof. We take $w = 0$ and write S_n for S_n^0. For $t \ge 1$ let y_{t-1} be the configuration obtained by applying C to X_{t-1} in every plane, so that $y_{t-1}(u, P') = C(X_{t-1}(\cdot, P'))(u)$.

Step 1: a tuple that fails the move round. Let $\mathcal{Q}$ be the set of pairs present at (P, t) whose positions lie in $\{z_1, \dots, z_j\}$. It has the first two properties of Lemma 3.3(b). If every member $(\mathbf{z}', \mathbf{b}')$ of $\mathcal{Q}$ had $\mathbf{b}' \in \mathrm{Timed}(S_{n-1})(\mathbf{z}')$, then Lemma 3.3(b) with $S = \mathrm{Timed}(S_{n-1})$ would give $\mathcal{Q} \subseteq \mathrm{Inst}(\mathrm{Timed}(S_{n-1})) = S_n$. This is false, since $(\mathbf{z}, \mathbf{b}) \in \mathcal{Q}$ and $\mathbf{b} \notin S_n(\mathbf{z})$. So some

$(\mathbf{z}^*, \mathbf{b}^*) \in \mathcal{Q}$ has $\mathbf{b}^* \notin \mathrm{Timed}(S_{n-1})(\mathbf{z}^*)$. Then either $\mathbf{b}^* \notin S_{n-1}(\mathbf{z}^*)$, or $\mathbf{b}^* \in S_{n-1}(\mathbf{z}^*)$ and the move rule with reference S_{n-1} deletes it.

Step 2: common providers. The positions of $\mathbf{z}^*$ form a set of at most j sites among $z_1, \dots, z_j$. Let z be one of them. Say that the index i *succeeds* at z if $y_{t-1}(z, P + c_i) = X_t(z, P)$. The point $((z, P), t)$ is not an event, because $(\mathbf{z}, \mathbf{b})$ is clean, so $X_t(z, P) = F(X_{t-1})(z, P)$, and this value is live. By the definition of F, at least k of the $k + 1$ indices succeed at z, so at most one fails. Let I be the set of indices that succeed at every position of $\mathbf{z}^*$. Then $|I| \geq k + 1 - j$. Let $i \in I$ and $1 \leq l \leq j$. Then $y_{t-1}(z_l^*, P + c_i) = X_t(z_l^*, P) = b_l^*$, which is live. By the definition of C, $X_{t-1}(z_l^*, P + c_i) = b_l^*$, and for every $d \in D$ the symbol $X_{t-1}(z_l^* + d, P + c_i)$ is live and compatible with b_l^* (the latter at z_l^*, the former at $z_l^* + d$).

Step 3: the parents. Suppose first that $\mathbf{b}^* \notin S_{n-1}(\mathbf{z}^*)$. Let Γ be the set of positions of $\mathbf{z}^*$ and let π be the identity. By Step 2, $(\mathbf{z}^*, \mathbf{b}^*)$ is present at $(P + c_i, t - 1)$ for every $i \in I$, and it violates S_{n-1}.

Otherwise the move rule deletes $\mathbf{b}^*$: there are l and $d \in D$ with $z_l^* + d \in V$ such that no live symbol b compatible with b_l^* (with b_l^* at z_l^* and b at $z_l^* + d$) has $\mathbf{b}^*[l \leftarrow b] \in S_{n-1}(\mathbf{z}')$, where $\mathbf{z}' = \mathbf{z}^*[l \leftarrow z_l^* + d]$. Let Γ be the set of positions of $\mathbf{z}'$. Put $\pi(z_l^* + d) = z_l^*$ and $\pi(z) = z$ for the other points z of Γ; then $z - \pi(z) \in \bar{D}$ for every $z \in \Gamma$. For $i \in I$ let $b_i = X_{t-1}(z_l^* + d, P + c_i)$. By Step 2, b_i is live and compatible with b_l^*, so $\mathbf{b}^*[l \leftarrow b_i] \notin S_{n-1}(\mathbf{z}')$. The pair $(\mathbf{z}', \mathbf{b}^*[l \leftarrow b_i])$ is present at $(P + c_i, t - 1)$: the position $z_{l'}^*$, $l' \neq l$, carries $b_{l'}^*$ by Step 2, and the position $z_l^* + d$ carries b_i. If $z_l^* + d$ equals some $z_{l'}^*$, both statements describe the one symbol there.

In both cases $\Gamma \subseteq V$ and $|\Gamma| \leq j$. The positions of $\mathbf{z}^*$ lie in $\{z_1, \dots, z_j\}$, so π maps into this set. The set I was fixed in Step 2, and Γ and π are determined by $\mathbf{z}^*$, $\mathbf{b}^*$ and, in the second case, l and d. All of these were chosen before any $i \in I$ was considered. □

The lemma is the only place where the quorum enters. Every provider that succeeds at all positions of the tuple carries a violating parent, and all these parents sit on the same set Γ.

3.3 A covering field computed from the noise

As in Toom's stability proofs [29, 10], we dominate the errors by a binary field that is driven by the noise alone.

Definition 3.5 (domination field)**.** Let $\eta = (\eta_t(v))_{t \geq 1,\, v \in \mathbb{Z}^{2+k}}$ be the replacement indicators. The *domination field* $E_t(v) \in \{0, 1\}$, for $t \geq 0$ and $v \in \mathbb{Z}^{2+k}$, is defined recursively by $E_0 \equiv 0$ and, for $t \geq 1$ and $v = (u, P)$, as follows. If $\eta_t(v) = 1$, then $E_t(v) = 1$. If $\eta_t(v) = 0$, then $E_t(v) = 1$ if and only if there are indices $i \neq i'$ in $\{0, \dots, k\}$ and offsets $d, d' \in \bar{D}$ with

$$E_{t-1}(u + d,\ P + c_i) = E_{t-1}(u + d',\ P + c_{i'}) = 1. \tag{5}$$

The field E is a function of the indicators η alone. It does not depend on R, on the initial configuration or on the replacement values ζ. In (5) the two points with value 1 lie in different planes, their planar offsets may differ, and the offset 0 is allowed.

Lemma 3.6 (cover)**.** *Let $k \geq 2j$. Fix a configuration x, a realisation of the noise and the run X from x. Let $0 \leq n \leq t$, $w \in \mathbb{Z}^2$ and $P \in \mathbb{Z}^k$. If a pair $(\mathbf{z}, \mathbf{b})$ with $\mathbf{z} \in (w + V)^j$ is present at (P, t) and violates S_n^w, then $E_t(z_l, P) = 1$ for some l.*

Proof. We prove by induction on $n \geq 0$ the statement

(C_n) for every $t \geq n$, every $w \in \mathbb{Z}^2$, every $P \in \mathbb{Z}^k$ and every pair $(\mathbf{z}, \mathbf{b})$ with $\mathbf{z} \in (w + V)^j$ that is present at (P, t) and violates S_n^w, there is an index l with $E_t(z_l, P) = 1$.

The configuration x and the realisation of the noise stay fixed throughout.

The case $n = 0$. Fix t, w and P. The family $\mathcal{Q}$ of all pairs $(\mathbf{z} - w, \mathbf{b})$ with $(\mathbf{z}, \mathbf{b})$ present at (P, t) has the first two properties of Lemma 3.3(b), and it lies in S_{all}. So every present pair lies in S_0^w. No pair satisfies the hypothesis of (C_0), and (C_0) holds.

From $n-1$ to n. Let $n \geq 1$ and assume (C_{n-1}). Fix $t \geq n$, w, P and a pair $(\mathbf{z}, \mathbf{b})$ with $\mathbf{z} \in (w + V)^j$ that is present at (P, t) and violates S_n^w.

If some point $((z_l, P), t)$ is an event, then $E_t(z_l, P) = 1$ by Definition 3.5, since $t \geq 1$. We are done in this case.

Otherwise the pair is clean. Since $t \geq 1$ and $n \geq 1$, Lemma 3.4 applies. It gives a set Γ, a map π and a set I as there. We fix them now; nothing below changes them. For each $i \in I$ the lemma gives a pair $(\mathbf{z}^{(i)}, \mathbf{b}^{(i)})$ with $\mathbf{z}^{(i)} \in \Gamma^j$ that is present at $(P + c_i, t - 1)$ and violates S_{n-1}^w; fix one such pair for each i. Since $0 \leq n - 1 \leq t - 1$, the statement (C_{n-1}), applied with time $t - 1$, translate w, plane $P + c_i$ and the pair $(\mathbf{z}^{(i)}, \mathbf{b}^{(i)})$, gives an index l_i with

$$E_{t-1}\big(z_i',\, P + c_i\big) = 1, \qquad \text{where } z_i' = z_{l_i}^{(i)} \in \Gamma.$$

Now consider the map $i \mapsto \pi(z_i')$ from I to $\{z_1, \dots, z_j\}$. Its domain has $|I| \geq k + 1 - j \geq j + 1$ elements, because $k \geq 2j$, and its target has at most j elements. So there are indices $i \neq i'$ in I with $\pi(z_i') = \pi(z_{i'}')$; call this point z_l. This comparison is possible because π is one map, fixed before the indices i were considered. Put $d = z_i' - z_l$ and $d' = z_{i'}' - z_l$. Since $z_l = \pi(z_i')$, the defining property of π gives $d \in \bar{D}$, and likewise $d' \in \bar{D}$. So

$$E_{t-1}(z_l + d,\, P + c_i) = E_{t-1}(z_l + d',\, P + c_{i'}) = 1, \qquad i \neq i'.$$

The point $((z_l, P), t)$ is not an event, because the pair is clean. By Definition 3.5, $E_t(z_l, P) = 1$. This proves (C_n). □

Corollary 3.7. *Let $k \geq 2j$, and let $S_\tau(0, \dots, 0) = \emptyset$ for the box V. Fix a configuration x, a realisation of the noise and the run X from x. For every $t \geq \tau$ and every site v, if $X_t(v)$ is live then $E_t(v) = 1$.*

Proof. Let $v = (u, P)$ and $X_t(v) = b$ live. The diagonal pair $((u, \dots, u), (b, \dots, b))$ is present at (P, t) on $u + V$. It violates S_τ^u, because $S_\tau^u(u, \dots, u) = S_\tau(0, \dots, 0) = \emptyset$. Lemma 3.6 with $n = \tau \leq t$ and $w = u$ gives $E_t(u, P) = 1$, since all positions equal u. □

The corollary holds for every initial configuration and every realisation of η and ζ, with the same field E. In particular it holds for both runs of the common-noise coupling.

3.4 Contours

Let $\mathcal{G}$ be the graph with vertex set $\mathbb{Z}^2 \times \mathbb{Z}^k \times \{0, 1, 2, \dots\}$ in which each vertex (u, P, t) with $t \geq 1$ is joined to the points $(u + d, P + c_i, t - 1)$, $d \in \bar{D}$, $0 \leq i \leq k$. We call $(u + d, P + c_i, t - 1)$ a *lower neighbour of (u, P, t) through provider i*. We identify (u, P, t) with the space-time point (v, t), $v = (u, P)$. Every edge of $\mathcal{G}$ joins two consecutive times. A vertex has at most $9q$ lower and $9q$ upper neighbours, so every degree is at most $18q \leq D_0$. By (1), $(v', t - 1)$ is a lower neighbour of (v, t) whenever $v' \in \mathcal{N}(v)$.

The next lemma is Toom's contour argument [29]; see also [10]. The poles, the charges, the linear functions that sum to zero, and the splitting of frontier components in Step 3 below follow [28, Theorem 33, Definitions 35 and 37, Lemma 36]. We state it for any set of vertices in which every non-fault has two lower neighbours through different providers. Only the sinks need to be distinct faults; arms may meet and may pass through the same vertices. We include the proof to keep the section self-contained.

Lemma 3.8 (contours)**.** *Let $\mathcal{U}$ be a set of vertices of $\mathcal{G}$, all of positive time, and let $\mathcal{U}_\mathrm{f} \subseteq \mathcal{U}$; we call the points of $\mathcal{U}_\mathrm{f}$* faults*. Suppose that every vertex of $\mathcal{U} \setminus \mathcal{U}_\mathrm{f}$ has two lower neighbours in $\mathcal{U}$ through different providers. Let $\rho = (v, t_\rho) \in \mathcal{U} \setminus \mathcal{U}_\mathrm{f}$. Then there are an integer $m \geq 2$, a set $\Sigma \subseteq \mathcal{U}_\mathrm{f}$ of m faults and a finite set K of edges of $\mathcal{G}$ with the following properties.*

(i) The edges of K form a connected graph. Each edge of K joins a vertex of $\mathcal{U} \setminus \mathcal{U}_\mathrm{f}$ to one of its lower neighbours in $\mathcal{U}$. The vertices of K lie in $\mathcal{U}$ and include ρ and every point of Σ.

(ii) $1 \leq |K| \leq q^2(m-1)$. In particular $|\Sigma| = m \geq 1 + |K|/q^2$.

(iii) Let s be the least time of a point of Σ. Every vertex of K has time in $[s, t_\rho]$, and $t_\rho - s \leq q^2(m-1)$.

In particular, fix a realisation of the indicators η, let E be the domination field of Definition 3.5, and let $\rho = (v, t_\rho)$ with $t_\rho \geq 1$, $E_{t_\rho}(v) = 1$ and $\eta_{t_\rho}(v) = 0$. Then there is a set K of $n = |K| \geq 1$ edges of $\mathcal{G}$, each joining a vertex to one of its lower neighbours, that forms a connected graph containing ρ, whose vertices include m distinct events, and with $n \leq q^2(m-1)$.

A vertex (v, t) is *marked* if $E_t(v) = 1$. The last assertion of Lemma 3.8 is the case in which $\mathcal{U}$ is the set of marked vertices and $\mathcal{U}_\mathrm{f}$ is the set of events. These sets satisfy the hypothesis of the lemma. Indeed, marked vertices have positive time because $E_0 \equiv 0$; events are marked by Definition 3.5; and by (5) a marked vertex that is not an event has two marked lower neighbours through different providers, at offsets in $\bar{D}$.

Proof of Lemma 3.8. The last assertion follows from the general one, applied to the sets of the preceding paragraph, which satisfy its hypothesis; we prove the general assertion.

Step 1: the support graph. For every vertex $\theta \in \mathcal{U} \setminus \mathcal{U}_\mathrm{f}$ fix once and for all two lower neighbours $\theta^{(1)}, \theta^{(2)} \in \mathcal{U}$ of θ through different providers, the *supports* of θ. They lie in different planes, so they are distinct. Let $\mathcal{B}$ be the set of vertices reached from ρ by following supports, including ρ. It is contained in $\mathcal{U}$. Times decrease by one along each support and stay positive, so $\mathcal{B}$ is finite. The faults in $\mathcal{B}$ are the *terminals*; let $\Phi = \mathcal{B} \cap \mathcal{U}_\mathrm{f}$ be the set of terminals. Every other vertex of $\mathcal{B}$, including ρ, has both supports in $\mathcal{B}$. For $\theta \in \mathcal{B}$ let $\Phi(\theta)$ be the set of terminals reached from θ. Following supports from θ ends at a terminal, so $\Phi(\theta) \neq \emptyset$. Moreover $\Phi(\rho) = \Phi$, and

$$\Phi(\varphi) = \{\varphi\} \text{ for } \varphi \in \Phi, \qquad \Phi(\theta) = \Phi(\theta^{(1)}) \cup \Phi(\theta^{(2)}) \text{ for } \theta \in \mathcal{B} \setminus \Phi. \tag{6}$$

Step 2: charges, arms and the potential. The *charges* are the indices $i \in \{0, \dots, k\}$. An edge from a vertex to its lower neighbour through provider i' is *good* for charge i if $i' \neq i$. The two supports of a non-terminal vertex θ come through different providers, so for each charge i at least one of them is reached by a good edge; fix such a support $g_i(\theta)$. A *pole* σ consists of a non-terminal vertex $\theta_\sigma \in \mathcal{B}$, its *location*, and a *first step* $f_i(\sigma) \in \{\theta_\sigma^{(1)}, \theta_\sigma^{(2)}\}$ for each charge i. Its *arm* of charge i is the path

$$\theta_\sigma,\ f_i(\sigma),\ g_i(f_i(\sigma)),\ g_i(g_i(f_i(\sigma))),\ \dots$$

stopped at its first terminal. Let $\mathrm{end}_i(\sigma) \in \Phi(f_i(\sigma))$ be the last vertex of this arm. Every arm has at least one edge, and every edge of an arm except possibly the first is good for its charge. The *root pole* σ_ρ has location ρ and first steps $f_i(\sigma_\rho) = g_i(\rho)$, so all its edges are good.

For $P \in \mathbb{Z}^k$ and $t \geq 0$ put

$$\psi_0(P, t) = q(P_1 + \cdots + P_k) + kt, \qquad \psi_i(P, t) = -qP_i - t \quad (1 \leq i \leq k),$$

and $\psi_i(\theta) = \psi_i(P,t)$ for $\theta = (u,P,t)$. Then

$$\sum_{i=0}^{k} \psi_i(\theta) = 0 \qquad \text{for every vertex } \theta. \tag{7}$$

Let θ' be a lower neighbour of θ through provider i'. The function ψ_0 changes by $-k$ if $i' = 0$ and by $q - k = 1$ otherwise. For $i \geq 1$, the function ψ_i changes by $1 - q = -k$ if $i' = i$ and by 1 otherwise. Hence

$$\psi_i(\theta') - \psi_i(\theta) = \begin{cases} 1 & \text{if } i' \neq i, \\ -k & \text{if } i' = i. \end{cases} \tag{8}$$

So a good edge raises the function of its charge by exactly 1, and any edge lowers it by at most k.

Step 3: the poles. List the non-terminal vertices of $\mathcal{B}$ as $\theta_1 = \rho, \theta_2, \ldots, \theta_N$ in order of non-increasing time. Put $W_0 = \{\rho\}$. For $1 \leq h \leq N$ let W_h be the set of vertices of $\mathcal{B} \setminus \{\theta_1, \ldots, \theta_h\}$ that are supports of at least one of $\theta_1, \ldots, \theta_h$. On W_h join θ and θ' if $\Phi(\theta) \cap \Phi(\theta') \neq \emptyset$. For a connected component $\mathcal{K}$ of this graph put $\Phi(\mathcal{K}) = \bigcup_{\theta \in \mathcal{K}} \Phi(\theta)$. We use four facts.

(a) For $0 \leq h < N$, we have $\theta_{h+1} \in W_h$ and $W_{h+1} = (W_h \setminus \{\theta_{h+1}\}) \cup \{\theta_{h+1}^{(1)}, \theta_{h+1}^{(2)}\}$. For $h = 0$ this is the definition. Let $h \geq 1$. The vertex $\theta_{h+1} \neq \rho$ is a support of some vertex of $\mathcal{B}$. That vertex is non-terminal and has larger time, so it is among $\theta_1, \ldots, \theta_h$. The supports of θ_{h+1} have smaller time than θ_{h+1}, so they are not among $\theta_1, \ldots, \theta_{h+1}$. The formula follows.

(b) For each h, the sets $\Phi(\mathcal{K})$, over the components $\mathcal{K}$ of W_h, partition Φ. They are disjoint by the definition of the graph. Their union $\bigcup_{\theta \in W_h} \Phi(\theta)$ equals $\Phi(\rho) = \Phi$ for $h = 0$, and by (a) and (6) it does not change from h to $h+1$.

(c) $W_N = \Phi$, and the components of W_N are single terminals. Indeed W_N contains no non-terminal vertex, so $\Phi(\theta) = \{\theta\}$ for $\theta \in W_N$, and (b) gives $W_N = \Phi$. Two distinct terminals are not joined.

(d) Let $\theta = \theta_{h+1}$, let $\mathcal{K}$ be its component in W_h, and let $\mathcal{K}' = (\mathcal{K} \setminus \{\theta\}) \cup \{\theta^{(1)}, \theta^{(2)}\}$. Then the components of W_{h+1} are the components of W_h other than $\mathcal{K}$, together with one or two components whose union is $\mathcal{K}'$. Each of the latter contains $\theta^{(1)}$ or $\theta^{(2)}$, and $\Phi(\mathcal{K}') = \Phi(\mathcal{K})$.

To prove (d), note first that a support $\theta^{(l)}$ lying in W_h lies in $\mathcal{K}$, because $\emptyset \neq \Phi(\theta^{(l)}) \subseteq \Phi(\theta)$. So W_{h+1} is the disjoint union of $W_h \setminus \mathcal{K}$ and $\mathcal{K}'$. By (6), $\Phi(\mathcal{K}') = \Phi(\mathcal{K})$, and this set is disjoint from $\Phi(\theta')$ for every $\theta' \in W_h \setminus \mathcal{K}$. So no vertex of $\mathcal{K}'$ is joined to a vertex outside $\mathcal{K}'$, and the components of W_h other than $\mathcal{K}$ remain components. Now let $\theta' \in \mathcal{K} \setminus \{\theta\}$, and take a shortest path $\theta' = \theta'_0, \theta'_1, \ldots, \theta'_p = \theta$ in $\mathcal{K}$. Its vertices other than θ lie in $\mathcal{K}'$, and $\Phi(\theta'_{p-1})$ meets $\Phi(\theta) = \Phi(\theta^{(1)}) \cup \Phi(\theta^{(2)})$. So θ' is joined inside $\mathcal{K}'$ to $\theta^{(1)}$ or to $\theta^{(2)}$.

We now choose finite sets $\mathcal{P}_0 \subseteq \mathcal{P}_1 \subseteq \cdots \subseteq \mathcal{P}_N$ of poles with the following property.

(I_h) For every component $\mathcal{K}$ of W_h and every charge i, exactly one pole $\sigma \in \mathcal{P}_h$ has $\mathrm{end}_i(\sigma) \in \Phi(\mathcal{K})$.

Put $\mathcal{P}_0 = \{\sigma_\rho\}$. Property ($\mathrm{I}_0$) holds, since W_0 has the single component $\{\rho\}$ and $\Phi(\rho) = \Phi$. Suppose (I_h) holds, and let θ, $\mathcal{K}$ and $\mathcal{K}'$ be as in (d). If $\mathcal{K}'$ is connected, put $\mathcal{P}_{h+1} = \mathcal{P}_h$. Otherwise $\mathcal{K}'$ has two components, $\mathcal{K}_1 \ni \theta^{(1)}$ and $\mathcal{K}_2 \ni \theta^{(2)}$. For each charge i, the unique pole of $\mathcal{P}_h$ whose arm of charge i ends in $\Phi(\mathcal{K}) = \Phi(\mathcal{K}_1) \cup \Phi(\mathcal{K}_2)$ ends in exactly one of these two sets; let $\iota(i) \in \{1,2\}$ be the index of the other one. Let $\mathcal{P}_{h+1}$ be $\mathcal{P}_h$ together with the pole σ_θ located at θ with first steps $f_i(\sigma_\theta) = \theta^{(\iota(i))}$. Its arm of charge i ends in $\Phi(\theta^{(\iota(i))}) \subseteq \Phi(\mathcal{K}_{\iota(i)})$. In both cases ($\mathrm{I}_{h+1}$) follows from (d): the other components keep their terminal sets and their poles, and the arms of the new pole end in $\Phi(\mathcal{K})$.

Let $\mathcal{P} = \mathcal{P}_N$. By (c) and (I_N),

$$\text{for every terminal } \varphi \text{ and every charge } i, \text{ exactly one pole } \sigma \in \mathcal{P} \text{ has } \mathrm{end}_i(\sigma) = \varphi. \tag{9}$$

Every pole other than σ_ρ was added at a split; its first steps are arbitrary supports, and all later edges of its arms are good.

Step 4: the contour. Build an abstract graph H as follows. Take a separate copy of every arm of every pole of $\mathcal{P}$, as a path graph whose vertices are labelled by the vertices of $\mathcal{B}$ that the arm visits. Identify the initial vertices of the q arms of each pole σ into one vertex, the *source* of σ. Identify all final vertices labelled by the same terminal φ into one vertex, the *sink* of φ. The other vertices of H are *interior*. Each edge of H joins two vertices whose labels are joined in $\mathcal{G}$.

Let H_ρ be the connected component of H that contains the source of σ_ρ. Let $\mathcal{P}_\rho$ be the set of poles whose sources lie in H_ρ, and Φ_ρ the set of terminals whose sinks lie in H_ρ. The arms of a pole in $\mathcal{P}_\rho$ lie in H_ρ, and every edge of H_ρ lies on such an arm. For each charge i, the map end_i restricts to a bijection from $\mathcal{P}_\rho$ onto Φ_ρ. Indeed, end_i is injective on $\mathcal{P}$ by (9); an arm ends in the component of its source; and by (9) the unique arm of charge i that ends at a sink of H_ρ starts at a source of H_ρ. Let $m = |\mathcal{P}_\rho| = |\Phi_\rho|$, and let e_ρ be the number of edges of H_ρ.

Consider the sum of $\psi_i(\mathrm{end}_i(\sigma)) - \psi_i(\theta_\sigma)$ over all $\sigma \in \mathcal{P}_\rho$ and all charges i. By the bijections, it equals

$$\sum_{\varphi \in \Phi_\rho} \sum_{i=0}^{k} \psi_i(\varphi) - \sum_{\sigma \in \mathcal{P}_\rho} \sum_{i=0}^{k} \psi_i(\theta_\sigma) = 0,$$

by (7). On the other hand, each difference is the sum of the increments (8) along the edges of the arm. A good edge contributes 1 and any other edge at least $-k$. Only the first edges of the poles in $\mathcal{P}_\rho \setminus \{\sigma_\rho\}$ can fail to be good, and there are at most $q(m-1)$ of them. Hence

$$0 \geq \big(e_\rho - q(m-1)\big) - kq(m-1) = e_\rho - q^2(m-1),$$

so $e_\rho \leq q^2(m-1)$. The arms of σ_ρ have at least one edge each, so $e_\rho \geq q \geq 1$, and therefore $m \geq 2$.

Let K be the set of edges of $\mathcal{G}$ that are labels of edges of H_ρ, and let $\Sigma = \Phi_\rho$. Then $|K| \leq e_\rho \leq q^2(m-1)$, and K contains the first edges of the arms of σ_ρ, so $|K| \geq 1$. This is (ii). The labelling maps the connected graph H_ρ onto the graph formed by K, so the latter is connected. Each edge of an arm goes from a non-terminal vertex $\theta \in \mathcal{B}$ to one of its supports, so each edge of K joins a vertex of $\mathcal{U} \setminus \mathcal{U}_\mathrm{f}$ to one of its lower neighbours in $\mathcal{U}$. The vertices of K are labels, so they lie in $\mathcal{B} \subseteq \mathcal{U}$. They include ρ, the label of the source of σ_ρ, and every point of Σ, the labels of the sinks. Distinct sinks carry distinct terminals, so Σ consists of m distinct faults. This proves (i). All labels lie in $\mathcal{B}$, so every vertex of K has time at most t_ρ. Let θ be a vertex of H_ρ whose label has the least time. If θ were a source or an interior vertex, an arm would continue from θ to a vertex of H_ρ whose label has time one less. So θ is a sink, and the least time of a vertex of K is s. Finally, a path in H_ρ from the source of σ_ρ to a sink labelled at time s uses distinct edges of H_ρ, so it has at most e_ρ edges, and the time changes by one along each of them. Hence $t_\rho - s \leq e_\rho \leq q^2(m-1)$, which is (iii). □

3.5 Packing contours along a discrepancy path

Lemma 3.9. *Fix a configuration x and a realisation of the noise, and let X and Y be the runs from x and from $\perp$. If $t \geq 1$ and $X_t(v) \neq Y_t(v)$, then there are sites $v_t = v, v_{t-1}, \ldots, v_0$ such that, for $1 \leq s \leq t$, the point (v_s, s) is not an event, $v_{s-1} \in \mathcal{N}(v_s)$, and $X_{s-1}(v_{s-1}) \neq Y_{s-1}(v_{s-1})$. In particular (v_s, s) and $(v_{s-1}, s-1)$ are joined in $\mathcal{G}$.*

Proof. Let $s \geq 1$ and $X_s(v_s) \neq Y_s(v_s)$. Then $\eta_s(v_s) = 0$, since otherwise both values equal $\zeta_s(v_s)$. So $F(X_{s-1})(v_s) \neq F(Y_{s-1})(v_s)$. By (1), X_{s-1} and Y_{s-1} differ at some $v_{s-1} \in \mathcal{N}(v_s)$. Repeat down to $s = 1$. □

Lemma 3.10 (packing). *Let $k \geq 2j$, and let $S_\tau(0, \dots, 0) = \emptyset$ for the box V, with $\tau \geq 1$. Fix a configuration x and a realisation of the noise, and let X and Y be the runs from x and from $\perp$. If $t \geq \tau$ and $X_t(v) \neq Y_t(v)$, then there are an integer $e \geq t$, a set K of e edges of $\mathcal{G}$ and a set Σ of events with the following properties: the edges of K form a connected graph; its vertices include (v, t) and every point of Σ; and $|\Sigma| \geq (e - \tau)/a$.*

Proof. Let $v_t, \dots, v_0$ be the sites of Lemma 3.9. For $\tau \leq s \leq t$ the point (v_s, s) is not an event, because $s \geq 1$. One of $X_s(v_s)$ and $Y_s(v_s)$ is live, because they differ. Corollary 3.7, applied to that run, gives $E_s(v_s) = 1$. So (v_s, s) is a marked vertex that is not an event.

Define times $t_1 > t_2 > \dots$ as follows. Put $t_1 = t$. Given $t_l \geq \tau$, apply Lemma 3.8 with $\mathcal{U}$ the set of marked vertices and $\mathcal{U}_{\mathrm{f}}$ the set of events, at $\rho_l = (v_{t_l}, t_l)$. It gives m_l, a set Σ_l of m_l events, a set K_l of edges, and the least time s_l of a point of Σ_l. If $s_l - 1 \geq \tau$, put $t_{l+1} = s_l - 1$. Otherwise stop, and let $L = l$. Since $s_l \leq t_l$, the times decrease and the construction stops.

By Lemma 3.8(iii), the vertices of K_l have times in $[s_l, t_l]$. These intervals are pairwise disjoint, since $t_{l+1} = s_l - 1$. Hence the sets Σ_l are pairwise disjoint, and $\Sigma = \Sigma_1 \cup \dots \cup \Sigma_L$ has $m = m_1 + \dots + m_L$ elements. One event may lie in the support graphs of many marked vertices; the disjoint slabs ensure that it is counted at most once. The intervals $[s_l, t_l]$ cover $[s_L, t]$, and $s_L \leq \tau$ by the stopping rule. By Lemma 3.8(iii),

$$t - \tau + 1 \leq t - s_L + 1 = \sum_{l=1}^{L} (t_l - s_l + 1) \leq \sum_{l=1}^{L} \bigl(q^2(m_l - 1) + 1\bigr) \leq q^2 m.$$

Let K consist of the t edges joining (v_s, s) and $(v_{s-1}, s-1)$, $1 \leq s \leq t$, together with $K_1, \dots, K_L$. The path is connected, and each K_l is connected and contains the path vertex ρ_l. So K forms a connected graph, and its vertices include (v, t) and Σ. Let $e = |K|$. The t path edges join different pairs of times, so they are distinct and $e \geq t$. By Lemma 3.8(ii),

$$e \leq t + \sum_{l=1}^{L} |K_l| \leq t + q^2 \sum_{l=1}^{L} (m_l - 1) \leq t + q^2 m \leq \tau - 1 + 2q^2 m.$$

So $2q^2 m > e - \tau$. If $e \geq \tau$ this gives $m \geq (e - \tau)/(2q^2) \geq (e - \tau)/a$. If $e < \tau$ the bound $m \geq (e - \tau)/a$ is trivial. □

3.6 Counting, and the proof of the theorem

Lemma 3.11. *Let θ_0 be a vertex of $\mathcal{G}$ and $e \geq 1$. There are at most D_0^{2e} sets K of e edges of $\mathcal{G}$ that form a connected graph with θ_0 among its vertices.*

Proof. Every vertex of $\mathcal{G}$ has degree at most D_0. Fix, for every vertex, a numbering of its neighbours by numbers in $\{1, \dots, D_0\}$. Let K be such a set of edges. Doubling every edge of K gives a connected multigraph in which every degree is even. By Euler's theorem it has a closed walk $p_0 = \theta_0, p_1, \dots, p_{2e} = \theta_0$ that uses each of its $2e$ edges once. Fix one such walk for each K, and encode it by the word $(\omega_1, \dots, \omega_{2e})$, where ω_s is the number of p_s among the neighbours of p_{s-1}. The word and θ_0 determine the walk, and the walk determines K as the set of edges it uses. So distinct sets K have distinct words, and there are at most D_0^{2e} words. □

Proof of Theorem 3.1. If $L_R = \varnothing$, then $\mathcal{A}_R = \{\perp\}$, so $X_t = Y_t$ and there is nothing to prove. Let $L_R \neq \varnothing$, and fix r with $S_\tau(0,\dots,0) = \varnothing$ for the box $V = \{-r,\dots,r\}^2$. For a live symbol b, the one-element family $\{((0,\dots,0),(b,\dots,b))\}$ has the three properties of Lemma 3.3(b) with $S = S_{\mathrm{all}}$. So $(b,\dots,b) \in S_0(0,\dots,0)$, and therefore $\tau \geq 1$.

Fix x, v and t. If $t < \tau$, the right side of (3) equals $2(2D_0^2)^t \varepsilon^{(t-\tau)/a}/(1-\beta) \geq 2$, because $0 < \varepsilon < 1$; so (3) holds. Let $t \geq \tau$. By Lemma 3.10, on the event $X_t(v) \neq Y_t(v)$ there are an integer $e \geq t$, a set K of e edges of $\mathcal{G}$ that forms a connected graph with (v,t) among its vertices, and a set Σ of vertices of K, all of positive time, such that $|\Sigma| \geq (e-\tau)/a$ and $\eta = 1$ on Σ. The pairs (K,Σ) range over a countable family of fixed sets, so the union bound applies. For a fixed set Σ of distinct points of positive time, independence gives $\mathbb{P}(\eta = 1 \text{ on } \Sigma) = \varepsilon^{|\Sigma|}$, and this is at most $\varepsilon^{(e-\tau)/a}$ when $|\Sigma| \geq (e-\tau)/a$, since $0 < \varepsilon < 1$. By Lemma 3.11 there are at most D_0^{2e} choices of K. A connected graph with e edges has at most $e+1$ vertices, so there are at most 2^{e+1} choices of Σ. Therefore

$$\mathbb{P}\big(X_t(v) \neq Y_t(v)\big) \leq \sum_{e \geq t} D_0^{2e}\, 2^{e+1}\, \varepsilon^{(e-\tau)/a} = 2\varepsilon^{-\tau/a} \sum_{e \geq t} \big(2D_0^2 \varepsilon^{1/a}\big)^e = 2\varepsilon^{-\tau/a}\, \frac{\beta^t}{1-\beta}.$$

The right side does not depend on x or v. This proves (3).

For a finite set O, the pair $(X_t|_O, Y_t|_O)$ is a coupling of $\mathrm{Law}_x(X_t|_O)$ and $\mathrm{Law}_\perp(X_t|_O)$. So

$$d_{\mathrm{TV}}\big(\mathrm{Law}_x(X_t|_O), \mathrm{Law}_\perp(X_t|_O)\big) \leq \mathbb{P}\big(X_t|_O \neq Y_t|_O\big) \leq \sum_{v \in O} \mathbb{P}\big(X_t(v) \neq Y_t(v)\big),$$

and (3) gives (4). Since $\beta < 1$, the right side of (4) tends to 0, so the rule forgets its initial configuration. Lemma 2.12 gives ergodicity. □

4 Tilings give memory

This section proves Theorem 4.2, which is part (i) of Theorem B. When R tiles the plane, copy one tiling onto every plane. The copy is a fixed point of $F_R^{[k]}$. Under noise of small rate, each site stays in the copy with high probability at every time. From the all-bottom start, each site stays bottom with high probability. So the process remembers where it started, and it is not ergodic.

The proof is Toom's contour argument [29]; see also [10]. A site that has left the copy, and was not hit by noise, has two such sites one step earlier, in two different planes. So the sites that have left the copy lie inside the binary field used in the proof of Theorem 3.1. The contour lemma for that field, Lemma 3.8, which builds contours from poles in the manner of [28, Theorem 33, Definitions 35 and 37, Lemma 36], then forces many noise events near such a site, and a Peierls count bounds the probability. Copying a rule onto parallel planes that correct one another by a Toom-type vote goes back to the repeated planes of [11]. Toom-stable clones of Wang-tile models appear in [22].

Throughout, $k \geq 2$, $q = k+1$, $a = 2q^2+1$ and $D_0 = 50q$. A site is $v = (u,P)$ with $u \in \mathbb{Z}^2$ and $P \in \mathbb{Z}^k$. A space-time point is (u,P,t) with $t \in \mathbb{Z}$. The providers are $c_0 = 0$ and $c_1,\dots,c_k$, the unit vectors of $\mathbb{Z}^k$.

Definition 4.1. Let $\theta\colon \mathbb{Z}^2 \to R$ be a tiling. For $u \in \mathbb{Z}^2$ let $\theta[u]$ be the block $w \mapsto \theta(u+w)$ on $\{-1,0,1\}^2$; it lies in T_R because θ matches on every edge. The *cloned tiling* of θ is the configuration $\xi \in \mathcal{A}_R^{\mathbb{Z}^2 \times \mathbb{Z}^k}$ given by $\xi(u,P) = (\theta[u],0)$ for all u and P.

Theorem 4.2 (Memory). *Let $k \geq 2$, let $0 < \varepsilon \leq (4D_0^2)^{-a}$, and put $\lambda = 2D_0^2\,\varepsilon^{1/q^2}$. Let R be a Wang set that tiles $\mathbb{Z}^2$ and let ξ be a cloned tiling. Let (X_t) be the noisy $F_R^{[k]}$ at rate ε started from ξ, and (Y_t) the same process started from $\perp$. Then for every site v and every time $t \geq 0$,*

$$\mathbb{P}\big(X_t(v) \neq \xi(v)\big) \leq \varepsilon\Big(1 + \frac{2\lambda}{1-\lambda}\Big) < 1.01\,\varepsilon \quad \text{and} \quad \mathbb{P}\big(Y_t(v) \neq \perp\big) \leq \varepsilon\Big(1 + \frac{2\lambda}{1-\lambda}\Big) < 1.01\,\varepsilon. \tag{10}$$

Hence the noisy $F_R^{[k]}$ has two distinct invariant measures, and it is not ergodic.

In particular Theorem 4.2 applies to every $\varepsilon \in (0, \varepsilon_k]$.

4.1 The rule and the cloned tiling

Write $\overline{C}$ for the planar checker applied in every plane, $(\overline{C}x)(u,P) = C(x(\cdot,P))(u)$. Write Q_k for the quorum step: $Q_k(y)(u,P) = s$ if s is live and $y(u, P + c_i) = s$ for at least k of the q indices i, and $Q_k(y)(u,P) = \perp$ otherwise. Thus $F_R^{[k]} = Q_k \circ \overline{C}$. Order $\mathcal{A}_R$ flatly: $\perp \preceq s$ for every symbol s, and distinct live symbols are incomparable. Order configurations site by site.

We realize the noise as follows. For every site v and time $t \geq 1$ there are an indicator $\eta_t(v)$, equal to 1 with probability ε, and a value $\zeta_t(v)$, uniform on $\mathcal{A}_R$; all of these are independent. The run from x is $X_0 = x$ and, for $t \geq 1$, $X_t(v) = \zeta_t(v)$ if $\eta_t(v) = 1$ and $X_t(v) = F_R^{[k]}(X_{t-1})(v)$ otherwise. The space-time point (v,t) is an *event* if $\eta_t(v) = 1$, whatever the value $\zeta_t(v)$. Runs from different starts with the same η and ζ form the *common-noise coupling*.

Lemma 4.3. *Let $k \geq 2$.*

(a) For every configuration y and site (u,P), at most one live symbol s satisfies the quorum condition. So Q_k and $F_R^{[k]}$ are well defined.

(b) C, $\overline{C}$ and Q_k are monotone for $\preceq$, and hence so is $F_R^{[k]}$.

(c) Under the common-noise coupling, $X_0 \preceq X_0'$ implies $X_t \preceq X_t'$ for every t. In particular the run from $\perp$ lies below every run.

Proof. (a) Two distinct symbols, each at k or more of the $k+1$ provider positions, would need $2k \leq k+1$ positions, that is $k \leq 1$.

(b) Let $x \preceq x'$ be planar configurations and let $C(x)(u) = s$ be live. Then $x(u) = s$, the symbols $x(u)$ and $x(u+d)$, $d \in D$, are live, and each $x(u+d)$ is compatible with $x(u)$. A live symbol lies below only itself, so x' agrees with x on these nine sites, and $C(x')(u) = s$. If $C(x)(u) = \perp$ there is nothing to prove. So C is monotone, and so is $\overline{C}$, plane by plane. If $y \preceq y'$ and $Q_k(y)(u,P) = s$ is live, the k or more provider positions with $y(u, P + c_i) = s$ also carry s in y', so $Q_k(y')(u,P) = s$.

(c) Induction on t. At a non-event the claim follows from (b). At an event both runs take the same value $\zeta_t(v)$. The last sentence holds because $\perp \preceq x$ for every x. □

Lemma 4.4. *Let ξ be a cloned tiling. If a configuration x agrees with ξ at the nine sites $(u+d,P)$, $d \in \{-1,0,1\}^2$, then $(\overline{C}x)(u,P) = \xi(u,P)$. In particular $\overline{C}\xi = \xi$ and $F_R^{[k]}(\xi) = \xi$.*

Proof. The nine symbols $\xi(u+d,P) = (\theta[u+d], 0)$ are live. For $d \in D$ the bits of $\xi(u+d,P)$ and $\xi(u,P)$ are both 0, and at a common cell z of $u + \{-1,0,1\}^2$ and $u + d + \{-1,0,1\}^2$ both blocks show $\theta(z)$. So every neighbour is compatible with the centre, and C keeps $\xi(u,P)$. For the last claim, $(\overline{C}\xi)(u, P + c_i) = \xi(u, P + c_i) = \xi(u,P)$ for all q indices i, and $\xi(u,P)$ is live. □

4.2 Two bad parents

Lemma 4.5 (Two bad parents)**.** *Let $k \geq 2$, and fix any realization of the events and of the replacement values.*

(a) Run from ξ, and call (u, P, t) bad *if $X_t(u, P) \neq \xi(u, P)$. No point of time 0 is bad. If (u, P, t) is bad and not an event, there are providers $c \neq c'$ and offsets $d, d' \in \{-1, 0, 1\}^2$ such that $(u + d, P + c, t - 1)$ and $(u + d', P + c', t - 1)$ are bad.*

(b) Run from $\perp$, and call (u, P, t) bad *if $Y_t(u, P)$ is live. No point of time 0 is bad. If (u, P, t) is bad and not an event, there are providers $c \neq c'$ such that $(u, P + c, t - 1)$ and $(u, P + c', t - 1)$ are bad.*

Proof. At time 0 the runs equal ξ and $\perp$, so no point is bad. Let $t \geq 1$.

(a) Put $x = X_{t-1}$, $y = \overline{C}x$ and $s = \xi(u, P)$, which is live. As (u, P, t) is not an event, $F_R^{[k]}(x)(u, P) = X_t(u, P) \neq s$. If $y(u, P + c_i) = s$ held for k or more indices i, the quorum step would output s. Hence $y(u, P + c_i) \neq s$ for at least two indices $i \neq i'$. Fix such an i. Since $\xi(u, P + c_i) = s$, Lemma 4.4 shows that x differs from ξ at some site $(u + d, P + c_i)$ with $d \in \{-1, 0, 1\}^2$; that is, $(u + d, P + c_i, t - 1)$ is bad. The same holds for i', and $c_i \neq c_{i'}$.

(b) Let $s = Y_t(u, P)$, which is live. As (u, P, t) is not an event, $(\overline{C}Y_{t-1})(u, P + c_i) = s$ for at least $k \geq 2$ indices i. The checker outputs a live symbol only by keeping its centre, so $Y_{t-1}(u, P + c_i) = s$ is live for these indices. □

In both cases one bad plane among the providers is outvoted; a bad point that is not an event needs two.

4.3 Domination by the binary field

Define a field E on space-time points from the event indicators alone. Put $E_0 \equiv 0$. For $t \geq 1$, put $E_t(u, P) = 1$ if (u, P, t) is an event; otherwise put $E_t(u, P) = 1$ if and only if there are providers $c \neq c'$ and offsets $d, d' \in \{-1, 0, 1\}^2$ with $E_{t-1}(u + d, P + c) = E_{t-1}(u + d', P + c') = 1$. This is the binary field used in the proof of Theorem 3.1.

Lemma 4.6. *For every realization of the events and of the replacement values, every bad point of either run in Lemma 4.5 has $E = 1$.*

Proof. Induction on time. At time 0 no point is bad. Let (u, P, t) be bad with $t \geq 1$. If it is an event, $E_t(u, P) = 1$. Otherwise Lemma 4.5 gives two bad points at time $t - 1$, at providers $c \neq c'$ and planar offsets in $\{-1, 0, 1\}^2$. By induction both have $E_{t-1} = 1$, so $E_t(u, P) = 1$. □

4.4 Counting, and the proof of the theorem

Let H_k be the graph on $\mathbb{Z}^2 \times \mathbb{Z}^k \times \mathbb{Z}$ in which each point (u, P, t) is joined to $(u + d, P + c, t - 1)$ for every $d \in \{-1, 0, 1\}^2$ and every provider c. Each point has $9q$ neighbours one step lower and $9q$ one step higher, so every vertex of H_k has $18q \leq D_0$ neighbours.

Lemma 4.7. *Let H be a graph in which every vertex has at most Δ neighbours, let z be a vertex, and let $n \geq 1$. At most Δ^{2n} sets I of n edges of H form a connected graph having z as an end of one of their edges.*

Proof. Number the edges at each vertex of H injectively by $\{1, \ldots, \Delta\}$. Double every edge of such a set I. The resulting multigraph is connected and all its degrees are even, so it has a

closed walk from z of length $2n$ that uses each of its edges once. For each step of the walk, record the number of the edge it uses at the vertex where the step starts. Starting at z, the sequence determines the walk, and the walk determines I. So distinct sets I give distinct sequences in $\{1,\dots,\Delta\}^{2n}$. □

Lemma 4.8. *Let $k \geq 2$, $0 < \varepsilon < 1$ and $\lambda = 2D_0^2\varepsilon^{1/q^2} < 1$. For every site v and time $t \geq 0$,*

$$\mathbb{P}\big(E_t(v) = 1\big) \leq \varepsilon\Big(1 + \frac{2\lambda}{1-\lambda}\Big).$$

Proof. For $t = 0$ the probability is 0; let $t \geq 1$. Fix a realization of the events. The field E has $E_0 \equiv 0$, and every point with $E = 1$ that is not an event has two points with $E = 1$ one step earlier, at distinct providers and planar offsets in $\{-1,0,1\}^2$; choose two such points for each of them. Suppose $E_t(v) = 1$ and (v,t) is not an event. Lemma 3.8 gives a set I of $n \geq 1$ edges, each joining a point with $E = 1$ to one of its two chosen points, such that I forms a connected graph with (v,t) as an end, and the ends of edges in I include m distinct events with $n \leq q^2(m-1)$. So I is a set of edges of H_k, and its ends include a set S of at least $1 + n/q^2$ events. A connected graph with n edges has at most $n+1$ vertices, so S is one of at most 2^{n+1} subsets of the ends of I. Hence

$$\{E_t(v) = 1\} \subseteq \{(v,t) \text{ is an event}\} \cup \bigcup_{n\geq 1} \bigcup_{(I,S)} \{\text{every point of } S \text{ is an event}\},$$

where (I,S) runs over the connected n-edge sets I of H_k having (v,t) as an end and the sets S of at least $1 + n/q^2$ ends of edges of I. By Lemma 4.7 there are at most $D_0^{2n}2^{n+1}$ such pairs. Events occur only at positive times, independently with probability ε, so each pair contributes at most $\varepsilon^{|S|} \leq \varepsilon^{1+n/q^2}$. Therefore

$$\mathbb{P}\big(E_t(v) = 1\big) \leq \varepsilon + \sum_{n\geq 1} D_0^{2n}\, 2^{n+1}\, \varepsilon^{1+n/q^2} = \varepsilon + 2\varepsilon \sum_{n\geq 1} \lambda^n = \varepsilon\Big(1 + \frac{2\lambda}{1-\lambda}\Big). \qquad \square$$

Proof of Theorem 4.2. Since $a/q^2 > 2$ and $4D_0^2 > 1$, the hypothesis $\varepsilon \leq (4D_0^2)^{-a}$ gives $\varepsilon^{1/q^2} \leq (4D_0^2)^{-2}$, so

$$\lambda \leq \frac{2D_0^2}{16D_0^4} = \frac{1}{8D_0^2} \leq \frac{1}{180000},$$

because $D_0 = 50q \geq 150$. Then $2\lambda/(1-\lambda) \leq 3\lambda < 0.01$. By Lemma 4.6, the sets $\{X_t(v) \neq \xi(v)\}$ and $\{Y_t(v) \neq \perp\}$ are contained in $\{E_t(v) = 1\}$, and Lemma 4.8 gives (10).

Put $\delta = 1 - 2.02\,\varepsilon$, which is positive. Fix a site v and let W be the set of configurations x with $x(v) = \xi(v)$. Since $\xi(v)$ is live, (10) gives, for every t,

$$\mathbb{P}(X_t \in W) > 1 - 1.01\,\varepsilon, \qquad \mathbb{P}(Y_t \in W) \leq \mathbb{P}\big(Y_t(v) \neq \perp\big) < 1.01\,\varepsilon. \tag{11}$$

So $d_{\mathrm{TV}}\big(\mathrm{Law}_\xi(X_t(v)), \mathrm{Law}_\perp(Y_t(v))\big) > \delta$ for every t: the process has memory.

By (11) the process has memory, and Lemma 2.13 gives two distinct invariant measures. □

Remark 4.9. The field E depends only on the event indicators, Lemma 4.6 holds for every realization of the replacement values, and Lemma 4.8 uses the noise only through the bound $\mathbb{P}(\text{every point of } S \text{ is an event}) \leq \varepsilon^{|S|}$ for finite sets S of points of positive time. So (10) holds for any law of the replacement values, which may depend on the events and on the past, as long as that bound holds.

5 A hierarchical scaffold read by a fixed number of tokens

This section builds one fixed Wang set, the scaffold, and shows that a fixed number of tokens can read its hierarchy at every depth. Hierarchical tilings go back to Berger [3] and Robinson [26]. Mozes showed that substitution tilings with unique derivation are enforced by local rules [24]; unique derivation (recognizability) of aperiodic substitutions is due to Mossé [23] and Solomyak [27]. We use Ollinger's 104-tile substitution [25], in the transcription of [18], and prove the facts we need from its tables, which are printed in Section A.

The plan is as follows. Section 5.1 recalls the 104 tiles and proves that every tiling has one parent tiling. Section 5.2 adds two bits to every tile, one constant on columns and one constant on rows. The result is a Wang set $\mathbb{A}$ of 276 letters with a substitution σ of expansion 4 (Proposition 5.11). The bits record the depth of each column and row modulo 2, like a ruler. Section 5.3 reads the full letter of a depth-n block from $\ell = \binom{276}{2} + 1$ fixed sites. Section 5.4 defines anchors and builds the type, edge and point certificates, graphs of bounded tree width that force the readout (Theorem 5.18). Section 5.5 carries their conclusions to every existential pebble family (Lemma 5.19) and shows that every such family contains anchors of every depth (Proposition 5.21).

Throughout, $\mathbf{e}_1 = (1,0)$ and $\mathbf{e}_2 = (0,1)$. Intervals are sets of integers: $[a,b] = \{a,\dots,b\}$ and $[0,m) = \{0,\dots,m-1\}$. Wang sets, legal patterns and tilings are as in Definition 2.1. Tiles t, t' *match in direction* $\mathbf{e}_1$ if $t_{\mathrm{E}} = t'_{\mathrm{W}}$, and *in direction* $\mathbf{e}_2$ if $t_{\mathrm{N}} = t'_{\mathrm{S}}$. A pattern on a finite set is *extendable* if it is the restriction of a tiling.

5.1 Ollinger's substitution

A tile of the base set is a triple (θ, κ, ω) of a *thin* type $\theta \in \{\mathsf{a},\mathsf{b},\mathsf{c},\mathsf{d}\}$, a *thick* type κ and a *black* type $\omega \in \{\mathsf{A},\dots,\mathsf{E}\}$. A thick type is one of four corners $\angle\mathsf{a}, \angle\mathsf{b}, \angle\mathsf{c}, \angle\mathsf{d}$ or one of sixteen lines $\langle \alpha, \gamma \rangle$, with a vertical atom $\alpha \in \{\mathsf{r}_0, \mathsf{r}_2, \mathsf{g}_0, \mathsf{g}_2\}$ and a horizontal atom $\gamma \in \{\mathsf{r}_1, \mathsf{r}_3, \mathsf{g}_1, \mathsf{g}_3\}$. Each component has an endpoint ∂_X on each side $X \in \{\mathrm{N},\mathrm{E},\mathrm{S},\mathrm{W}\}$, given in Table 1.

Definition 5.1 (The base set)**.** $\mathbb{O}$ is the Wang set of the 104 triples listed in Table 2. The colour of (θ, κ, ω) on side X is the triple $(\partial_X\theta, \partial_X\kappa, \partial_X\omega)$.

This is the 104-tile set of the Lean development [18], a transcription of Ollinger's set [25], in a compact presentation. No agreement with the published figures of [25] is claimed or used: the proofs below use only the tables of Section A. The four exclusions in Table 2 replace a list of 104 cases. Distinct triples give distinct tiles. Indeed, the four thin types have distinct north endpoints. A line $\langle \alpha, \gamma \rangle$ has equal north and south endpoints α and east endpoint γ, while each corner has distinct north and south endpoints and the four corners have distinct north endpoints. The five black types have distinct bit tuples.

Lemma 5.2 (Thin phase)**.** *Let f be a tiling of $\mathbb{O}$. There is a unique coset $c + 2\mathbb{Z}^2$ such that the thin type of $f(c + 2v + \varsigma)$ is $\mathsf{a},\mathsf{c},\mathsf{d},\mathsf{b}$ for $\varsigma = (0,0),(1,0),(0,1),(1,1)$ respectively, for every $v \in \mathbb{Z}^2$.*

Proof. The east endpoints of the thin types are distinct, and so are the west endpoints. By Table 1, the thin type of the east neighbour is therefore determined: $\mathsf{a} \leftrightarrow \mathsf{c}$ and $\mathsf{b} \leftrightarrow \mathsf{d}$. In the same way the north neighbour is determined: $\mathsf{a} \leftrightarrow \mathsf{d}$ and $\mathsf{b} \leftrightarrow \mathsf{c}$. These two involutions commute; their composite exchanges $\mathsf{a} \leftrightarrow \mathsf{b}$ and $\mathsf{c} \leftrightarrow \mathsf{d}$. Hence the thin type at $u + (x_1, x_2)$ is obtained from the thin type at u by applying the first involution x_1 times and the second x_2 times. The sites of thin type a form one coset $c + 2\mathbb{Z}^2$, and the other three types sit as stated. □

Definition 5.3 (The substitution)**.** For a triple $t = (\theta, \kappa, \omega)$, the pattern $\varphi(t)$ on $[0,2)^2$ is given by Table 3 (left). It uses the maps F, se and nw of Section A. For a pattern w on a set $U \subseteq \mathbb{Z}^2$, the pattern $\varphi(w)$ on $2U + [0,2)^2$ is $\varphi(w)(2v + \varsigma) = \varphi(w(v))(\varsigma)$.

The definition makes sense for every triple of a thin type, a thick type and a black type, whether or not it lies in $\mathbb{O}$. We call such a triple a *formal triple*.

Lemma 5.4 (The substitution is sound)**.** *(a) For every $t \in \mathbb{O}$, $\varphi(t)$ is a legal pattern of tiles of $\mathbb{O}$.*

(b) For every pattern w of tiles of $\mathbb{O}$ on a set $U \subseteq \mathbb{Z}^2$, $\varphi(w)$ is legal if and only if w is legal.

(c) φ is injective on formal triples.

Proof. (a) Let $t = (\theta, \kappa, \omega) \in \mathbb{O}$. Child $(0,0)$ is $(\mathsf{a}, \kappa, \omega)$. It lies in $\mathbb{O}$ because every pair (κ, ω) allowed for some thin type is allowed for a (Table 2). Child $(1,1)$ is $(\mathsf{b}, F(\theta), \mathsf{A})$, a corner with black A, which is allowed. Child $(1,0)$ is the line $\langle \partial_S\theta, \partial_E\kappa \rangle$ with thin c and black $\mathrm{se}(\omega) \in \{\mathsf{B}, \mathsf{C}\}$. It is excluded only if $\mathrm{se}(\omega) = \mathsf{B}$ and $\partial_S\theta = \mathsf{r}_0$, or $\mathrm{se}(\omega) = \mathsf{C}$ and $\partial_S\theta = \mathsf{r}_2$. Now $\partial_S\theta = \mathsf{r}_0$ only for $\theta = \mathsf{d}$, where $\omega \in \{\mathsf{D}, \mathsf{E}\}$ and $\mathrm{se}(\omega) = \mathsf{C}$; and $\partial_S\theta = \mathsf{r}_2$ only for $\theta = \mathsf{b}$, where $\omega = \mathsf{A}$ and $\mathrm{se}(\omega) = \mathsf{B}$. Child $(0,1)$ is the line $\langle \partial_N\kappa, \partial_W\theta \rangle$ with thin d and black $\mathrm{nw}(\omega) \in \{\mathsf{D}, \mathsf{E}\}$. It is excluded only if $\mathrm{nw}(\omega) = \mathsf{D}$ and $\partial_W\theta = \mathsf{r}_3$, which needs $\theta = \mathsf{b}$ and $\omega = \mathsf{A}$, where $\mathrm{nw}(\omega) = \mathsf{E}$; or $\mathrm{nw}(\omega) = \mathsf{E}$ and $\partial_W\theta = \mathsf{r}_1$, which needs $\theta = \mathsf{c}$ and $\omega \in \{\mathsf{B}, \mathsf{C}\}$, where $\mathrm{nw}(\omega) = \mathsf{D}$. So all four children lie in $\mathbb{O}$.

The four internal edges match. The thin pattern $\mathsf{a}, \mathsf{c}, \mathsf{d}, \mathsf{b}$ matches by the thin matching rules listed in Section A. For the thick components, the edge from $(0,0)$ to $(1,0)$ carries $\partial_E\kappa$ on both sides, the edge from $(0,0)$ to $(0,1)$ carries $\partial_N\kappa$, the edge from $(1,0)$ to $(1,1)$ carries $\partial_S\theta = \partial_S F(\theta)$, and the edge from $(0,1)$ to $(1,1)$ carries $\partial_W\theta = \partial_W F(\theta)$. Here we use that $F(\theta)$ has the four endpoints of θ. For the black components, $\partial_W \mathrm{se}(\omega) = \partial_E\omega$, $\partial_S \mathrm{nw}(\omega) = \partial_N\omega$, and the two edges at child $(1,1)$ carry $\partial_N \mathrm{se}(\omega) = 1 = \partial_S\mathsf{A}$ and $\partial_E \mathrm{nw}(\omega) = 1 = \partial_W\mathsf{A}$.

(b) By (a) the edges inside each block $\varphi(w(v))$ match. Let $v, v + \mathbf{e}_1 \in U$ with $w(v) = (\theta, \kappa, \omega)$ and $w(v + \mathbf{e}_1) = (\theta', \kappa', \omega')$. Two fine edges cross the seam between their blocks. The lower one joins child $(1,0)$ of the first block to child $(0,0)$ of the second. Its thin endpoints are g_3 and g_3; its thick endpoints are $\partial_E\kappa$ and $\partial_W\kappa'$; its black bits are $\partial_E \mathrm{se}(\omega) = \partial_E\omega$ and $\partial_W\omega'$. The upper one joins child $(1,1)$ to child $(0,1)$. Its thin endpoints are g_1 and g_1; its thick endpoints are $\partial_E F(\theta) = \partial_E\theta$ and $\partial_W\theta'$; its black bits are $\partial_E\mathsf{A} = 0$ and $\partial_W \mathrm{nw}(\omega') = 0$. So both fine edges match if and only if the three components of the coarse edge match. For $v, v + \mathbf{e}_2 \in U$ the argument is the same: the left fine edge joins child $(0,1)$ to child $(0,0)$ and records $\partial_N\kappa = \partial_S\kappa'$ and $\partial_N \mathrm{nw}(\omega) = \partial_N\omega = \partial_S\omega'$, with thin endpoints $\mathsf{g}_2, \mathsf{g}_2$; the right fine edge joins child $(1,1)$ to child $(1,0)$ and records $\partial_N\theta = \partial_S\theta'$, with thin endpoints $\mathsf{g}_0, \mathsf{g}_0$ and black bits $0, 0$.

(c) Child $(0,0)$ shows κ and ω, and child $(1,1)$ shows $F(\theta)$. The map F is a bijection from thin types to corners. □

The proof of (b) compares components, so no list of the 104^2 pairs is needed.

Lemma 5.5 (Parents)**.** *Let w be a legal pattern of tiles of $\mathbb{O}$ on $[0,2)^2$ whose thin types are $\mathsf{a}, \mathsf{c}, \mathsf{d}, \mathsf{b}$ at $(0,0), (1,0), (0,1), (1,1)$. Then $w = \varphi(t)$ for exactly one formal triple t. Exactly 128 formal triples arise in this way: the 104 tiles of $\mathbb{O}$ and 24 others (Table 4). If w extends to a legal pattern on $[0,2)^2 \cup \{(0,-1), (1,-1), (-1,0), (-1,1)\}$, then $t \in \mathbb{O}$.*

Proof. The tile $w(1,1)$ has thin type b, so it is a corner with black A; let $\theta = F^{-1}$ of that corner. Let κ and ω be the thick and black types of $w(0,0)$, and $t = (\theta, \kappa, \omega)$. The tile $w(1,0)$ has thin

type c, so it is a line with black B or C. Its west and north thick endpoints must equal $\partial_E \kappa$ and $\partial_S F(\theta) = \partial_S \theta$; so it is the line $\langle \partial_S \theta, \partial_E \kappa \rangle$. Its black type has west bit $\partial_E \omega$, and the west bits of B and C are 0 and 1; so it is $\mathrm{se}(\omega)$. In the same way $w(0,1)$ is $(\mathsf{d}, \langle \partial_N \kappa, \partial_W \theta \rangle, \mathrm{nw}(\omega))$, because the south bits of D and E are 1 and 0. Hence $w = \varphi(t)$, and t is unique by Lemma 5.4(c).

Which t occur? Child $(0,0)$ must lie in $\mathbb{O}$, which leaves exactly the 52 pairs (κ, ω) of thin type a in Table 2. Child $(1,1)$ is always allowed. Child $(1,0)$ is allowed unless $\mathrm{se}(\omega) = \mathsf{B}$ and $\partial_S \theta = \mathsf{r}_0$, or $\mathrm{se}(\omega) = \mathsf{C}$ and $\partial_S \theta = \mathsf{r}_2$. For $\theta = \mathsf{b}$ ($\partial_S \theta = \mathsf{r}_2$) this excludes $\omega \in \{\mathsf{C}, \mathsf{D}, \mathsf{E}\}$; for $\theta = \mathsf{d}$ ($\partial_S \theta = \mathsf{r}_0$) it excludes $\omega \in \{\mathsf{A}, \mathsf{B}\}$. Child $(0,1)$ is allowed unless $\mathrm{nw}(\omega) = \mathsf{D}$ and $\partial_W \theta = \mathsf{r}_3$, or $\mathrm{nw}(\omega) = \mathsf{E}$ and $\partial_W \theta = \mathsf{r}_1$. For $\theta = \mathsf{b}$ this excludes $\omega \in \{\mathsf{B}, \mathsf{C}, \mathsf{D}\}$; for $\theta = \mathsf{c}$ it excludes $\omega \in \{\mathsf{A}, \mathsf{E}\}$. For $\theta = \mathsf{a}$ the new atoms g_2 and g_3 are green and exclude nothing. This gives the second column of Table 4. Counting the allowed pairs (κ, ω) in each case gives $52 + 4 + 36 + 36 = 128$ formal triples. Comparing with Table 2, the triples outside $\mathbb{O}$ are of two kinds: $\theta = \mathsf{c}$, $\omega = \mathsf{D}$ and κ a line with $\gamma \neq \mathsf{r}_3$; and $\theta = \mathsf{d}$, $\omega = \mathsf{C}$ and κ a line with $\alpha \neq \mathsf{r}_2$. There are 12 of each.

Let t be of the first kind, $t = (\mathsf{c}, \langle \alpha, \gamma \rangle, \mathsf{D})$ with $\gamma \neq \mathsf{r}_3$, and suppose w extends below. Then $w(1,0)$ is the line $\langle \mathsf{g}_0, \gamma \rangle$. By the thin matching rules (Section A), the tile at $(1,-1)$ has thin type b, so it is a corner; its north endpoint is g_0, so it is $\angle\mathsf{d}$, with west endpoint r_3. The tile at $(0,-1)$ has thin type d, so it is a line with black D or E. Its north bit equals the south bit of D, which is 1, so its black is D. Its east endpoint equals the west endpoint r_3 of $\angle\mathsf{d}$, so its horizontal atom is r_3. But thin d with black D forbids $\gamma = \mathsf{r}_3$. So the extension does not exist.

Let t be of the second kind, $t = (\mathsf{d}, \langle \alpha, \gamma \rangle, \mathsf{C})$ with $\alpha \neq \mathsf{r}_2$, and suppose w extends to the left. Then $w(0,1)$ is the line $\langle \alpha, \mathsf{g}_1 \rangle$. The tile at $(-1,1)$ has thin type b, so it is a corner; its east endpoint is g_1, so it is $\angle\mathsf{d}$, with south endpoint r_2. The tile at $(-1,0)$ has thin type c, so it is a line with black B or C. Its east bit equals the west bit of C, which is 1, so its black is C. Its north endpoint equals r_2, so its vertical atom is r_2. But thin c with black C forbids $\alpha = \mathsf{r}_2$. So this extension does not exist either. □

Proposition 5.6 (Every tiling has a parent)**.** *$\mathbb{O}$ tiles. For every tiling f of $\mathbb{O}$ there are a unique coset $c + 2\mathbb{Z}^2$ and, for each representative c, a unique tiling z of $\mathbb{O}$ with $f(c + 2v + \varsigma) = \varphi(z(v))(\varsigma)$ for all $v \in \mathbb{Z}^2$ and $\varsigma \in [0,2)^2$.*

Proof. Let $c + 2\mathbb{Z}^2$ be the coset of Lemma 5.2; no other coset has thin type a at all its sites, so the coset is unique. Fix a representative c. For each v, the block $f|_{c+2v+[0,2)^2}$ has the thin pattern of Lemma 5.5, and the tiling supplies the four cells below and to the left. So it equals $\varphi(z(v))$ for a unique $z(v) \in \mathbb{O}$. By Lemma 5.4(b), z is legal, since $\varphi(z)$ is a translate of f. For existence, $\varphi^m(t)$ is a legal pattern on $[0, 2^m)^2$ for every $t \in \mathbb{O}$ and $m \geq 0$, by Lemma 5.4(a),(b). So every square carries a legal pattern, and $\mathbb{O}$ tiles by König's lemma, as in the proof of Lemma 2.2. □

The proposition holds for every tiling. No assumption is made that f lies in the closure of the substituted patterns, so tilings with faults along lines or quadrants are covered.

5.2 The ruler decoration

We add a column bit and a row bit to every tile. A *red crossing* is a line $\langle \alpha, \gamma \rangle$ with both atoms red, that is $\alpha \in \{\mathsf{r}_0, \mathsf{r}_2\}$ and $\gamma \in \{\mathsf{r}_1, \mathsf{r}_3\}$.

Definition 5.7 (The scaffold)**.** The scaffold $\mathbb{A}$ is the set of tuples $\mathfrak{a} = (\theta, \kappa, \omega, b_1, b_2)$ with $(\theta, \kappa, \omega) \in \mathbb{O}$ and $b_1, b_2 \in \{0,1\}$ that satisfy:

(R1) if $\theta \in \{\mathsf{b}, \mathsf{c}\}$ then $b_1 = 0$;

(R2) if $\theta \in \{\mathsf{b}, \mathsf{d}\}$ then $b_2 = 0$;

(R3) if κ is a corner then $b_1 = b_2$;

(R4) if κ is a red crossing then $b_1 \neq b_2$.

The *column bit* is $\chi_1(\mathfrak{a}) = b_1$ and the *row bit* is $\chi_2(\mathfrak{a}) = b_2$. The north and south colours of $\mathfrak{a}$ are the colours of (θ, κ, ω) in $\mathbb{O}$ together with b_1; the east and west colours are those colours together with b_2. The *base* of $\mathfrak{a}$ is $\bar{\mathfrak{a}} = (\theta, \kappa, \omega)$.

So $\mathbb{A}$ is a Wang set. Table 5 counts it: $184 + 4 + 44 + 44 = 276$ letters. By construction, two letters that match in direction $\mathbf{e}_2$ have the same column bit, and two letters that match in direction $\mathbf{e}_1$ have the same row bit. We call this the *edgewise bit property*. It holds for every matching pair, not only for pairs that occur in tilings, and the certificates of Section 5.4 use it in that form.

Definition 5.8 (Decorated substitution)**.** For $\mathfrak{a} \in \mathbb{A}$ with bits (b_1, b_2), the pattern $\varphi'(\mathfrak{a})$ on $[0,2)^2$ has base $\varphi(\bar{\mathfrak{a}})$ and the bits of Table 3 (right). Patterns $\varphi'(w)$ are defined blockwise as in Definition 5.3. Put $\sigma = \varphi' \circ \varphi'$, that is, $\sigma(\mathfrak{a})(2\tau + \varsigma) = \varphi'\big(\varphi'(\mathfrak{a})(\tau)\big)(\varsigma)$ for $\tau, \varsigma \in [0,2)^2$. Then $\sigma(\mathfrak{a})$ is a pattern on $[0,4)^2$, and $\sigma^n(\mathfrak{a})$ is a pattern on $[0,4^n)^2$ with $\sigma^{n+1}(\mathfrak{a})(4^n d + y) = \sigma^n\big(\sigma(\mathfrak{a})(d)\big)(y)$ for $d \in [0,4)^2$ and $y \in [0,4^n)^2$.

Lemma 5.9 (Decorated substitution is sound)**.** *(a) For every $\mathfrak{a} \in \mathbb{A}$, $\varphi'(\mathfrak{a})$ is a legal pattern of letters of $\mathbb{A}$.*

(b) For every pattern w of $\mathbb{A}$ on a set $U \subseteq \mathbb{Z}^2$, $\varphi'(w)$ is legal if and only if w is legal.

(c) φ' is injective.

Proof. (a) By Lemma 5.4(a) the four bases lie in $\mathbb{O}$ and match. Let $\mathfrak{a} = (\theta, \kappa, \omega, b_1, b_2)$. Child $(0,0)$ has thin type a, so (R1) and (R2) are void; it has thick type κ and bits $(1-b_1, 1-b_2)$, and (R3), (R4) are unchanged when both bits are complemented. Child $(1,1)$ has thin type b, a corner and bits $(0,0)$, which satisfies (R1)–(R3). Child $(1,0)$ has thin type c and $b_1 = 0$, which gives (R1). Its thick type $\langle \partial_{\mathrm{S}}\theta, \partial_{\mathrm{E}}\kappa \rangle$ is a red crossing only if $\partial_{\mathrm{S}}\theta$ is red, that is $\theta \in \{\mathsf{b}, \mathsf{d}\}$. Then (R2) for $\mathfrak{a}$ gives $b_2 = 0$, so the child has bits $(0,1)$ and satisfies (R4). Child $(0,1)$ is symmetric: thin type d and row bit 0; its thick type $\langle \partial_{\mathrm{N}}\kappa, \partial_{\mathrm{W}}\theta \rangle$ is a red crossing only if $\theta \in \{\mathsf{b}, \mathsf{c}\}$, and then $b_1 = 0$ and the child has bits $(1,0)$. Inside the block, the vertical edges carry column bits $1-b_1, 1-b_1$ and $0,0$, and the horizontal edges carry row bits $1-b_2, 1-b_2$ and $0,0$.

(b) The bases match or fail as in Lemma 5.4(b). Across a vertical seam between the blocks of $\mathfrak{a}$ (left) and $\mathfrak{a}'$ (right), the lower fine edge carries row bits $1-b_2$ and $1-b_2'$, and the upper one carries 0 and 0. Across a horizontal seam, the left fine edge carries column bits $1-b_1$ and $1-b_1'$, and the right one carries 0 and 0. So the fine bits match if and only if the coarse bits match.

(c) Child $(0,0)$ shows κ, ω and the complements of both bits, and child $(1,1)$ shows θ. □

Lemma 5.10 (Every decorated tiling has a parent)**.** *For every tiling f of $\mathbb{A}$ there are a unique coset $c + 2\mathbb{Z}^2$ and, for each representative c, a unique tiling z of $\mathbb{A}$ with $f(c + 2v + \varsigma) = \varphi'(z(v))(\varsigma)$ for all $v \in \mathbb{Z}^2$ and $\varsigma \in [0,2)^2$.*

Proof. The bases of f form a tiling $\bar{f}$ of $\mathbb{O}$. Let c and $\bar{z}$ be given by Proposition 5.6 for $\bar{f}$. By the edgewise bit property, $\chi_1(f(x))$ depends only on the column x_1 and $\chi_2(f(x))$ only on the row x_2; write them $X(x_1)$ and $Y(x_2)$. A column $c_1 + 2v_1 + 1$ has thin types c and b, so $X = 0$ there by (R1). A row $c_2 + 2v_2 + 1$ has thin types d and b, so $Y = 0$ there by (R2). Define

$$z(v) = \big(\bar{z}(v),\ 1 - X(c_1 + 2v_1),\ 1 - Y(c_2 + 2v_2)\big).$$

Comparing with Table 3, $f(c+2v+\varsigma) = \varphi'(z(v))(\varsigma)$ for all v and ς. The bits of z are constant on columns and on rows and the bases form the tiling $\bar{z}$, so z is legal once each $z(v)$ lies in $\mathbb{A}$.

Conditions (R3) and (R4) for $z(v)$ follow from the same conditions for $f(c+2v)$, which has the thick type of $\bar{z}(v)$ and both bits complemented. For (R1) and (R2) we go down two levels. Apply Proposition 5.6 to $\bar{z}$: there are c' and a tiling $\bar{w}$ of $\mathbb{O}$ with $\bar{z}(c'+2m+\tau) = \varphi(\bar{w}(m))(\tau)$. Put $o = c+2c'$. Then $\bar{f}(o+4m+y) = \varphi^2(\bar{w}(m))(y)$ for $y \in [0,4)^2$. In this depth-two block, Table 6 shows that the cells $(2,1)$ and $(2,3)$ have thin type d and that exactly one of them is a red crossing, whatever the thin type of $\bar{w}(m)$. The row bits of both cells are 0 by (R2). By (R4) the red crossing has column bit 1, so $X(o_1+4m_1+2) = 1$. In the same way the cells $(1,2)$ and $(3,2)$ have thin type c, hence column bit 0, and one of them is a red crossing; so $Y(o_2+4m_2+2) = 1$.

Now let $v = c'+2m+\tau$. The letter $z(v)$ has thin type b or c exactly when $\tau_1 = 1$. Then $c_1+2v_1 = o_1+4m_1+2$, so $\chi_1(z(v)) = 1 - X(o_1+4m_1+2) = 0$, which is (R1). In the same way, thin type b or d means $\tau_2 = 1$, and then $\chi_2(z(v)) = 0$, which is (R2).

Uniqueness: the coset is the thin-a coset of $\bar{f}$, the bases of z are unique by Proposition 5.6, and the bits of z are read off from f at child $(0,0)$. □

One level does not suffice for (R1) and (R2). The witnesses for a coarse column of thin type b or c sit in the depth-two block, and the offset $o = c+2c'$ combines both levels.

Proposition 5.11 (The scaffold)**.** *The scaffold $\mathbb{A}$ is a Wang set of 276 letters, and it tiles. Its substitution $\sigma = (\varphi')^2$ has expansion 4 and the following properties.*

(i) σ is injective, and $\sigma^n(\mathfrak{a})$ is a legal pattern on $[0,4^n)^2$ for every $\mathfrak{a} \in \mathbb{A}$ and $n \geq 0$.

(ii) For every pattern w of $\mathbb{A}$ on a set $U \subseteq \mathbb{Z}^2$, $\sigma(w)$ is legal if and only if w is legal.

(iii) For every tiling f of $\mathbb{A}$ there is a unique class $\pi(f) \in (\mathbb{Z}/4\mathbb{Z})^2$, the phase *of f, such that for every representative p of $\pi(f)$ there is a unique tiling y of $\mathbb{A}$ with $f(p+4v+i) = \sigma(y(v))(i)$ for all $v \in \mathbb{Z}^2$ and $i \in [0,4)^2$.*

(iv) For every $\mathfrak{a} \in \mathbb{A}$ and $i = (i_1, i_2) \in [0,4)^2$,

$$\chi_1\big(\sigma(\mathfrak{a})(i)\big) = \begin{cases} \chi_1(\mathfrak{a}) & i_1 = 0, \\ 0 & i_1 \in \{1,3\}, \\ 1 & i_1 = 2, \end{cases} \qquad \chi_2\big(\sigma(\mathfrak{a})(i)\big) = \begin{cases} \chi_2(\mathfrak{a}) & i_2 = 0, \\ 0 & i_2 \in \{1,3\}, \\ 1 & i_2 = 2. \end{cases}$$

(v) Letters that match in direction $\mathbf{e}_2$ have equal column bits, and letters that match in direction $\mathbf{e}_1$ have equal row bits.

Proof. (i) and (ii) follow from Lemma 5.9 applied twice, and (v) is the edgewise bit property. By (i), every square carries a legal pattern, so $\mathbb{A}$ tiles by König's lemma.

(iii) Let f be a tiling. Lemma 5.10 gives $f(c+2v+\varsigma) = \varphi'(z(v))(\varsigma)$, and again $z(c'+2m+\tau) = \varphi'(y(m))(\tau)$. Substituting, $f(c+2c'+4m+2\tau+\varsigma) = \sigma(y(m))(2\tau+\varsigma)$, so the class of $p = c+2c'$ works; for another representative $p+4m_0$ use $y(\cdot+m_0)$. Conversely, suppose $f(p+4m+i) = \sigma(y(m))(i)$ for a tiling y. Then $f(p+2v+\varsigma) = \varphi'(z'(v))(\varsigma)$ with $z'(2m+\tau) = \varphi'(y(m))(\tau)$, and z' is a tiling by Lemma 5.9. By the uniqueness in Lemma 5.10, $p \equiv c \pmod 2$. Write $p = c+2d$; uniqueness gives $z' = z(\cdot+d)$. So $z(d+2m+\tau) = \varphi'(y(m))(\tau)$, and uniqueness for z gives $d \equiv c' \pmod 2$. Hence $p \equiv c+2c' \pmod 4$. For a fixed representative, y is unique because σ is injective.

(iv) Write $i_1 = 2\tau_1 + \varsigma_1$. By Table 3, the column bit of $\varphi'(\mathfrak{b})(\varsigma)$ is $1-\chi_1(\mathfrak{b})$ if $\varsigma_1 = 0$ and 0 if $\varsigma_1 = 1$. Applying this twice: for $i_1 = 0$ we get $1-(1-\chi_1(\mathfrak{a})) = \chi_1(\mathfrak{a})$; for $i_1 \in \{1,3\}$ we

get 0; for $i_1 = 2$ the middle letter has column bit 0, so we get 1. The row bit is the same with τ_2, ς_2. □

Iterating (iv), the letter $\sigma^n(\mathfrak{a})(0,0)$ has the bits of $\mathfrak{a}$ for every n. In a tiling of phase π, the column bit of a column x_1 is 0 if $x_1 - p_1 \equiv 1, 3 \pmod 4$ and 1 if $x_1 - p_1 \equiv 2 \pmod 4$, for any representative p of π. Only the columns $x_1 \equiv p_1 \pmod 4$ inherit a bit from the parent. So the column bits form a ruler sequence, like the parity of the 2-adic valuation of a column index. The rows behave the same way.

5.3 Reading a block from a fixed number of sites

A depth-n block $\sigma^n(\mathfrak{a})$ has 16^n cells, but its root letter can be read from at most ℓ of them, at every depth.

Let $\mathcal{P}$ be the set of unordered pairs of distinct letters of $\mathbb{A}$; it has $\binom{276}{2} = 37950$ elements. For each $p = \{\mathfrak{a}, \mathfrak{b}\} \in \mathcal{P}$ fix once and for all a child $\delta(p) \in [0,4)^2$ with $\sigma(\mathfrak{a})(\delta(p)) \neq \sigma(\mathfrak{b})(\delta(p))$. It exists because σ is injective. Put $\nu(p) = \{\sigma(\mathfrak{a})(\delta(p)), \sigma(\mathfrak{b})(\delta(p))\}$, again an element of $\mathcal{P}$. Define offsets $y_n(p) \in [0, 4^n)^2$ by

$$y_0(p) = (0,0), \qquad y_{n+1}(p) = 4^n \delta(p) + y_n(\nu(p)),$$

and put

$$P_n = \{(0,0)\} \cup \{y_n(p) : p \in \mathcal{P}\} \subseteq [0,4^n)^2, \qquad \mathrm{Port}_n(u) = u + P_n, \qquad \ell = \tbinom{276}{2} + 1 = 37951.$$

So $|\mathrm{Port}_n(u)| \leq \ell$; coincident offsets are counted once. The *readout* of $\mathfrak{a}$ at depth n is the restriction of $\sigma^n(\mathfrak{a})$ to P_n. A map g into $\mathbb{A}$ defined on $\mathrm{Port}_n(v)$ *reads* $\mathfrak{a}$ if $g(v+y) = \sigma^n(\mathfrak{a})(y)$ for every $y \in P_n$.

Lemma 5.12 (Pair ports)**.** *(a) For every $n \geq 0$, the readout at depth n is injective on $\mathbb{A}$.*

(b) Every $y \in P_{n+1}$ has the form $y = 4^n d + y'$ with $d \in [0,4)^2$ and $y' \in P_n$, and then $\sigma^{n+1}(\mathfrak{a})(y) = \sigma^n\big(\sigma(\mathfrak{a})(d)\big)(y')$ for every $\mathfrak{a}$.

(c) $(0,0) \in P_n$, and $\sigma^n(\mathfrak{a})(0,0)$ has the column bit and the row bit of $\mathfrak{a}$.

Proof. (a) We show $\sigma^n(\mathfrak{a})(y_n(p)) \neq \sigma^n(\mathfrak{b})(y_n(p))$ for every $p = \{\mathfrak{a}, \mathfrak{b}\}$ and every n. For $n = 0$ the two values are $\mathfrak{a}$ and $\mathfrak{b}$. For $n+1$, Definition 5.8 gives $\sigma^{n+1}(\mathfrak{a})(y_{n+1}(p)) = \sigma^n(\mathfrak{a}')(y_n(\nu(p)))$ with $\mathfrak{a}' = \sigma(\mathfrak{a})(\delta(p))$, and the same for $\mathfrak{b}$ with $\mathfrak{b}' = \sigma(\mathfrak{b})(\delta(p))$. Since $\nu(p) = \{\mathfrak{a}', \mathfrak{b}'\}$, the induction hypothesis applies. So the offset $y_n(\{\mathfrak{a}, \mathfrak{b}\})$ separates any two distinct letters.

(b) The offset $(0,0)$ is $4^n \cdot (0,0) + (0,0)$, and $y_{n+1}(p) = 4^n\delta(p) + y_n(\nu(p))$ with $y_n(\nu(p)) \in P_n$. The identity is Definition 5.8.

(c) This is Proposition 5.11(iv) at $i = (0,0)$, applied n times. □

Part (b) says that each port of a depth-$(n+1)$ block is a port of one of its 16 children. So the readouts of all depths are nested and their number of sites does not grow with n. The same construction works for any injective substitution on m letters, with $\binom{m}{2} + 1$ sites.

5.4 Anchors and certificates

For $u \in \mathbb{Z}^2$ and $n \geq 0$ put

$$\mathrm{Anc}_n(u) = \{u,\ u + 4^n \mathbf{e}_1,\ u + 4^n \mathbf{e}_2\}.$$

Definition 5.13 (Anchors)**.** A map g into $\mathbb{A}$ whose domain contains $\mathrm{Anc}_n(u)$ is a *good n-anchor at u* if $\chi_1(g(u)) \neq \chi_1(g(u + 4^n\mathbf{e}_1))$ and $\chi_2(g(u)) \neq \chi_2(g(u + 4^n\mathbf{e}_2))$.

The condition involves three sites and their labels only. It does not assume that the labels come from a block of depth n; the certificates below derive this.

Definition 5.14 (Certificates)**.** A *certificate* Γ consists of a finite vertex set V, a map $\rho\colon V \to \mathbb{Z}^2$ (its *grid map*), a set of *edges*, and a finite set $\partial\Gamma \subseteq \mathbb{Z}^2$ of sites, its *boundary*. Each edge is an ordered pair (v, v') of vertices with a label $e \in \{\mathbf{e}_1, \mathbf{e}_2\}$ such that $\rho(v') = \rho(v) + e$. Each boundary site x has exactly one *boundary vertex* v_x, with $\rho(v_x) = x$. A *homomorphism* of Γ is a map $\psi\colon V \to \mathbb{A}$ such that $\psi(v)$ and $\psi(v')$ match in direction e for every edge (v, v') with label e. For a boundary site x we write $\psi(x)$ for $\psi(v_x)$.

A *rooted tree decomposition* of Γ is a finite rooted tree with a *bag* $\beta_t \subseteq V$ at each node t such that every vertex lies in some bag, the nodes whose bags contain a given vertex form a connected subtree, both ends of every edge lie in a common bag, and the root bag contains every boundary vertex. Its *width* is the largest bag size.

Distinct vertices may have the same image under ρ, and a homomorphism may give them different letters. Only the edges and the boundary tie vertices together. This freedom is what keeps the width bounded: a certificate may use the same site in many independent sub-certificates without identifying the copies. Certificates of bounded tree width are the structures for which the existential pebble game is decisive, by the work of Dalmau, Kolaitis and Vardi [6]; Lemma 5.19 below is the form we need.

A finite recognition chart. The next lemma replaces an explicit table of local phases by compactness. We need only that the constants exist; they are never computed.

Lemma 5.15 (Recognition chart)**.** *(a) For every finite set $D \subseteq \mathbb{Z}^2$ there is $L(D) \geq 0$ such that, for every $u \in \mathbb{Z}^2$, every legal pattern of $\mathbb{A}$ on $u + D + [-L(D), L(D)]^2$ restricts to an extendable pattern on $u + D$.*

(b) There is $L_0 \geq 8$ such that any two tilings of $\mathbb{A}$ that agree on $[-L_0, L_0]^2$ have the same phase.

Proof. (a) Legality and extendability are invariant under translation, so take $u = 0$. Let w be a pattern on D that is not extendable. Suppose that for every L it extends to a legal pattern on $D + [-L, L]^2$. These extensions form a finitely branching tree under restriction, so König's lemma gives a legal pattern on $\mathbb{Z}^2$ that extends w, a contradiction. So w has a radius L_w beyond which it has no legal extension. There are finitely many patterns on D; let $L(D)$ be the largest L_w.

(b) For a class $\pi \in (\mathbb{Z}/4\mathbb{Z})^2$, the set of tilings of phase π is closed in $\mathbb{A}^{\mathbb{Z}^2}$. Indeed, let $f_k \to f$ be tilings of phase π and fix a representative p. Write $f_k(p + 4v + i) = \sigma(y_k(v))(i)$. Since σ is injective, $y_k(v)$ is determined by f_k on the block $p + 4v + [0, 4)^2$, so y_k converges to a map y with $f(p + 4v + i) = \sigma(y(v))(i)$. The pattern $\sigma(y)$ is a translate of f, hence legal, so y is a tiling by Proposition 5.11(ii). Now suppose (b) fails. Then for every L there are tilings f_L, f'_L that agree on $[-L, L]^2$ and have different phases. Passing to a subsequence, the phases are constant classes $\pi \neq \pi'$, and $f_L \to f$, $f'_L \to f'$ by compactness of $\mathbb{A}^{\mathbb{Z}^2}$. The limits agree on every box, so $f = f'$, and by closedness f has both phases π and π'. This contradicts Proposition 5.11(iii). Enlarging L_0 keeps (b) true, so we may take $L_0 \geq 8$. □

The phase of a translate changes by the translation: if $f(p + 4v + i) = \sigma(y(v))(i)$ then $f(\cdot + u)$ has phase $\pi(f) - u$. So by (b), two tilings that agree on $u + [-L_0, L_0]^2$ have the same phase, for every u.

Definition 5.16 (Constants)**.** Fix L_0 as in Lemma 5.15(b). Put $\mathsf{D}_0 = [-L_0, L_0]^2$, $Q = [-L_0 - 1, L_0 + 1]^2$, $L_1 = L(Q)$ and $\varrho = L_0 + 1 + L_1$. Put $L_2 = L([0,5)^2)$ and $L_3 = L([0,32)^2)$. Define

$$B = \max\Big\{\ell(2\varrho+3)^2 + 2\ell + 26,\ \ (2L_2+5)^2,\ \ 3 + \ell\,(2L_3+32)^2\Big\}.$$

The integer B depends only on $\mathbb{A}$, σ and the fixed choice of δ. It does not depend on any depth, site, distance or machine. Its first term bounds the bags of the certificates; the other two are the sizes of the fixed boxes used in Proposition 5.21. In particular $B \geq \ell + 4$. We only use that B exists.

Corridors. A *corridor walk* is a sequence $w_0, \dots, w_m \in \mathbb{Z}^2$ in which each step $w_{k+1} - w_k$ is one of $\pm\mathbf{e}_1, \pm\mathbf{e}_2$ and each coordinate of w_k is monotone in k. Its *corridor* is $S = \bigcup_k (w_k + [-\varrho, \varrho]^2)$.

Lemma 5.17 (Phase transport)**.** *Let $w_0, \dots, w_m$ be a corridor walk with corridor S, and let $\mathfrak{c}$ be a legal pattern of $\mathbb{A}$ on S. Suppose that $\chi_1(\mathfrak{c}(w_0)) \neq \chi_1(\mathfrak{c}(w_0 + 4\mathbf{e}_1))$ and $\chi_2(\mathfrak{c}(w_0)) \neq \chi_2(\mathfrak{c}(w_0 + 4\mathbf{e}_2))$. Let $v \equiv w_0 \pmod 4$ with $v + [0,8)^2 \subseteq w_k + \mathsf{D}_0$ for some k. Then there are letters $\mathfrak{a}, \mathfrak{b}_1, \mathfrak{b}_2$ such that, for $i \in [0,4)^2$ and $e \in \{\mathbf{e}_1, \mathbf{e}_2\}$,*

$$\mathfrak{c}(v+i) = \sigma(\mathfrak{a})(i), \qquad \mathfrak{c}(v + 4e + i) = \sigma(\mathfrak{b}_e)(i),$$

and $\mathfrak{a}$ and $\mathfrak{b}_e$ match in direction e.

Proof. For every k the pattern $\mathfrak{c}$ is legal on $w_k + Q + [-L_1, L_1]^2 = w_k + [-\varrho, \varrho]^2$, so by Lemma 5.15(a) its restriction to $w_k + Q$ is extendable. Let g_k be a tiling that agrees with $\mathfrak{c}$ on $w_k + Q$, and let π_k be its phase. By the remark after Lemma 5.15, π_k depends only on $\mathfrak{c}|_{w_k + \mathsf{D}_0}$. Since $w_{k+1} + \mathsf{D}_0 \subseteq w_k + Q$, the tiling g_k also agrees with $\mathfrak{c}$ on $w_{k+1} + \mathsf{D}_0$; hence $\pi_{k+1} = \pi_k$. So all π_k equal π_0.

The sites w_0, $w_0 + 4\mathbf{e}_1$ and $w_0 + 4\mathbf{e}_2$ lie in $w_0 + \mathsf{D}_0$ because $L_0 \geq 8$. Let p represent π_0. If $w_{0,1} \not\equiv p_1 \pmod 4$, then the columns $w_{0,1}$ and $w_{0,1} + 4$ have the same nonzero residue modulo 4 relative to p_1, so by Proposition 5.11(iii),(iv) they have the same column bit in g_0; this contradicts the hypothesis. So $w_{0,1} \equiv p_1$, and in the same way $w_{0,2} \equiv p_2$, using rows and row bits. Thus π_k is the class of w_0 for every k, and also the class of v.

Take k with $v + [0,8)^2 \subseteq w_k + \mathsf{D}_0$ and use the representative v: by Proposition 5.11(iii) there is a tiling y with $g_k(v + 4m + i) = \sigma(y(m))(i)$. Put $\mathfrak{a} = y(0)$ and $\mathfrak{b}_e = y(e)$. The cells $v + i$ and $v + 4e + i$ lie in $v + [0,8)^2$, where g_k agrees with $\mathfrak{c}$. Since y is a tiling, $\mathfrak{a}$ and $\mathfrak{b}_e$ match in direction e. □

The walk carries the phase step by step through overlapping windows. This is the only way the phase fixed at the anchor reaches a distant block, and every certificate below contains such a walk.

The parser. For $n \geq 0$ and $u, z \in \mathbb{Z}^2$, the *block* of z at depth n relative to u is $u + 4^n z + [0, 4^n)^2$, and its ports are $\mathrm{Port}_n(u + 4^n z)$.

Theorem 5.18 (Parser)**.** *Let B be as in Definition 5.16. For all $n \geq 0$, $u, z \in \mathbb{Z}^2$, $e \in \{\mathbf{e}_1, \mathbf{e}_2\}$ and $s \in u + 4^n z + [0, 4^n)^2$ there are three certificates:*

- *the* type certificate $\Gamma^{\mathrm{ty}}_n(u, z)$*, with boundary* $\mathrm{Anc}_n(u) \cup \mathrm{Port}_n(u + 4^n z)$*;*
- *the* edge certificate $\Gamma^{\mathrm{ed}}_n(u, z, e)$*, with boundary* $\mathrm{Anc}_n(u) \cup \mathrm{Port}_n(u + 4^n z) \cup \mathrm{Port}_n(u + 4^n(z + e))$*;*

- *the* point certificate $\Gamma_n^{\mathrm{pt}}(u, z, s)$*, with boundary* $\mathrm{Anc}_n(u) \cup \mathrm{Port}_n(u + 4^n z) \cup \{s\}$.

Each has a rooted tree decomposition of width at most B. Let ψ be a homomorphism of one of them whose values on $\mathrm{Anc}_n(u)$ form a good n-anchor at u. Then:

(T) there is $\mathfrak{a} \in \mathbb{A}$ such that ψ reads $\mathfrak{a}$ on $\mathrm{Port}_n(u + 4^n z)$;

(E) for the edge certificate, moreover ψ reads some $\mathfrak{b}$ on $\mathrm{Port}_n(u + 4^n(z + e))$, and $\mathfrak{a}$ and $\mathfrak{b}$ match in direction e;

(P) for the point certificate, moreover $\psi(s) = \sigma^n(\mathfrak{a})(s - u - 4^n z)$.

The bound B does not depend on n, u, z, e, s, on the distance from the anchor to the block, or on any machine.

By Lemma 5.12(a) the letters $\mathfrak{a}$ and $\mathfrak{b}$ are unique. The query block may be arbitrarily far from the anchor.

Proof. We build the three certificates together, by induction on n, and check the decomposition and the conclusions at each step.

Depth 0. Here $P_0 = \{(0,0)\}$, so the ports of z form the single site $u + z$, and $s = u + z$. The type and point certificates have one vertex for each boundary site and no edges. The edge certificate adds one edge from $u + z$ to $u + z + e$ with label e. One bag holds everything; it has at most 5 vertices. Conclusions (T) and (P) hold with $\mathfrak{a} = \psi(u + z)$, and (E) holds because of the edge. No anchor condition is needed at depth 0.

From depth n to n + 1: the skeleton. Fix u, z, and e or s as required. Coarse sites $x \in \mathbb{Z}^2$ stand for the depth-n blocks $u + 4^n x + [0, 4^n)^2$; these blocks tile the plane. The depth-$(n+1)$ block of z is the union of the depth-n blocks of $4z + d$, $d \in [0,4)^2$. Choose a corridor walk from $w_0 = (0,0)$ to $w_m = 4z$, say first along $\mathbf{e}_1$ and then along $\mathbf{e}_2$, and let S be its corridor. The *skeleton* has one vertex for each site of

$$\bigcup_{x \in S} \mathrm{Port}_n(u + 4^n x),$$

and, for the point certificate, one more vertex for s if s is not already among these sites. Its grid map is the identity on sites. The skeleton contains the 25 *midpoint* sites $u + 4^n(\iota, \kappa_0)$, $\iota, \kappa_0 \in [0,4]$, because each is the corner port of a coarse site of $[0,4]^2 \subseteq S$. Write $\mathcal{M}_0$ for this set. The boundary lies in the skeleton: $\mathrm{Anc}_{n+1}(u) \subseteq \mathcal{M}_0$, and $\mathrm{Port}_{n+1}(u + 4^{n+1} z)$ is contained in the union of the ports of the coarse sites $4z + d$ by Lemma 5.12(b). The same holds for $z + e$ in the edge certificate, since $4z + 4e + [0,4)^2 \subseteq 4z + \mathsf{D}_0 \subseteq S$.

Bit paths. For $\iota \in [0,4]$ and $\kappa_0 \in [0,3]$ attach a path of fresh vertices along the column of $u + 4^n(\iota, \kappa_0)$ from that site to $u + 4^n(\iota, \kappa_0 + 1)$, with edges labelled $\mathbf{e}_2$; its two end vertices are the skeleton vertices of these midpoints. For $\kappa_0 \in [0,4]$ and $\iota \in [0,3]$ attach in the same way a path along the row from $u + 4^n(\iota, \kappa_0)$ to $u + 4^n(\iota + 1, \kappa_0)$, with edges labelled $\mathbf{e}_1$. There are 40 paths.

The sixteen candidates. For each $c \in [0,4)^2$ put $v_c = u + 4^n c$, so that $\mathrm{Anc}_n(v_c) \subseteq \mathcal{M}_0$. For each c attach a fresh copy of each of the following depth-n certificates:

- $\Gamma_n^{\mathrm{ty}}(v_c, x - c)$ for every $x \in S$;
- $\Gamma_n^{\mathrm{ed}}(v_c, x - c, e')$ for every x and $e' \in \{\mathbf{e}_1, \mathbf{e}_2\}$ with $x, x + e' \in S$;
- for the point certificate only, $\Gamma_n^{\mathrm{pt}}(v_c, 4z + d_s - c, s)$, where $d_s \in [0,4)^2$ is the child whose block contains s.

The block of $x - c$ relative to v_c is the block of x relative to u. So the boundary of each copy consists of skeleton sites, and we identify each boundary vertex of the copy with the skeleton vertex of the same site. All other vertices of the copy are fresh. In particular, the sixteen candidates share the same port vertices: they read the same physical blocks.

The conclusions. Let ψ be a homomorphism of the depth-$(n+1)$ certificate whose values on $\mathrm{Anc}_{n+1}(u)$ form a good $(n+1)$-anchor at u. Write $m_{\iota\kappa_0} = \psi(u + 4^n(\iota, \kappa_0))$.

Step 1: one good candidate. Along a column path every edge joins letters that match in direction $\mathbf{e}_2$, so all letters on the path have the same column bit (Proposition 5.11(v)). Hence $\chi_1(m_{\iota\kappa_0}) = \chi_1(m_{\iota 0})$ for all $\iota, \kappa_0 \in [0,4]$, and in the same way $\chi_2(m_{\iota\kappa_0}) = \chi_2(m_{0\kappa_0})$. Since $\chi_1(m_{00}) \neq \chi_1(m_{40})$, some $\iota \in [0,3]$ has $\chi_1(m_{\iota 0}) \neq \chi_1(m_{\iota+1,0})$. Since $\chi_2(m_{00}) \neq \chi_2(m_{04})$, some $\kappa_0 \in [0,3]$ has $\chi_2(m_{0\kappa_0}) \neq \chi_2(m_{0,\kappa_0+1})$. Then $\chi_1(m_{\iota\kappa_0}) \neq \chi_1(m_{\iota+1,\kappa_0})$ and $\chi_2(m_{\iota\kappa_0}) \neq \chi_2(m_{\iota,\kappa_0+1})$. So ψ is a good n-anchor at v_c for $c = (\iota, \kappa_0)$. The construction attached all sixteen candidates in advance; which one is good depends on ψ, and the argument below uses only that one.

Step 2: a common coarse pattern. The restriction of ψ to a copy attached for c is a homomorphism of that copy, with the same values on its boundary. By the induction hypothesis (T), for every $x \in S$ the map ψ reads some letter on $\mathrm{Port}_n(u + 4^n x)$; by Lemma 5.12(a) it is unique, and we call it $\mathfrak{c}(x)$. It depends only on the values of ψ at the skeleton vertices of these ports, so every copy that meets these ports sees the same letter. By (E), $\mathfrak{c}(x)$ and $\mathfrak{c}(x + e')$ match in direction e' whenever $x, x + e' \in S$. So $\mathfrak{c}$ is a legal pattern on S.

Step 3: the phase. By Lemma 5.12(c), $m_{\iota\kappa_0} = \psi(u + 4^n(\iota, \kappa_0)) = \sigma^n(\mathfrak{c}(\iota, \kappa_0))(0,0)$ has the bits of $\mathfrak{c}(\iota, \kappa_0)$. The good $(n+1)$-anchor therefore gives $\chi_1(\mathfrak{c}(0,0)) \neq \chi_1(\mathfrak{c}(4,0))$ and $\chi_2(\mathfrak{c}(0,0)) \neq \chi_2(\mathfrak{c}(0,4))$. Apply Lemma 5.17 with $v = 4z$, which is congruent to $w_0 = (0,0)$ modulo 4 and satisfies $4z + [0,8)^2 \subseteq w_m + \mathsf{D}_0$ because $L_0 \geq 8$. It gives a letter $\mathfrak{a}$ with $\mathfrak{c}(4z + d) = \sigma(\mathfrak{a})(d)$ for $d \in [0,4)^2$, and, for the edge certificate, a letter $\mathfrak{b}$ with $\mathfrak{c}(4z + 4e + d) = \sigma(\mathfrak{b})(d)$, where $\mathfrak{a}$ and $\mathfrak{b}$ match in direction e.

Step 4: the readout. Let $y \in P_{n+1}$ and write $y = 4^n d + y'$ with $y' \in P_n$, by Lemma 5.12(b). Then

$$\begin{aligned}\psi(u + 4^{n+1}z + y) &= \psi\big(u + 4^n(4z + d) + y'\big) = \sigma^n\big(\mathfrak{c}(4z + d)\big)(y') \\ &= \sigma^n\big(\sigma(\mathfrak{a})(d)\big)(y') = \sigma^{n+1}(\mathfrak{a})(y).\end{aligned}$$

So ψ reads $\mathfrak{a}$ on $\mathrm{Port}_{n+1}(u + 4^{n+1}z)$, which is (T). The same computation for $z + e$ gives (E).

Step 5: the point. For the good candidate c, the induction hypothesis (P) for the attached copy of $\Gamma_n^{\mathrm{pt}}(v_c, 4z + d_s - c, s)$ gives a letter $\mathfrak{a}''$ read on the ports of $4z + d_s$, with $\psi(s) = \sigma^n(\mathfrak{a}'')(s - u - 4^n(4z + d_s))$. By uniqueness of the readout, $\mathfrak{a}'' = \mathfrak{c}(4z + d_s) = \sigma(\mathfrak{a})(d_s)$. Since $s - u - 4^{n+1}z = 4^n d_s + \big(s - u - 4^n(4z + d_s)\big)$, Definition 5.8 gives $\psi(s) = \sigma^{n+1}(\mathfrak{a})(s - u - 4^{n+1}z)$, which is (P).

The decomposition. Let ∂ be the boundary of the depth-$(n+1)$ certificate. For $k \in [0, m]$ let β_k consist of the skeleton vertices at the sites of $\mathcal{M}_0$, of ∂, and of $\mathrm{Port}_n(u + 4^n x)$ for every $x \in S \cap \big(w_k + [-\varrho - 1, \varrho + 1]^2\big)$. Arrange $\beta_0, \ldots, \beta_m$ in a path with root β_0. A skeleton vertex at a site of $\mathcal{M}_0 \cup \partial$ lies in every bag. Any other skeleton vertex lies in the ports of exactly one coarse site x, because the depth-n blocks are disjoint; the set of k with $x \in w_k + [-\varrho - 1, \varrho + 1]^2$ is an interval, because each coordinate of w_k is monotone in k. So the skeleton bags satisfy the connectivity condition. Every $x \in S$ lies in $w_k + [-\varrho, \varrho]^2$ for some k, and then its ports and those of every neighbour $x + e' \in S$ lie in β_k.

Attach the decomposition of each child copy by joining its root bag to a bag that contains the copy's boundary. For a type copy at x or an edge copy at $x, x + e'$, take β_k with $x \in w_k + [-\varrho, \varrho]^2$; the anchor sites of the copy lie in $\mathcal{M}_0$. For the point copy, take β_m: the coarse

site $4z + d_s$ lies in $w_m + [-\varrho, \varrho]^2$, and $s \in \partial$. For a bit path with vertices $x_0, \dots, x_N$, use the bags $\{x_0, x_N, x_k, x_{k+1}\}$, $k < N$, in a path, and join its first bag to β_0, which contains the midpoints x_0 and x_N. The fresh vertices of a copy or a path lie only in its own bags, and each shared vertex lies in a connected set of skeleton bags together with the connected set of bags of each copy that contains it, joined at the attachment. Every edge lies inside a copy or a path, where a bag contains it. So this is a rooted tree decomposition, and $\beta_0 \supseteq \partial$.

Each β_k has at most $\ell(2\varrho + 3)^2$ port vertices, 25 midpoints, and at most $2\ell + 1$ further boundary vertices, so $|\beta_k| \le \ell(2\varrho + 3)^2 + 2\ell + 26 \le B$. The bags of the paths have 4 vertices, and the bags of the copies have at most B vertices by induction. So the width is at most B. The bound is a maximum over finitely many kinds of bags, not a sum over depths, and the length of the corridor changes only the number of bags. □

Two features of the construction carry the proof. First, all sixteen candidates are attached to the same port vertices, so the letter decoded at a coarse site is one letter, whatever certificate looks at it. Second, the point certificate contains the whole corridor from the anchor to the block that holds s. A window near the block alone would fix some phase of the decoded pattern, but nothing would tie that phase to the anchor's grid; inside the recursion the child block can lie at coarse distance $3, 15, 63, \dots$ from the candidate anchors.

The bit paths use the edgewise bit property of Proposition 5.11(v). If the bits were only known to be constant along the columns and rows of tilings, a bare path would force nothing, since a homomorphism of a certificate need not extend to a tiling. The literal bit colours of Definition 5.7 avoid this.

5.5 Transfer to pebble families, and anchors of every depth

We use E_K families for the Wang set $\mathbb{A}$ as in Definition 2.22: nonempty sets of legal patterns of $\mathbb{A}$ on finite subsets of $\mathbb{Z}^2$, each with at most K sites, closed under restriction and with the extension property for maps on fewer than K sites. By Remark 2.23, a member h extends to a member defined on $\operatorname{dom} h \cup U'$ whenever $|\operatorname{dom} h \cup U'| \le K$.

Lemma 5.19 (Transfer)**.** *Let H be an E_K family for $\mathbb{A}$. Let Γ be a certificate with a rooted tree decomposition of width at most B', and let $\Pi \subseteq \mathbb{Z}^2$ be a finite set of* pinned *sites with $|\Pi| \le K - B'$. Let $h \in H$ have domain $\partial\Gamma \cup \Pi$. Then there is a homomorphism ψ of Γ such that $\psi(x) = h(x)$ for every boundary site x, and $\psi(v) = h(\rho(v))$ for every vertex v with $\rho(v) \in \Pi$.*

So a use of a certificate of width at most B' may retain up to $K - B'$ pinned sites beyond its bags, and the certificate then respects the pins wherever it touches them. With $B' = B$ and $K = j = 10(B + \ell + 100)$, as in Theorem 7.1, this allows $9B + 10\ell + 1000$ pinned sites. The applications in this paper need fewer: Proposition 5.21 and Corollary 5.20 below take $\Pi = \emptyset$, and Section 7 applies certificates only to restrictions of its maps to the certificate boundary, which is again the case $\Pi = \emptyset$. All maps there have at most $\ell + 6 \le B$ sites, so the single condition $K \ge B$ covers the fixed boxes of Proposition 5.21, the certificates, and the pins and sibling sites of Section 7.

Proof. Let (β_t) be the decomposition, with root t_0; it has width at most B', so $|\rho(\beta_t)| \le B'$. We choose maps $g_t \in H$ with domain $\rho(\beta_t) \cup \Pi$ and $g_t|_\Pi = h|_\Pi$, from the root down. For the root, $\partial\Gamma \subseteq \rho(\beta_{t_0})$, and h extends to g_{t_0} on $\rho(\beta_{t_0}) \cup \Pi$, a set of at most K sites. For a child t of a node t', restrict $g_{t'}$ to $\rho(\beta_t \cap \beta_{t'}) \cup \Pi$ and extend the restriction to $\rho(\beta_t) \cup \Pi$. Put $\psi(v) = g_t(\rho(v))$ for any t with $v \in \beta_t$. This does not depend on t: the nodes whose bags contain v form a subtree, and along each of its edges the two maps agree at $\rho(v)$ by construction. Every edge (v, v') of

Γ with label e lies in a bag β_t; then $\psi(v) = g_t(\rho(v))$ and $\psi(v') = g_t(\rho(v) + e)$ are values of one legal pattern at sites $\rho(v)$ and $\rho(v) + e$, so they match in direction e. At a boundary site x, $\psi(x) = g_{t_0}(x) = h(x)$. If $\rho(v) \in \Pi$, then $\psi(v) = g_t(\rho(v)) = h(\rho(v))$. □

This is the standard argument showing that the existential K-pebble game decides homomorphism from structures of tree width less than K [16, 6]. Two vertices with one site in a common bag receive one letter. Vertices with one site in unrelated branches may receive different letters, and Theorem 5.18 never needs them to agree.

Corollary 5.20 (Readout in a family)**.** *Let H be an E_K family for $\mathbb{A}$ with $K \geq B$. Let $n \geq 0$, $u \in \mathbb{Z}^2$, and let $g \in H$ be defined on $\mathrm{Anc}_n(u) \cup \mathrm{Port}_n(u)$ and be a good n-anchor at u. Then g reads a letter $\mathfrak{a}$ on $\mathrm{Port}_n(u)$, that is $g(y) = \sigma^n(\mathfrak{a})(y - u)$ for every $y \in \mathrm{Port}_n(u)$; the letter is unique. If g is also defined at a site $s \in u + [0, 4^n)^2$, then $g(s) = \sigma^n(\mathfrak{a})(s - u)$. More generally, for every $z \in \mathbb{Z}^2$: if g is defined on $\mathrm{Anc}_n(u) \cup \mathrm{Port}_n(u + 4^n z)$ with a good anchor, it reads a letter on $\mathrm{Port}_n(u + 4^n z)$, and the value at any s in the block of z is the corresponding entry of σ^n of that letter.*

Proof. Restrict g to the boundary of the type certificate $\Gamma_n^{\mathrm{ty}}(u, z)$, or of the point certificate $\Gamma_n^{\mathrm{pt}}(u, z, s)$, and apply Lemma 5.19 with $\Pi = \emptyset$ and $B' = B$. The resulting homomorphism has a good anchor, and Theorem 5.18(T),(P) give the claims. The letter given by the point certificate is the one read on the ports, by Lemma 5.12(a). □

Proposition 5.21 (Anchors of every depth)**.** *Let H be an E_K family for $\mathbb{A}$ with $K \geq B$. For every $n \geq 0$ there are $u \in \mathbb{Z}^2$ and $h \in H$ with domain $\mathrm{Anc}_n(u)$ that is a good n-anchor at u.*

Proof. We induct on n. The site u may depend on H and on n.

Depth 0. The box $Y = [-L_2, 4 + L_2]^2$ has $(2L_2 + 5)^2 \leq B \leq K$ sites, so H has a member g with domain Y (Remark 2.23). It is a legal pattern, so by Lemma 5.15(a) its restriction to $[0, 5)^2$ is the restriction of a tiling f. Let p represent the phase of f, and let $x \in [0, 4)^2$ satisfy $x \equiv p + (1, 1) \pmod 4$. Then x, $x + \mathbf{e}_1$ and $x + \mathbf{e}_2$ lie in $[0, 5)^2$. By Proposition 5.11(iii),(iv), the column of x has residue 1 relative to p_1 and column bit 0, and the column of $x + \mathbf{e}_1$ has residue 2 and column bit 1. In the same way $\chi_2(f(x)) = 0$ and $\chi_2(f(x + \mathbf{e}_2)) = 1$. So $h = g|_{\mathrm{Anc}_0(x)}$ is a good 0-anchor at x, and $h \in H$.

From depth n to $n + 1$. Let $h_n \in H$ be a good n-anchor at u, with domain $\mathrm{Anc}_n(u)$. Let $U = [-L_3, 31 + L_3]^2$ and

$$\Sigma = \mathrm{Anc}_n(u) \cup \bigcup_{x \in U} \mathrm{Port}_n(u + 4^n x).$$

Then $|\Sigma| \leq 3 + \ell(2L_3 + 32)^2 \leq B \leq K$, so h_n extends to a member $g \in H$ with domain Σ. By Corollary 5.20, for every $x \in U$ the map g reads a unique letter $\mathfrak{c}(x)$ on $\mathrm{Port}_n(u + 4^n x)$. For $x, x + e \in U$, the edge certificate $\Gamma_n^{\mathrm{ed}}(u, x, e)$ with Lemma 5.19, applied to the restriction of g to its boundary, shows that $\mathfrak{c}(x)$ and $\mathfrak{c}(x + e)$ match in direction e. So $\mathfrak{c}$ is a legal pattern on U, and by Lemma 5.15(a) its restriction to $[0, 32)^2$ is the restriction of a tiling f.

Apply Proposition 5.11(iii) to f and then to its parent: there are $o \in [0, 16)^2$ and a tiling y' with $f(o + 16m + i) = \sigma^2(y'(m))(i)$ for $i \in [0, 16)^2$. By Proposition 5.11(iv) applied twice, every letter of $\sigma^2(\mathfrak{b})$ in column 4 has column bit $\chi_1(\sigma(\mathfrak{b})(1, \cdot)) = 0$, and every letter in column 8 has column bit $\chi_1(\sigma(\mathfrak{b})(2, \cdot)) = 1$; rows 4 and 8 have row bits 0 and 1. Put $w = o + (4, 4)$. The sites w, $w + 4\mathbf{e}_1 = o + (8, 4)$ and $w + 4\mathbf{e}_2 = o + (4, 8)$ lie in $o + [0, 16)^2 \subseteq [0, 32)^2$. So

$$\chi_1(\mathfrak{c}(w)) = 0 \neq 1 = \chi_1(\mathfrak{c}(w + 4\mathbf{e}_1)), \qquad \chi_2(\mathfrak{c}(w)) = 0 \neq 1 = \chi_2(\mathfrak{c}(w + 4\mathbf{e}_2)).$$

By Lemma 5.12(c), $g(u+4^n x) = \sigma^n(\mathfrak{c}(x))(0,0)$ has the bits of $\mathfrak{c}(x)$. Put $u' = u + 4^n w$. Then $\mathrm{Anc}_{n+1}(u') = \{u+4^n w,\ u+4^n(w+4\mathbf{e}_1),\ u+4^n(w+4\mathbf{e}_2)\} \subseteq \Sigma$, and $h_{n+1} = g|_{\mathrm{Anc}_{n+1}(u')}$ is a good $(n+1)$-anchor at u'. It lies in H by restriction. □

The old anchor is dropped as soon as the new one is found, so no pins accumulate over the depths: each step uses one member of H on at most B sites. The anchor of depth n may sit anywhere, and different families may place it differently. Together, Proposition 5.21 and Corollary 5.20 give, in every E_K family with $K \geq B$ and at every depth n, a member on $\mathrm{Anc}_n(u) \cup \mathrm{Port}_n(u)$, at most $\ell + 3$ sites, that reads one root letter $\mathfrak{a}$; and every member that agrees with it there and is defined at a site of the block $u + [0,4^n)^2$ takes the value prescribed by the word $\sigma^n(\mathfrak{a})$.

6 Routing a machine onto the scaffold

This section builds the family $M \mapsto R_M$. A tile of R_M sits on one quarter of a letter of the scaffold $\mathbb{A}$ of Section 5. It carries the scaffold letter and its quadrant, two flows that read the light red walls of the scaffold, and a payload. The flows mark the free rows and columns of nested boards, and the payload carries a computation of M along them. Routing a computation through the free rows and columns of nested boards is Robinson's method [26]; Berger embedded Turing machines in Wang sets to show that tiling is undecidable [3]. Our payload is a system of Horn clauses. Its local constraints are closed under intersection, the property that lets Section 7 rebuild a payload by local consistency [14, 8].

Section 6.1 classifies the red paths of the scaffold. Section 6.2 defines the quarters, the walls, the flows and the roles, and proves that walls force the carriers (Proposition 6.11). Section 6.3 studies the light frames. Section 6.4 builds a routing model on the whole plane, and Section 6.5 finds an initialized board inside every word $\sigma^n(\mathfrak{a})$ (Proposition 6.15). Section 6.6 builds the Horn payload (Proposition 6.16), and Section 6.7 defines R_M and shows that it tiles when M does not halt (Definition 6.17 and Proposition 6.18).

We keep the notation of Section 5: letters $\mathfrak{a} = (\theta, \kappa, \omega, b_1, b_2)$ of $\mathbb{A}$, column bit χ_1, row bit χ_2, the substitutions φ, φ' and $\sigma = (\varphi')^2$, and the endpoints $\partial_{\mathrm{N}}, \partial_{\mathrm{E}}, \partial_{\mathrm{S}}, \partial_{\mathrm{W}}$ of Table 1. Integer intervals are $[a,b] = \{a,\dots,b\}$, and $z \bmod m \in [0,m)$. An atom r_i is *red* and an atom g_i is *green*. The base tile $(\mathsf{a}, \angle\mathsf{b}, \mathsf{A})$ is the *index*-0 *tile*, and a letter with this base is an *index*-0 *letter*. Ruler bits $b_1 = b_2 = 0$ are *dark*.

6.1 Towers and red paths

Definition 6.1 (Towers)**.** Let $h \in \{1,2,\dots\} \cup \{\infty\}$ and $v_m \in \{0,1\}^2$ for $0 \leq m < h$. A *tower of height h with digits (v_m)* is a family of legal patterns f_m of $\mathbb{A}$ on sets $U_m \subseteq \mathbb{Z}^2$, for integers $0 \leq m \leq h$, such that for every $m < h$

$$U_m = v_m + 2U_{m+1} + [0,2)^2, \qquad f_m(v_m + 2v + \varsigma) = \varphi'\big(f_{m+1}(v)\big)(\varsigma) \quad (v \in U_{m+1},\ \varsigma \in [0,2)^2).$$

If $h = \infty$ we require $U_m = \mathbb{Z}^2$ for all m, so every f_m is a tiling. The *phase* of the tower is $\varpi = \sum_{m<h} 2^m v_m$, an element of $\mathbb{Z}^2$ if $h < \infty$ and of the 2-adic integers $\mathbb{Z}_2^2$ if $h = \infty$.

For a letter $\mathfrak{a}$ and $n \geq 1$, the patterns $f_m = (\varphi')^{2n-m}(\mathfrak{a})$ on $[0, 2^{2n-m})^2$, $0 \leq m \leq 2n$, form a tower of height $2n$ with all digits 0 and phase 0. Its bottom pattern is $f_0 = (\varphi')^{2n}(\mathfrak{a}) = \sigma^n(\mathfrak{a})$, because blockwise substitution of patterns composes. Its patterns are legal by Lemma 5.9.

Lemma 6.2 (Towers of every phase)**.** *For every $\varpi \in \mathbb{Z}_2^2$ there is a tower of height ∞ with phase ϖ.*

Proof. Write $\varpi = \sum_{m\ge 0} 2^m v_m$ with $v_m \in \{0,1\}^2$. For a tiling f of $\mathbb{A}$ and $v \in \{0,1\}^2$, the pattern $\mathrm{sub}_v(f)(v + 2v + \varsigma) = \varphi'(f(v))(\varsigma)$ is a tiling by Lemma 5.9(b). Let $\mathcal{T}_l$ be the set of sequences $(f_m)_{m\ge 0}$ of tilings with $f_m = \mathrm{sub}_{v_m}(f_{m+1})$ for $m < l$. It is nonempty: take any tiling f (Proposition 5.11), put $f_m = f$ for $m \ge l$ and $f_m = \mathrm{sub}_{v_m}(f_{m+1})$ for $m < l$. Each condition $f_m = \mathrm{sub}_{v_m}(f_{m+1})$ is a family of conditions on finitely many sites each, so $\mathcal{T}_l$ is closed in the product of the compact spaces $\mathbb{A}^{\mathbb{Z}^2}$. The sets $\mathcal{T}_l$ decrease, so their intersection is nonempty. An element of it is a tower with phase ϖ. □

Fix a tower. For $x \in U_0$ put $x^{(0)} = x$, and for $m < h$ let $\varsigma_m(x) \in \{0,1\}^2$ and $x^{(m+1)} \in U_{m+1}$ be the unique pair with

$$x^{(m)} = v_m + 2x^{(m+1)} + \varsigma_m(x).$$

Then $f_m(x^{(m)}) = \varphi'(f_{m+1}(x^{(m+1)}))(\varsigma_m(x))$, and by induction

$$x - \varpi \equiv \sum_{m<l} 2^m \varsigma_m(x) \pmod{2^l} \qquad (l \le h,\ l < \infty). \tag{12}$$

The first coordinate $\varsigma_{m,1}(x)$ depends only on x_1, and the second only on x_2. The *rank* of a column x_1 is the least $m < h$ with $\varsigma_{m,1}(x) = 1$, if there is one; the rank of a row x_2 is defined with $\varsigma_{m,2}$. By (12), a column of rank r satisfies $x_1 - \varpi_1 \equiv 2^r \pmod{2^{r+1}}$. If $h = \infty$ and $\varpi_1 \notin \mathbb{Z}$, every column has a finite rank, namely the 2-adic valuation of $x_1 - \varpi_1$.

Lemma 6.3 (Ruler bits)**.** *If the column x_1 has rank r, then $\chi_1(f_0(x)) \equiv r \pmod 2$ for every $x \in U_0$ in this column. If the row x_2 has rank r, then $\chi_2(f_0(x)) \equiv r \pmod 2$.*

Proof. By Table 3, a child ς of a letter has column bit 0 if $\varsigma_1 = 1$, and the complement of the column bit of the parent if $\varsigma_1 = 0$. The letter $f_r(x^{(r)})$ is a child with $\varsigma_{r,1}(x) = 1$, so its column bit is 0. For $m < r$ we have $\varsigma_{m,1}(x) = 0$, so each of the r steps down to $f_0(x)$ complements the bit. Rows are the same. □

A thick type has a *vertical red atom* if its north or its south endpoint is red, and a *horizontal red atom* if its east or its west endpoint is red. A line $\langle \alpha, \gamma \rangle$ has a vertical red atom exactly when α is red. Every corner has exactly one red endpoint in each direction (Table 1): $\angle$b has red north r_0 and red east r_1, $\angle$c red north r_2 and red west r_1, $\angle$a red south r_0 and red east r_3, and $\angle$d red south r_2 and red west r_3. So $\angle$b is the south-west corner of a red rectangle, $\angle$c the south-east, $\angle$a the north-west and $\angle$d the north-east corner.

Lemma 6.4 (Red paths)**.** *Let (f_m) be a tower of height h, and let $r \ge 0$ with $r + 2 \le h$. For $i = 1, 2$ and $z \in \mathbb{Z}$ put $s_i(z) = (z - \varpi_i) \bmod 2^{r+2}$. Let x_1 be a column of rank r. Then $s_1(x_1) \in \{2^r, 3 \cdot 2^r\}$, and for every $x = (x_1, x_2) \in U_0$:*

(a) the thick type of $f_0(x)$ has a vertical red atom if and only if $s_2(x_2) \in [2^r, 3 \cdot 2^r]$;

(b) this atom is r_0 if $s_1(x_1) = 2^r$ and r_2 if $s_1(x_1) = 3 \cdot 2^r$;

(c) the thick type is the corner with red north endpoint ($\angle$b or $\angle$c) if $s_2(x_2) = 2^r$, the corner with red south endpoint ($\angle$a or $\angle$d) if $s_2(x_2) = 3 \cdot 2^r$, and a line if $2^r < s_2(x_2) < 3 \cdot 2^r$.

The same holds for a row x_2 of rank r with the coordinates exchanged: $s_2(x_2) \in \{2^r, 3 \cdot 2^r\}$; a horizontal red atom occurs if and only if $s_1(x_1) \in [2^r, 3 \cdot 2^r]$; it is r_1 if $s_2(x_2) = 2^r$ and r_3 if $s_2(x_2) = 3 \cdot 2^r$; and the thick type is the corner with red east endpoint ($\angle$b or $\angle$a) if $s_1(x_1) = 2^r$, the corner with red west endpoint ($\angle$c or $\angle$d) if $s_1(x_1) = 3 \cdot 2^r$, and a line in between.

Proof. By (12) with $l = r+2$, $s_1(x_1) = 2^r + 2^{r+1}\varsigma_{r+1,1}(x)$, which is 2^r or $3\cdot 2^r$. We prove (a)–(c) by induction on r, for all towers of height at least $r+2$ at once.

Let $r = 0$, so $\varsigma_{0,1}(x) = 1$. The letter $f_0(x)$ is child $(1, \varsigma_{0,2})$ of $P = f_1(x^{(1)})$, and P is child $\varsigma_1(x)$ of $f_2(x^{(2)})$; this uses $h \geq 2$. By Table 3 the thin type θ of P is $\mathsf{a}, \mathsf{c}, \mathsf{d}, \mathsf{b}$ for $\varsigma_1(x) = (0,0), (1,0), (0,1), (1,1)$, and the thick type of $f_0(x)$ is the line $\langle \partial_S\theta, \cdot\rangle$ if $\varsigma_{0,2} = 0$ and the corner $F(\theta)$ if $\varsigma_{0,2} = 1$. Here $s_2(x_2) = \varsigma_{0,2} + 2\varsigma_{1,2}$ and $s_1(x_1) = 1 + 2\varsigma_{1,1}$. Table 1 and $F(\mathsf{a},\mathsf{b},\mathsf{c},\mathsf{d}) = (\angle\mathsf{b}, \angle\mathsf{d}, \angle\mathsf{c}, \angle\mathsf{a})$ give:

$s_2(x_2)$	0	1	2	3
$s_1(x_1) = 1$	$\partial_S\mathsf{a} = \mathsf{g}_2$	$F(\mathsf{a}) = \angle\mathsf{b}$	$\partial_S\mathsf{d} = \mathsf{r}_0$	$F(\mathsf{d}) = \angle\mathsf{a}$
$s_1(x_1) = 3$	$\partial_S\mathsf{c} = \mathsf{g}_0$	$F(\mathsf{c}) = \angle\mathsf{c}$	$\partial_S\mathsf{b} = \mathsf{r}_2$	$F(\mathsf{b}) = \angle\mathsf{d}$

This is (a)–(c) for $r = 0$.

Let $r \geq 1$, so $\varsigma_{0,1}(x) = 0$. The family $(f_{m+1})_m$ is a tower of height $h-1 \geq r+1$ with digits (v_{m+1}) and phase ϖ', where $\varpi = v_0 + 2\varpi'$. In it, the column of $x^{(1)}$ has rank $r-1$, the digits of $x^{(1)}$ are $\varsigma_{m+1}(x)$, and $x - \varpi = 2(x^{(1)} - \varpi') + \varsigma_0(x)$. With $s_i'(z) = (z - \varpi_i') \bmod 2^{r+1}$ this gives $s_i(x_i) = 2s_i'(x_i^{(1)}) + \varsigma_{0,i}(x)$ and $s_1(x_1) = 2s_1'(x_1^{(1)})$. The letter $f_0(x)$ is child $(0, \varsigma_{0,2})$ of $P = f_1(x^{(1)})$, with thick type κ_P. By Table 3, child $(0,0)$ has thick type κ_P, and child $(0,1)$ has the line $\langle \partial_N\kappa_P, \cdot\rangle$. Apply the induction hypothesis to P. If $\varsigma_{0,2} = 0$, then $s_2 = 2s_2'$ and $f_0(x)$ has the thick type of P; (a)–(c) follow, since $s_2' \in [2^{r-1}, 3\cdot 2^{r-1}]$ if and only if $2s_2' \in [2^r, 3\cdot 2^r]$, and the end values correspond. If $\varsigma_{0,2} = 1$, then $s_2 = 2s_2' + 1$ is odd, and $f_0(x)$ is a line whose vertical atom is $\partial_N\kappa_P$. This endpoint is red exactly when κ_P is a line with a red vertical atom or the corner with red north endpoint, that is when $s_2' \in [2^{r-1}, 3\cdot 2^{r-1})$. Since $r \geq 1$, the odd numbers $2s_2' + 1$ with this property are exactly the odd numbers in $[2^r, 3\cdot 2^r]$, and they lie strictly between 2^r and $3\cdot 2^r$. In both cases the atom is that of P, so (b) follows from $s_1(x_1) = 2s_1'(x_1^{(1)})$.

For rows the proof is the same with the coordinates exchanged. For $r = 0$ the letter is child $(\varsigma_{0,1}, 1)$ of P, that is the line $\langle \cdot, \partial_W\theta\rangle$ or the corner $F(\theta)$, and the table reads, for $s_1(x_1) = 0,1,2,3$: $\partial_W\mathsf{a} = \mathsf{g}_3$, $F(\mathsf{a}) = \angle\mathsf{b}$, $\partial_W\mathsf{c} = \mathsf{r}_1$, $F(\mathsf{c}) = \angle\mathsf{c}$ when $s_2(x_2) = 1$, and $\partial_W\mathsf{d} = \mathsf{g}_1$, $F(\mathsf{d}) = \angle\mathsf{a}$, $\partial_W\mathsf{b} = \mathsf{r}_3$, $F(\mathsf{b}) = \angle\mathsf{d}$ when $s_2(x_2) = 3$. In the induction step child $(1,0)$ is the line $\langle \cdot, \partial_E\kappa_P\rangle$, and $\partial_E\kappa_P$ is red exactly for a line with red horizontal atom or the corner with red east endpoint. □

So the red atoms in the columns and rows of rank r form the boundaries of squares: for a column x_1 with $s_1(x_1) = 2^r$, the column $x_1 + 2^{r+1}$ has $s_1 = 3\cdot 2^r$ and the same rank, and the two sides run over the same rows. By Lemma 6.3 every letter on a vertical side of such a boundary has column bit r mod 2, every letter on a horizontal side has row bit r mod 2, and the four corners have both bits equal to r mod 2.

6.2 Quarters, walls, flows and roles

Each letter of $\mathbb{A}$ is cut into four quarters. A *scaffold pair* is a pair $(\varsigma, \mathfrak{a}) \in \{0,1\}^2 \times \mathbb{A}$; the first entry is its *quadrant* and the second its *parent*. Let $\circ$ and $\bullet$ be two new symbols.

Definition 6.5 (Quarters)**.** The quarter set $\mathbb{A}^q$ is the Wang set of the $4\cdot 276$ scaffold pairs. The colours of $(\varsigma, \mathfrak{a})$, $\varsigma = (\varsigma_1, \varsigma_2)$, are

east:	$(\circ, \mathfrak{a}, \varsigma_2)$ if $\varsigma_1 = 0$,	$(\bullet, \mathfrak{a}_E, \varsigma_2)$ if $\varsigma_1 = 1$;
west:	$(\circ, \mathfrak{a}, \varsigma_2)$ if $\varsigma_1 = 1$,	$(\bullet, \mathfrak{a}_W, \varsigma_2)$ if $\varsigma_1 = 0$;
north:	$(\circ, \mathfrak{a}, \varsigma_1)$ if $\varsigma_2 = 0$,	$(\bullet, \mathfrak{a}_N, \varsigma_1)$ if $\varsigma_2 = 1$;
south:	$(\circ, \mathfrak{a}, \varsigma_1)$ if $\varsigma_2 = 1$,	$(\bullet, \mathfrak{a}_S, \varsigma_1)$ if $\varsigma_2 = 0$.

Here $\mathfrak{a}_E, \dots$ are the colours of $\mathfrak{a}$ in $\mathbb{A}$. The *quartering* of a pattern f of $\mathbb{A}$ on U is the pattern f^{q} on $2U + [0,2)^2$ with $f^{\mathrm{q}}(2x + \varsigma) = (\varsigma, f(x))$. Its value at ξ is the *quarter of f at ξ*.

Lemma 6.6 (Quarter encoding)**.** *(a) Scaffold pairs $(\varsigma, \mathfrak{a})$ and $(\varsigma', \mathfrak{a}')$ match horizontally if and only if $\varsigma' \equiv \varsigma + \mathbf{e}_1 \pmod 2$ and either $\varsigma_1 = 0$ and $\mathfrak{a}' = \mathfrak{a}$, or $\varsigma_1 = 1$ and $\mathfrak{a}, \mathfrak{a}'$ match horizontally in $\mathbb{A}$. They match vertically if and only if $\varsigma' \equiv \varsigma + \mathbf{e}_2 \pmod 2$ and either $\varsigma_2 = 0$ and $\mathfrak{a}' = \mathfrak{a}$, or $\varsigma_2 = 1$ and $\mathfrak{a}, \mathfrak{a}'$ match vertically in $\mathbb{A}$.*

(b) A pattern f of $\mathbb{A}$ is legal if and only if f^{q} is legal. A map $\mathcal{Y}$ on $\mathbb{Z}^2$ is a tiling of $\mathbb{A}^{\mathrm{q}}$ if and only if $\mathcal{Y}(\xi) = f^{\mathrm{q}}(\xi - c)$ for some $c \in \{0,1\}^2$ and some tiling f of $\mathbb{A}$.

Proof. (a) If $\varsigma_1 = 0$, the east colour $(\circ, \mathfrak{a}, \varsigma_2)$ equals a west colour only if that colour is internal; so $\varsigma'_1 = 1$, $\mathfrak{a}' = \mathfrak{a}$ and $\varsigma'_2 = \varsigma_2$. If $\varsigma_1 = 1$, the east colour is external; so $\varsigma'_1 = 0$, $\varsigma'_2 = \varsigma_2$ and $\mathfrak{a}_E = \mathfrak{a}'_W$. The converse is read off the same colours. The vertical case is the same.

(b) The first claim follows from (a): inside a block $2x + [0,2)^2$ the four quarters have one parent, and across blocks the external colours compare the parents. For the second, let $\mathcal{Y}$ be a tiling. By (a), the quadrant of $\mathcal{Y}(\xi)$ minus ξ is constant modulo 2 along rows and along columns, so it is a constant c modulo 2. By (a) again, the four quarters of each block $c + 2x + [0,2)^2$ have a common parent $f(x)$, and adjacent blocks have matching parents. The converse follows from the first claim. □

Definition 6.7 (Walls)**.** Let $(\varsigma, \mathfrak{a})$ be a scaffold pair whose parent has thick type κ.

- The pair lies on a *vertical red path* if κ has a vertical red atom, $\varsigma_1 = 1$ if this atom is r_0 and $\varsigma_1 = 0$ if it is r_2, and, when κ is a corner, $\varsigma_2 = 1$ if its north endpoint is red and $\varsigma_2 = 0$ if its south endpoint is red. It is then a *left side* if the atom is r_0 and a *right side* if it is r_2. The path is *light* at the pair if $\chi_1(\mathfrak{a}) = 1$ and *dark* if $\chi_1(\mathfrak{a}) = 0$. A pair on a light vertical red path lies on a *light vertical wall*.
- The pair lies on a *horizontal red path* if κ has a horizontal red atom, $\varsigma_2 = 1$ if this atom is r_1 and $\varsigma_2 = 0$ if it is r_3, and, when κ is a corner, $\varsigma_1 = 1$ if its east endpoint is red and $\varsigma_1 = 0$ if its west endpoint is red. It is then a *bottom side* if the atom is r_1 and a *top side* if it is r_3. The path is light if $\chi_2(\mathfrak{a}) = 1$, and the pair then lies on a *light horizontal wall*.

So the lane r_0 runs in the east half of a letter and r_2 in the west half; r_1 runs in the north half and r_3 in the south half. A corner turns in one quarter: $\angle\mathsf{b}$ in quadrant $(1,1)$, $\angle\mathsf{c}$ in $(0,1)$, $\angle\mathsf{a}$ in $(1,0)$ and $\angle\mathsf{d}$ in $(0,0)$. That quarter is on a vertical and on a horizontal path, and both have the shade $b_1 = b_2$ by (R3). A pair lies on at most one vertical and at most one horizontal red path, so the side of a wall is well defined.

Definition 6.8 (Frames)**.** Let integers a_d, $d \geq 1$, satisfy $a_{d+1} \equiv a_d \pmod{4^{d+1}}$. The *frames of depth d* are the intervals

$$I_d(m) = \left[a_d + (4m+1)4^d + 1,\ a_d + (4m+3)4^d\right], \qquad m \in \mathbb{Z}.$$

The *left endpoint* of $I_d(m)$ is $a_d + (4m+1)4^d + 1$ and its *right endpoint* is $a_d + (4m+3)4^d$. We write $\mathring{I}$ for the interior of an interval I, that is I without its two endpoints.

For a tower whose phase (ϖ_0, ϖ_0) has equal coordinates we use the frames with $a_d \equiv 2\varpi_0 \pmod{4^{d+1}}$. They are the light red squares of the tower, seen in quarter coordinates.

Lemma 6.9 (Light walls)**.** *Let (f_m) be a tower of height h with phase (ϖ_0, ϖ_0), take the frames with $a_d \equiv 2\varpi_0 \pmod{4^{d+1}}$, and let $\xi \in 2U_0 + [0,2)^2$. Suppose the column $\lfloor \xi_1/2 \rfloor$ has rank at*

most $h-2$. Then the quarter of f_0 at ξ lies on a light vertical wall if and only if there are $d \geq 1$ and $m, m' \in \mathbb{Z}$ such that ξ_1 is an endpoint of $I_d(m)$ and $\xi_2 \in I_d(m')$. It is a left side when ξ_1 is the left endpoint and a right side when ξ_1 is the right endpoint, and the rank of the column is then $2d-1$. Dually, if the row $\lfloor \xi_2/2 \rfloor$ has rank at most $h-2$, the quarter lies on a light horizontal wall if and only if ξ_2 is an endpoint of some $I_d(m')$ and $\xi_1 \in I_d(m)$ for some m; it is a bottom side at the left endpoint and a top side at the right endpoint.

Proof. Write $\xi = 2x + \varsigma$ and let $r \leq h-2$ be the rank of the column x_1. With s_i as in Lemma 6.4, put $\hat{s}_i = (\xi_i - 2\varpi_0) \bmod 2^{r+3}$, so that $\hat{s}_i = 2s_i(x_i) + \varsigma_i$. By Lemma 6.4(a)–(c) and Definition 6.7, the quarter lies on a vertical red path if and only if three conditions hold: $s_2(x_2) \in [2^r, 3\cdot 2^r]$; $\varsigma_1 = 1$ if $s_1(x_1) = 2^r$ and $\varsigma_1 = 0$ if $s_1(x_1) = 3\cdot 2^r$; and $\varsigma_2 = 1$ if $s_2(x_2) = 2^r$, $\varsigma_2 = 0$ if $s_2(x_2) = 3\cdot 2^r$. Since $s_1(x_1) \in \{2^r, 3\cdot 2^r\}$, this says

$$\hat{s}_1 \in \{2^{r+1}+1,\ 3\cdot 2^{r+1}\} \quad \text{and} \quad \hat{s}_2 \in [2^{r+1}+1,\ 3\cdot 2^{r+1}],$$

with a left side exactly when $\hat{s}_1 = 2^{r+1}+1$. By Lemma 6.3 the path is light if and only if r is odd. For $r = 2d-1$ we have $2^{r+1} = 4^d$, $2^{r+3} = 4^{d+1}$ and $\hat{s}_i \equiv \xi_i - a_d \pmod{4^{d+1}}$, and the two conditions say that ξ_1 is an endpoint of a frame of depth d and that ξ_2 lies in a frame of depth d. Conversely, if ξ_1 is an endpoint of a frame of depth d, then $\xi_1 - 2\varpi_0 \equiv 4^d + 1$ or $3\cdot 4^d \pmod{4^{d+1}}$. Hence $x_1 - \varpi_0 \equiv 2^{2d-1}$ or $3\cdot 2^{2d-1} \pmod{2^{2d+1}}$, the column has rank $2d-1$, and the first part applies. Rows are the same, with the second half of Lemma 6.4. □

Flows and roles. Each edge of a tile of R_M carries a *flow*: a subset of $\{\text{left}, \text{right}\}$ on the west and east edges, and a subset of $\{\text{down}, \text{up}\}$ on the south and north edges.

Definition 6.10 (Routed scaffold)**.** The *routed scaffold* $\mathbb{A}^{\mathrm{r}}$ is the Wang set of tuples

$$(\varsigma, \mathfrak{a}, \phi_{\mathrm{W}}, \phi_{\mathrm{E}}, \phi_{\mathrm{S}}, \phi_{\mathrm{N}})$$

with $(\varsigma, \mathfrak{a}) \in \mathbb{A}^{\mathrm{q}}$, $\phi_{\mathrm{W}}, \phi_{\mathrm{E}} \subseteq \{\text{left}, \text{right}\}$ and $\phi_{\mathrm{S}}, \phi_{\mathrm{N}} \subseteq \{\text{down}, \text{up}\}$ that satisfy the *flow rules*:

(F1) if $(\varsigma, \mathfrak{a})$ lies on no light vertical wall, then $\phi_{\mathrm{W}} = \phi_{\mathrm{E}}$;

(F2) on a left side of a light vertical wall, $\phi_{\mathrm{W}} \neq \emptyset$ and $\phi_{\mathrm{E}} \subseteq \{\text{left}\}$;

(F3) on a right side of a light vertical wall, $\phi_{\mathrm{E}} \neq \emptyset$ and $\phi_{\mathrm{W}} \subseteq \{\text{right}\}$;

(F4) if $(\varsigma, \mathfrak{a})$ lies on no light horizontal wall, then $\phi_{\mathrm{S}} = \phi_{\mathrm{N}}$;

(F5) on a bottom side of a light horizontal wall, $\phi_{\mathrm{S}} \neq \emptyset$ and $\phi_{\mathrm{N}} \subseteq \{\text{down}\}$;

(F6) on a top side of a light horizontal wall, $\phi_{\mathrm{N}} \neq \emptyset$ and $\phi_{\mathrm{S}} \subseteq \{\text{up}\}$.

The colour of a tile on a side $X \in \{\mathrm{N}, \mathrm{E}, \mathrm{S}, \mathrm{W}\}$ is the pair of the colour of $(\varsigma, \mathfrak{a})$ in $\mathbb{A}^{\mathrm{q}}$ on side X and the flow ϕ_X.

A corner quarter on a light wall obeys one horizontal and one vertical rule, and the two concern different edges. A tile is a *horizontal carrier* if $\phi_{\mathrm{W}} = \phi_{\mathrm{E}} = \emptyset$, and a *vertical carrier* if $\phi_{\mathrm{S}} = \phi_{\mathrm{N}} = \emptyset$. Its *role* is *inactive* if it is neither, *horizontal* if it is only a horizontal carrier, *vertical* if it is only a vertical carrier, and *active* if it is both, except that an active tile whose scaffold pair is $((1,1), \mathfrak{a})$, with $\mathfrak{a}$ an index-0 letter with dark ruler bits, has the *seed* role. The role is a function of the tile.

Proposition 6.11 (Walls force carriers)**.** *Let P be a legal pattern of $\mathbb{A}^{\mathrm{r}}$ whose domain contains the sites (x, z) with $w_1 \leq x \leq w_2$, where $w_1 < w_2$. Suppose that the scaffold pairs of $P(w_1, z)$ and $P(w_2, z)$ lie on light vertical walls and that those of $P(x, z)$, $w_1 < x < w_2$, do not. Then:*

(a) *the flows* $\phi_{\mathrm{E}}(P(w_1,z))$, $\phi_{\mathrm{W}}(P(x,z))$ *and* $\phi_{\mathrm{E}}(P(x,z))$ *for* $w_1 < x < w_2$, *and* $\phi_{\mathrm{W}}(P(w_2,z))$ *are all equal to one set* ψ;

(b) $\psi = \emptyset$ *if and only if* (w_1,z) *carries a left side and* (w_2,z) *a right side;*

(c) *for* $w_1 < x < w_2$, *the tile* $P(x,z)$ *is a horizontal carrier if and only if* (w_1,z) *carries a left side and* (w_2,z) *a right side. No tile on a light vertical wall is a horizontal carrier.*

The same holds for a vertical segment between two quarters on light horizontal walls, with the flows $\phi_{\mathrm{S}}, \phi_{\mathrm{N}}$, *bottom and top sides, and vertical carriers. It also holds for legal patterns of* R_M *(Definition 6.17), whose tiles carry scaffold pairs and flows with the same rules.*

Proof. (a) Matching edges carry equal flows, and (F1) makes the west and east flows of each tile strictly between the walls equal. (b) The set ψ is the east flow of the tile at w_1 and the west flow of the tile at w_2. If w_1 carries a left side then $\psi \subseteq \{\text{left}\}$ by (F2); if a right side, $\psi \neq \emptyset$ by (F3). If w_2 carries a right side then $\psi \subseteq \{\text{right}\}$ by (F3); if a left side, $\psi \neq \emptyset$ by (F2). So $\psi = \emptyset$ when the walls are a left side and a right side in this order, since $\{\text{left}\} \cap \{\text{right}\} = \emptyset$. In the three other cases one of the walls forces $\psi \neq \emptyset$. (c) The first claim is (a) and (b). A tile on a left side has $\phi_{\mathrm{W}} \neq \emptyset$, and a tile on a right side has $\phi_{\mathrm{E}} \neq \emptyset$. The vertical case is the same with (F4)–(F6). □

Only the emptiness of a flow is forced; a nonempty flow may take several values. We use an elementary fact about nested intervals.

Lemma 6.12 (Laminar walls)**.** *Let* $\mathcal{J}$ *be a set of integer intervals with at least two points each, such that any two distinct members are disjoint or one lies in the interior of the other. Let* E *be the set of their endpoints. Then every* $e \in E$ *is an endpoint of exactly one member, so it is a left or a right endpoint. If* $w_1 < w_2$ *are consecutive elements of* E, w_1 *is a left endpoint and* w_2 *is a right endpoint, then* $[w_1, w_2] \in \mathcal{J}$. *Conversely, if a member* I *contains no other member, its endpoints are consecutive in* E.

Proof. If e were an endpoint of two members, they would meet, so one would lie in the interior of the other, and that one would not have e as an endpoint. Now let $w_1 = \min I$ and $w_2 = \max I'$ be consecutive, with $I, I' \in \mathcal{J}$, and suppose $I \neq I'$. If $I \cap I' = \emptyset$, then $\max I < \min I'$ because $\min I < \max I'$, so $\max I$ lies strictly between w_1 and w_2. If I lies in the interior of I', then $\max I$ lies strictly between w_1 and w_2. If I' lies in the interior of I, then $\min I'$ does. Each case contradicts consecutiveness, so $I = I' = [w_1, w_2]$. Conversely, let I contain no other member, and let $I' \neq I$. If $I' \cap I = \emptyset$, no endpoint of I' lies in I. Otherwise I lies in the interior of I', since the reverse is excluded, and then both endpoints of I' lie outside I. So no element of E lies strictly between the endpoints of I. □

6.3 Frames and free coordinates

Fix frames as in Definition 6.8. For $d \geq 1$ and $m \in \mathbb{Z}$, the *free coordinates* of $I_d(m)$ are the elements of

$$\mathcal{F}_d(m) = \big\{ z \in \mathring{I}_d(m) : \; z \notin I_e(m') \text{ for all } 1 \leq e < d \text{ and } m' \in \mathbb{Z} \big\},$$

and its *centre* is $\mathrm{cen}_d(m) = a_d + (4m+2)4^d + 1$. Lower frames are excluded together with their endpoints.

Lemma 6.13 (Frames)**.** *(a) Frames of the same depth are disjoint. If* $e < d$, *no endpoint of a frame of depth* d *lies in a frame of depth* e. *Hence two frames of different depths are disjoint, or the one of smaller depth lies in the interior of the other.*

(b) *If $e < d$, every frame of depth d contains a frame of depth e.*

(c) *$|\mathcal{F}_d(m)| = 2^{d+2} - 2$. The centre $\mathrm{cen}_d(m)$ lies in $\mathcal{F}_d(m)$, and $\mathcal{F}_d(m)$ has exactly $2^{d+1} - 1$ elements smaller than $\mathrm{cen}_d(m)$ and $2^{d+1} - 2$ elements larger.*

(d) *Suppose $a_d \equiv 2/3 \pmod{4^{d+1}}$ in $\mathbb{Z}_2$ for every d. Then every integer lies in the interior of some frame.*

Proof. (a) The frame $I_d(m)$ lies in $[a_d + 4m \cdot 4^d,\ a_d + 4(m+1)4^d)$, and these windows are disjoint. Let $e < d$. Then $a_d \equiv a_e \pmod{4^{e+1}}$ and 4^{e+1} divides 4^d, so the endpoints of a frame of depth d are congruent to $a_e + 1$ and a_e modulo 4^{e+1}. An integer z lies in a frame of depth e exactly when $(z - a_e) \bmod 4^{e+1} \in [4^e + 1, 3 \cdot 4^e]$, which excludes 0 and 1. If a frame I of depth e meets a frame I' of depth d and contains neither endpoint of I', then I lies in the interior of I'.

(b) Let $I_d(m) = [L, L + 2 \cdot 4^d - 1]$. Left endpoints of frames of depth e form one residue class modulo 4^{e+1}, so one of them, p, lies in $[L, L + 4^{e+1} - 1]$. Its frame $[p, p + 2 \cdot 4^e - 1]$ ends before $L + 4^{e+1} + 2 \cdot 4^e - 1 \le L + 2 \cdot 4^d - 1$, because $4^{e+1} + 2 \cdot 4^e \le 2 \cdot 4^{e+1} \le 2 \cdot 4^d$.

(c) Call a residue $s \in [0, 4^d)$ *allowed* if $s \bmod 4^{e+1} \notin [4^e + 1, 3 \cdot 4^e]$ for every $1 \le e < d$, and let N_d be the number of allowed residues. Since $a_e \equiv a_d \pmod{4^{e+1}}$, an integer z lies in no frame of depth below d exactly when $(z - a_d) \bmod 4^d$ is allowed. Clearly $N_1 = 4$. Write $s \in [0, 4^{d+1})$ as $s = t \cdot 4^d + s'$ with $t \in [0, 4)$ and $s' \in [0, 4^d)$; the conditions for $e < d$ concern only s', and the condition for $e = d$ is $s \notin [4^d + 1, 3 \cdot 4^d]$. For $t = 0$ all N_d allowed s' remain; for $t = 1$ only $s' = 0$; for $t = 2$ none; for $t = 3$ all allowed s' except $s' = 0$. The residue 0 is allowed, so $N_{d+1} = N_d + 1 + 0 + (N_d - 1) = 2N_d$, and $N_d = 2^{d+1}$.

Now write $z = a_d + 4m \cdot 4^d + t \cdot 4^d + s'$. Then $z \in \mathring{I}_d(m)$ exactly when $t = 1$ and $s' \ge 2$, or $t = 2$. The residues $s' = 0$ and $s' = 1$ are both allowed, since $1 < 4^e + 1$. So $\mathcal{F}_d(m)$ has $N_d - 2$ elements with $t = 1$ and N_d elements with $t = 2$, in total $2^{d+2} - 2$. The centre has $t = 2$ and $s' = 1$, so it is free. The free coordinates below it are the $N_d - 2$ with $t = 1$ and the one with $t = 2$, $s' = 0$; those above it are the $N_d - 2$ with $t = 2$, $s' \ge 2$.

(d) In $\mathbb{Z}_2$ we have $-1/3 = \sum_{l \ge 0} 4^l$, so $-2/3 = \sum_{l \ge 0} 2 \cdot 4^l$ has every base-4 digit equal to 2. Adding a nonnegative integer z changes only finitely many digits, because a carry into a digit 2 stops there; subtracting a positive integer changes only finitely many digits, because a borrow from a digit 2 stops there. So $z - 2/3$ has base-4 digit 2 in every position $d \ge d_0$. For such $d \ge 1$, $(z - a_d) \bmod 4^{d+1} \in [2 \cdot 4^d, 3 \cdot 4^d - 1]$, so z lies in the interior of a frame of depth d. □

When every integer lies in a frame, the *owner depth* $\mathrm{own}(z)$ of an integer z is the least $d \ge 1$ such that z lies in a frame of depth d. By definition, $z \in \mathcal{F}_d(m)$ if and only if $z \in \mathring{I}_d(m)$ and $\mathrm{own}(z) = d$. The centres are the only free coordinates with residue $a_d + 2 \cdot 4^d + 1$ modulo 4^{d+1}.

6.4 A routing model on the plane

We now take the phase $(1/3, 1/3)$. Both coordinates are 2-adic integers that are not integers, so every column and row has a finite rank, and $2\varpi_0 = 2/3$.

Lemma 6.14 (Routing on the plane)**.** *Let (f_m) be a tower of height ∞ with phase $(1/3, 1/3)$ (Lemma 6.2). Take the frames with $a_d \equiv 2/3 \pmod{4^{d+1}}$, and let $\mathcal{Y} = f_0^{\mathrm{q}}$, a tiling of $\mathbb{A}^{\mathrm{q}}$.*

(a) There are flows on the edges of $\mathbb{Z}^2$ such that $\mathcal{Y}$ together with these flows is a tiling ϑ of $\mathbb{A}^{\mathrm{r}}$.

(b) *Let ϑ be any tiling of $\mathbb{A}^{\mathrm{r}}$ with scaffold pairs $\mathcal{Y}$. Then $\vartheta(\xi)$ is a horizontal carrier if and only if $\xi_1 \in \mathring{I}_d(m)$ for $d = \mathrm{own}(\xi_2)$ and some m, and it is a vertical carrier if and only if $\xi_2 \in \mathring{I}_{d'}(m')$ for $d' = \mathrm{own}(\xi_1)$ and some m'.*

(c) *In such a tiling, $\vartheta(\xi)$ is active or seed if and only if $\xi \in \mathcal{F}_d(m) \times \mathcal{F}_d(m')$ for some $d \geq 1$ and $m, m' \in \mathbb{Z}$. Every seed tile sits at a pair of centres $(\mathrm{cen}_d(m), \mathrm{cen}_d(m'))$.*

Proof. By Lemma 6.13(d) the owner depth is defined everywhere. Fix a row ξ_2, let D be the set of depths d such that ξ_2 lies in a frame of depth d, and let $d_0 = \mathrm{own}(\xi_2) = \min D$. Let $\mathcal{J}$ be the set of all frames of the depths in D. By Lemma 6.9, applied with $h = \infty$, the quarters of the row ξ_2 that lie on light vertical walls are exactly the endpoints of the members of $\mathcal{J}$, left endpoints carrying left sides and right endpoints right sides. By Lemma 6.13(a), $\mathcal{J}$ satisfies the hypothesis of Lemma 6.12. Its members that contain no other member are exactly the frames of depth d_0: a frame of a larger depth in $\mathcal{J}$ contains a frame of depth d_0 by Lemma 6.13(b), and a member inside a frame of depth d_0 would have a smaller depth by Lemma 6.13(a), which D does not contain. The frames of depth d_0 are unbounded in both directions, so every horizontal edge of the row lies between two consecutive walls.

(a) Each horizontal edge of the row lies between a nearest wall to its west and a nearest wall to its east. Give it the flow $\emptyset$ if these are a left side and a right side, $\{\mathrm{left}\}$ if both are left sides, and $\{\mathrm{right}\}$ if the western one is a right side. A tile on no light vertical wall has both edges between the same two walls, so (F1) holds. At a left side, the edge to the west ends at a left side, so its flow is $\{\mathrm{left}\}$ or $\{\mathrm{right}\}$, and the edge to the east starts at a left side, so its flow is $\emptyset$ or $\{\mathrm{left}\}$; this is (F2). At a right side, the edge to the east starts at a right side and has flow $\{\mathrm{right}\}$, and the edge to the west has flow $\emptyset$ or $\{\mathrm{right}\}$; this is (F3). Columns are treated in the same way with (F4)–(F6). Matching edges carry one flow, and $\mathcal{Y}$ is legal by Lemma 6.6(b), so ϑ is a tiling.

(b) If ξ_1 is a wall of the row, $\vartheta(\xi)$ is not a horizontal carrier by Proposition 6.11(c), and ξ_1 lies in the interior of no frame of depth d_0, because the endpoints of such a frame are consecutive walls (Lemma 6.12). Otherwise let $w_1 < \xi_1 < w_2$ be the nearest walls. By Proposition 6.11(c), $\vartheta(\xi)$ is a horizontal carrier if and only if w_1 is a left side and w_2 a right side. By Lemma 6.12 this holds if and only if $[w_1, w_2]$ is a member of $\mathcal{J}$ whose endpoints are consecutive, that is a member containing no other member, that is a frame of depth d_0. The vertical claim is the same along the column ξ_1.

(c) If $\vartheta(\xi)$ is both a horizontal and a vertical carrier, then ξ_1 lies in a frame of depth $\mathrm{own}(\xi_2)$, so $\mathrm{own}(\xi_1) \leq \mathrm{own}(\xi_2)$, and symmetrically $\mathrm{own}(\xi_2) \leq \mathrm{own}(\xi_1)$. With d the common value, $\xi_1 \in \mathring{I}_d(m)$ and $\mathrm{own}(\xi_1) = d$, that is $\xi_1 \in \mathcal{F}_d(m)$; likewise $\xi_2 \in \mathcal{F}_d(m')$. The converse follows from (b).

Let $\vartheta(\xi)$ be a seed tile, with $\xi = 2x + (1,1)$ and $f_0(x)$ an index-0 letter with dark bits. Its thick type $\angle\mathsf{b}$ has red north endpoint r_0. Let r be the rank of the column x_1. By Lemma 6.4(b),(c), $s_1(x_1) = 2^r$ and $s_2(x_2) = 2^r$. By Lemma 6.3, r is even. If $r = 0$, then $\varsigma_{0,1}(x) = 1$ and $f_0(x)$ is a child of thin type c or b, not a. So $r = 2d$ with $d \geq 1$, and $x_i - 1/3 \equiv 4^d \pmod{2^{2d+2}}$ for $i = 1, 2$. Hence $\xi_i - 2/3 \equiv 2 \cdot 4^d + 1 \pmod{4^{d+1}}$, so ξ_i is a centre of a frame of depth d. □

Distinct seed tiles therefore never share a product $\mathcal{F}_d(m) \times \mathcal{F}_d(m')$: each product contains at most one pair of centres.

6.5 The initialized board in every word

The refutation of Section 7 cannot use a plane tiling. It reads one finite word $\sigma^n(\mathfrak{a})$, with $\mathfrak{a}$ unknown, and needs a complete initialized board inside it. Every such word has one, at the same place.

Proposition 6.15 (The initialized board)**.** *Let $n \geq 2$, $\mathfrak{a} \in \mathbb{A}$ and $W = \sigma^n(\mathfrak{a})$, a pattern on $[0, 4^n)^2$. Put $\Delta_n = 4^{n-1}$, $J = [\Delta_n + 1, 3\Delta_n]$ and $J^\circ = [\Delta_n + 2, 3\Delta_n - 1]$. Let $\mathcal{F}$ be the set of $z \in J^\circ$ that lie in no interval $[(4m+1)4^e + 1, (4m+3)4^e]$ with $1 \leq e \leq n-2$ and $m \in \mathbb{Z}$; these are the* free coordinates*. Then:*

(a) $2\Delta_n + 1 \in \mathcal{F}$ and $|\mathcal{F}| = 2^{n+1} - 2$; exactly $2^n - 1$ elements of $\mathcal{F}$ are smaller than $2\Delta_n + 1$ and $2^n - 2$ are larger. In particular at least $2^n - 1$ elements of $\mathcal{F}$ are $\geq 2\Delta_n + 1$.

(b) The quarter of W at $(2\Delta_n + 1, 2\Delta_n + 1)$ is $((1,1), \mathfrak{b})$, where $\mathfrak{b}$ is an index-0 letter with dark ruler bits.

(c) Let $Z_W = (\mathcal{F} \times J^\circ) \cup (J^\circ \times \mathcal{F})$ and $\xi \in Z_W$. The quarter of W at ξ lies on no light wall. In the row of ξ, the nearest quarters of W to the left and to the right of ξ that lie on a light vertical wall both lie in $J \times \{\xi_2\}$, and they are a left side and a right side, in this order, if and only if $\xi_2 \in \mathcal{F}$. In the column of ξ, the nearest quarters below and above ξ that lie on a light horizontal wall both lie in $\{\xi_1\} \times J$, and they are a bottom side and a top side, in this order, if and only if $\xi_1 \in \mathcal{F}$.

(d) Let P be a legal pattern of $\mathbb{A}^{\mathrm{r}}$ or of R_M on $J \times J$ whose scaffold pairs are the quarters of W. For $\xi \in Z_W$, $P(\xi)$ is a horizontal carrier if and only if $\xi_2 \in \mathcal{F}$, and a vertical carrier if and only if $\xi_1 \in \mathcal{F}$. So the role of $P(\xi)$ is seed at $(2\Delta_n + 1, 2\Delta_n + 1)$, active at the other sites of $\mathcal{F} \times \mathcal{F}$, horizontal if $\xi_2 \in \mathcal{F}$ and $\xi_1 \notin \mathcal{F}$, and vertical if $\xi_1 \in \mathcal{F}$ and $\xi_2 \notin \mathcal{F}$.

None of this depends on $\mathfrak{a}$.

Proof. Use the tower $f_m = (\varphi')^{2n-m}(\mathfrak{a})$ of height $h = 2n$, with all digits 0 and phase 0, so that $f_0 = W$ and $U_0 = [0, 4^n)^2$. Take the frames with $a_d = 0$. Then $J = I_{n-1}(0)$, $\mathcal{F} = \mathcal{F}_{n-1}(0)$ and $2\Delta_n + 1 = \mathrm{cen}_{n-1}(0)$, so (a) is Lemma 6.13(c) with $d = n - 1$.

Walls in the board. Let $\xi \in J \times J$. Its index coordinates $\lfloor \xi_i / 2 \rfloor$ lie in $[2^{2n-3}, 3 \cdot 2^{2n-3}]$, so they are positive and below 2^{2n-1}. With phase 0, (12) shows that the rank of an index coordinate $x \in (0, 4^n)$ is the 2-adic valuation of x; here it is at most $2n - 2 = h - 2$. So Lemma 6.9 applies to ξ in both directions. A light wall at ξ comes from a frame of a depth e with $2e - 1 \leq 2n - 2$, that is $e \leq n - 1$; so every light wall in the board has rank at most $2n - 3$, and it is determined by the word. A frame of depth $n - 1$ that meets J is J itself, and a frame of smaller depth that meets J lies in $\mathring{J}$, by Lemma 6.13(a). Hence, for $\xi_2 \in J$, the quarters (z, ξ_2) with $z \in J$ that lie on light vertical walls are exactly the endpoints of the members of

$$\mathcal{J}(\xi_2) = \{J\} \cup \big\{ I_e(m) \subseteq J : \ e < n - 1, \ \xi_2 \text{ lies in a frame of depth } e \big\},$$

with left endpoints carrying left sides and right endpoints right sides. The same holds for columns. This family satisfies the hypothesis of Lemma 6.12.

(b) The letter $W(x)$ at $x = (\Delta_n, \Delta_n)$ gives the quarter at $2x + (1,1)$. The column $x_1 = 2^{2n-2}$ has rank $r = 2n - 2$, and $s_1(x_1) = s_2(x_2) = 2^r$ in Lemma 6.4. By Lemma 6.4(b),(c) the thick type of $W(x)$ is the corner with red north endpoint r_0, which is $\angle\mathsf{b}$. Since $\varsigma_0(x) = (0,0)$, the letter is a child $(0,0)$ and has thin type a. Every corner of $\mathbb{O}$ has black type A (Table 2). By Lemma 6.3 both bits equal $r \bmod 2 = 0$.

(c) Let $\xi \in Z_W$ and suppose first $\xi_1 \in \mathcal{F}$. A light vertical wall at ξ would make ξ_1 an endpoint of a frame of depth $e \leq n - 1$ meeting J. For $e < n - 1$ this puts ξ_1 in a lower

frame, which $\mathcal{F}$ excludes; for $e = n-1$ the frame is J, whose endpoints are not in J°. A light horizontal wall at ξ would need ξ_1 in a frame of depth e and ξ_2 an endpoint of a frame of the same depth. For $e < n-1$ the first is excluded; for $e = n-1$ the second frame is J and $\xi_2 \in J^\circ$ is not its endpoint. The case $\xi_2 \in \mathcal{F}$ is symmetric. So no light wall lies at ξ.

The endpoints $\Delta_n + 1$ and $3\Delta_n$ of J carry a left and a right side in every row $\xi_2 \in J$, and $\Delta_n + 1 < \xi_1 < 3\Delta_n$. So the nearest walls $w_1 < \xi_1 < w_2$ in the row exist and lie in J. If $\xi_2 \in \mathcal{F}$, no frame of depth below $n-1$ contains ξ_2, so $\mathcal{J}(\xi_2) = \{J\}$, and the nearest walls are $\Delta_n + 1$ and $3\Delta_n$: a left side and a right side. If $\xi_2 \notin \mathcal{F}$, then $\xi_1 \in \mathcal{F}$, and ξ_2 lies in a frame of some depth $e < n-1$. Suppose w_1 were a left side and w_2 a right side. By Lemma 6.12, $[w_1, w_2]$ is a member of $\mathcal{J}(\xi_2)$. It is not J, because J contains a frame of depth e (Lemma 6.13(b)), which is a member, and whose endpoints lie strictly between those of J. So $[w_1, w_2]$ is a frame of depth below $n-1$ containing ξ_1, which contradicts $\xi_1 \in \mathcal{F}$. The column statement is the same with rows and columns exchanged.

(d) By (c), the segment from ξ to its nearest walls in its row lies in $J \times \{\xi_2\}$, and Proposition 6.11(c) gives the horizontal claim; the vertical one is the same. Hence $P(\xi)$ is active or seed on $\mathcal{F} \times \mathcal{F}$, and its role is as stated off $\mathcal{F} \times \mathcal{F}$. At $(2\Delta_n + 1, 2\Delta_n + 1)$ it is seed by (b). Let $\xi \in \mathcal{F} \times \mathcal{F}$ be a quarter $((1,1), W(x))$ with $W(x)$ an index-0 letter with dark bits. As in the proof of Lemma 6.14(c), the column x_1 has an even rank $r = 2e \geq 2$ and $s_1(x_1) = s_2(x_2) = 2^r$; here $r \leq 2n-2$, so $e \leq n-1$. With phase 0 this gives $\xi_i \equiv 2 \cdot 4^e + 1 \pmod{4^{e+1}}$, so ξ_i is the centre of a frame of depth e. For $e < n-1$, ξ_i would lie in a lower frame; so $e = n-1$, and the only centre of depth $n-1$ in J is $2\Delta_n + 1$. Thus the seed role occurs on $\mathcal{F} \times \mathcal{F}$ only at $(2\Delta_n + 1, 2\Delta_n + 1)$. □

Every role in (d) is certified by two straight corridors from ξ to walls inside $J \times J$, together with the flow rules along them. The corridors may be long, but each step involves only two adjacent quarters. Section 7 walks along them with a bounded number of tokens.

6.6 The pointed Horn layer

Machines. A Turing machine M has finitely many states, among them an initial state and a set of halting states, a finite tape alphabet with a blank symbol $\sqcup$, and a transition function. It starts on a blank two-sided tape and *halts* if it enters a halting state. A *one-sided* machine has tape cells $1, 2, \ldots$, starts with its head on cell 1, moves its head one cell to the left or to the right at every step, and never moves left from cell 1. A standard simulation computes from M a one-sided machine M° with a nonhalting initial state that halts if and only if M halts: M° keeps the two halves of the tape of M on two tracks, marks cell 1 at its first step, and performs a stay of M, or a change of track at cell 1, as a step to the right followed by a step to the left. Let $T(M) \in \{1, 2, \ldots\} \cup \{\infty\}$ be the number of steps M° makes before it enters a halting state; we call $T(M)$ the *halting time* of M.

A cellular automaton. Let Γ, Q and tr be the tape alphabet, the state set and the transition function of M°, with initial state q_0. Let $\mathcal{S}_M = \Gamma \cup (\mathsf{Q} \times \Gamma) \cup \{\mathsf{w}\}$, where w marks the wall at cell 0. Define next: $\mathcal{S}_M^3 \to \mathcal{S}_M$ by

- $\mathrm{next}(a, \mathsf{w}, c) = \mathsf{w}$;
- $\mathrm{next}(a, (\mathsf{q}, \gamma), c) = \gamma'$ if q is not halting and $\mathrm{tr}(\mathsf{q}, \gamma) = (\mathsf{q}', \gamma', \pm 1)$;
- for $\gamma \in \Gamma$: $\mathrm{next}((\mathsf{q}, \gamma_a), \gamma, c) = (\mathsf{q}', \gamma)$ if q is not halting and $\mathrm{tr}(\mathsf{q}, \gamma_a) = (\mathsf{q}', \cdot, +1)$; otherwise $\mathrm{next}(a, \gamma, (\mathsf{q}, \gamma_c)) = (\mathsf{q}', \gamma)$ if q is not halting and $\mathrm{tr}(\mathsf{q}, \gamma_c) = (\mathsf{q}', \cdot, -1)$;

- $\mathrm{next}(a,b,c) = b$ in all other cases.

Let $\mathcal{C}_t\colon \{0,1,2,\dots\} \to \mathcal{S}_M$ be given by $\mathcal{C}_t(0) = \mathsf{w}$ for all t, $\mathcal{C}_0(1) = (\mathsf{q}_0, \sqcup)$, $\mathcal{C}_0(x) = \sqcup$ for $x \geq 2$, and $\mathcal{C}_{t+1}(x) = \mathrm{next}(\mathcal{C}_t(x-1), \mathcal{C}_t(x), \mathcal{C}_t(x+1))$ for $x \geq 1$. For $t \leq T(M)$, $\mathcal{C}_t$ is the configuration of M° at time t: the head cell holds the pair of the state and the scanned symbol, and the other cells hold their symbols. Indeed, in a configuration with one head, the head cell writes, the neighbour in the direction of the move receives the head, and every other cell keeps its symbol; the head never reaches cell 0. If M halts, the head at time $T = T(M)$ is at a cell $x_T \leq T+1$, and $\mathcal{C}_T(x_T) = (\mathsf{q}, \gamma)$ with q halting.

Facts and clauses. Let $\mathcal{H}_M = \{\mathsf{Seed}, \mathsf{Start}, \mathsf{Ray}\} \cup \{\mathsf{P}_b : b \in \mathcal{S}_M\}$. A *fact field* on a set $V \subseteq \mathbb{Z}^2$ is a map $\mathsf{G}\colon V \to 2^{\mathcal{H}_M}$; we write $\mathsf{F}(\xi)$ for $\mathsf{F} \in \mathsf{G}(\xi)$. The clauses, for every $\xi \in \mathbb{Z}^2$, are:

$$
\begin{array}{ll}
\text{(H1) } \mathsf{Seed}(\xi) \to \mathsf{P}_{\mathsf{w}}(\xi), & \text{(H2) } \mathsf{Seed}(\xi) \to \mathsf{Start}(\xi + \mathbf{e}_1), \\
\text{(H3) } \mathsf{Start}(\xi) \to \mathsf{P}_{(\mathsf{q}_0,\sqcup)}(\xi), & \text{(H4) } \mathsf{Start}(\xi) \to \mathsf{Ray}(\xi + \mathbf{e}_1), \\
\text{(H5) } \mathsf{Ray}(\xi) \to \mathsf{P}_{\sqcup}(\xi), & \text{(H6) } \mathsf{Ray}(\xi) \to \mathsf{Ray}(\xi + \mathbf{e}_1), \\
\text{(H7) } \mathsf{P}_{\mathsf{w}}(\xi) \to \mathsf{P}_{\mathsf{w}}(\xi + \mathbf{e}_2), & \\
\text{(H8) } \mathsf{P}_a(\xi - \mathbf{e}_1) \wedge \mathsf{P}_b(\xi) \wedge \mathsf{P}_c(\xi + \mathbf{e}_1) \to \mathsf{P}_{\mathrm{next}(a,b,c)}(\xi + \mathbf{e}_2) \quad (a,b,c \in \mathcal{S}_M), & \\
\text{(H9) } \neg \mathsf{P}_b(\xi) \quad \text{for every } b = (\mathsf{q}, \gamma) \text{ with } \mathsf{q} \text{ halting}. &
\end{array}
$$

Each clause has at most one positive literal, so it is a Horn clause, and the sites it mentions lie in $\xi + \Lambda$, with Λ as in Section 2. A fact field on V *satisfies* a clause instance if the instance mentions only sites of V and holds there. A *model* is a fact field on $\mathbb{Z}^2$ that satisfies every instance.

The Wang presentation. A *legal fact block* is a fact field Θ on Λ that satisfies every clause instance mentioning only sites of Λ. Let $\mathcal{U}_M = \{\mathrm{h}, \mathrm{v}\} \times \Lambda \times \mathcal{H}_M$ and $C_M = 2^{\mathcal{U}_M}$, with $\wedge$ the intersection of sets. The tile of Θ is $\mathrm{tile}(\Theta) = (\Theta_{\mathrm{W}}, \Theta_{\mathrm{E}}, \Theta_{\mathrm{S}}, \Theta_{\mathrm{N}}) \in C_M^4$, where

$$
\begin{aligned}
\Theta_{\mathrm{E}} &= \{(\mathrm{h}, y, \mathsf{F}) : y \in \Lambda,\ y_1 \geq 0,\ \mathsf{F} \in \Theta(y)\}, \\
\Theta_{\mathrm{W}} &= \{(\mathrm{h}, y + \mathbf{e}_1, \mathsf{F}) : y \in \Lambda,\ y_1 \leq 0,\ \mathsf{F} \in \Theta(y)\}, \\
\Theta_{\mathrm{N}} &= \{(\mathrm{v}, y, \mathsf{F}) : y \in \Lambda,\ y_2 \geq 0,\ \mathsf{F} \in \Theta(y)\}, \\
\Theta_{\mathrm{S}} &= \{(\mathrm{v}, y + \mathbf{e}_2, \mathsf{F}) : y \in \Lambda,\ y_2 \leq 0,\ \mathsf{F} \in \Theta(y)\}.
\end{aligned}
$$

T_M is the Wang set of the tiles of all legal fact blocks, with these colours. Thus $\mathrm{tile}(\Theta)_{\mathrm{E}} = \mathrm{tile}(\Theta')_{\mathrm{W}}$ exactly when $\Theta(y) = \Theta'(y - \mathbf{e}_1)$ for every $y \in \Lambda$ with $y_1 \geq 0$: the two blocks agree on their overlap of two columns. The same holds vertically. Since the east and west colours together record all of Θ, the map $\Theta \mapsto \mathrm{tile}(\Theta)$ is injective.

The *least field* G_M on $\mathbb{Z}^2$ is $\mathsf{G}_M(0,0) = \{\mathsf{Seed}, \mathsf{P}_{\mathsf{w}}\}$, $\mathsf{G}_M(1,0) = \{\mathsf{Start}, \mathsf{P}_{(\mathsf{q}_0,\sqcup)}\}$, $\mathsf{G}_M(x,0) = \{\mathsf{Ray}, \mathsf{P}_{\sqcup}\}$ for $x \geq 2$, $\mathsf{G}_M(0,t) = \{\mathsf{P}_{\mathsf{w}}\}$ for $t \geq 1$, $\mathsf{G}_M(x,t) = \{\mathsf{P}_{\mathcal{C}_t(x)}\}$ for $x, t \geq 1$, and $\mathsf{G}_M(\xi) = \emptyset$ at all other sites. The *seed tile* is $s_M = \mathrm{tile}(\mathsf{G}_M|_\Lambda)$.

Proposition 6.16 (The pointed Horn layer)**.** *The map $M \mapsto (T_M, s_M)$ is total computable, C_M is a finite set with the associative, commutative and idempotent operation $\wedge$, and:*

(a) $T_M \subseteq C_M^4$ is closed under coordinatewise $\wedge$, and $s_M \in T_M$. The palettes C_M^4, $\{\mathbf{c} \in C_M^4 : \mathbf{c}_{\mathrm{W}} = \mathbf{c}_{\mathrm{E}}\}$, $\{\mathbf{c} \in C_M^4 : \mathbf{c}_{\mathrm{S}} = \mathbf{c}_{\mathrm{N}}\}$, T_M and $\{s_M\}$ are all closed under $\wedge$.

(b) If M does not halt, T_M has a tiling Υ with $\Upsilon(0,0) = s_M$.

(c) If M halts, with halting time T, then for every $N \geq 2T+2$ there is no legal pattern Ξ of T_M on $[0,N)^2$ with $\Xi(0,0) = s_M$. In particular there is none on $[0, 2T+10)^2$.

Proof. Computability. M°, $\mathcal{S}_M$, next and $\mathcal{H}_M$ are computed from M. Legal fact blocks are found by checking the finitely many clause instances inside Λ for each of the finitely many fact fields on Λ, and s_M needs only $\mathcal{C}_0$ and $\mathcal{C}_1$ on cells $0, 1, 2$.

(a) Let Θ, Θ' be legal blocks and $\Theta \cap \Theta'$ their pointwise intersection. If the antecedents of an instance of (H1)–(H8) hold in $\Theta \cap \Theta'$, they hold in Θ and in Θ', so the consequent holds in both and in $\Theta \cap \Theta'$. A fact forbidden by (H9) is absent from Θ, hence from $\Theta \cap \Theta'$. So $\Theta \cap \Theta'$ is legal, and $\mathrm{tile}(\Theta \cap \Theta') = \mathrm{tile}(\Theta) \wedge \mathrm{tile}(\Theta')$ because every colour is a set of facts read off Θ. Thus T_M is closed under $\wedge$. For the palettes: C_M^4 is closed, $\{s_M\}$ is closed by idempotence, and if $\mathbf{c}_\mathrm{W} = \mathbf{c}_\mathrm{E}$ and $\mathbf{c}'_\mathrm{W} = \mathbf{c}'_\mathrm{E}$ then $(\mathbf{c} \wedge \mathbf{c}')_\mathrm{W} = (\mathbf{c} \wedge \mathbf{c}')_\mathrm{E}$; the vertical palette is the same.

We show that G_M satisfies every instance of (H1)–(H8). The instances of (H1)–(H6) hold by the definition of row 0. (H7) holds because P_w occurs exactly at the sites $(0, t)$, $t \geq 0$. In (H8) the three antecedents hold only if $\xi = (x, t)$ with $x \geq 1$ and $t \geq 0$, since no site with negative coordinates carries a fact; then the antecedents are $\mathsf{P}_{\mathcal{C}_t(x-1)}$, $\mathsf{P}_{\mathcal{C}_t(x)}$, $\mathsf{P}_{\mathcal{C}_t(x+1)}$, and the consequent $\mathsf{P}_{\mathcal{C}_{t+1}(x)}$ holds at $(x, t+1)$. The block $\mathsf{G}_M|_\Lambda$ carries P-facts only for $\mathcal{C}_0(0) = \mathcal{C}_1(0) = \mathsf{w}$, $\mathcal{C}_0(1) = (\mathsf{q}_0, \sqcup)$ and $\mathcal{C}_1(1)$. The state q_0 is not halting, and $\mathcal{C}_1(1) \in \Gamma$ because the head leaves cell 1 at the first step. So (H9) holds on Λ, and $s_M \in T_M$.

(b) If M does not halt, no $\mathcal{C}_t(x)$ contains a halting state, so G_M also satisfies (H9) and is a model. Put $\mathrm{Y}(u) = \mathrm{tile}(\Theta_u)$ with $\Theta_u(y) = \mathsf{G}_M(u + y)$ for $y \in \Lambda$. Each Θ_u is legal, and neighbouring blocks agree on their overlaps, so Y is a tiling with $\mathrm{Y}(0, 0) = s_M$.

(c) Let $\Xi(u) = \mathrm{tile}(\Theta_u)$, $u \in [0, N)^2$, be a legal pattern with $\Xi(0, 0) = s_M$. For a site ξ, the set of $u \in [0, N)^2$ with $\xi - u \in \Lambda$ is a product of two integer intervals; along each step inside it the two blocks agree at ξ, by the overlap property. So $\mathsf{G}(\xi) = \Theta_u(\xi - u)$ is a well defined fact field on $[-1, N]^2$, and G satisfies every instance that mentions only sites of $u + \Lambda$ for some $u \in [0, N)^2$. Every instance used below has this property with $u = \xi$ for its site $\xi \in [0, 2T+1] \times [0, T] \subseteq [0, N)^2$.

Since $s_M = \mathrm{tile}(\mathsf{G}_M|_\Lambda)$, $\mathsf{Seed}(0, 0)$ holds in G. By (H1)–(H6) at the sites $(x, 0)$ with $0 \leq x \leq 2T+1$, G contains $\mathsf{P}_\mathsf{w}(0, 0)$ and $\mathsf{P}_{\mathcal{C}_0(x)}(x, 0)$ for $1 \leq x \leq 2T+1$. By (H7), $\mathsf{P}_\mathsf{w}(0, t)$ holds for $0 \leq t \leq T$. By induction on $t \leq T$, $\mathsf{P}_{\mathcal{C}_t(x)}(x, t)$ holds for $1 \leq x \leq 2T+1-t$: the instance of (H8) at (x, t), with $x \leq 2T - t$, has its antecedents at $(x-1, t)$, (x, t) and $(x+1, t)$, which hold by the induction hypothesis and by $\mathsf{P}_\mathsf{w}(0, t)$. At $t = T$ this covers $1 \leq x \leq T+1$, and in particular x_T. So $\mathsf{P}_{\mathcal{C}_T(x_T)}(x_T, T)$ holds with a halting state, which violates (H9) at (x_T, T). Hence no such Ξ exists. Since $2T + 10 \geq 2T + 2$, the last claim follows. □

A whole-plane model always exists, the empty one; only the seed forces the computation. Relations closed under a semilattice operation such as $\wedge$ form a classical tractable class of constraints, decided by local consistency [14, 8, 2]. Section 7 uses this through the closure in (a).

6.7 The family

Definition 6.17 (The family R_M)**.** For a Turing machine M, R_M is the Wang set of tuples

$$\vartheta = \big(\mathrm{quad}(\vartheta), \mathrm{par}(\vartheta), \phi_\mathrm{W}(\vartheta), \phi_\mathrm{E}(\vartheta), \phi_\mathrm{S}(\vartheta), \phi_\mathrm{N}(\vartheta), \mathrm{pay}(\vartheta)\big)$$

such that the first six entries form a tile of the routed scaffold $\mathbb{A}^\mathrm{r}$ (Definition 6.10), and the payload $\mathrm{pay}(\vartheta) = (\mathbf{c}_\mathrm{W}, \mathbf{c}_\mathrm{E}, \mathbf{c}_\mathrm{S}, \mathbf{c}_\mathrm{N}) \in C_M^4$ lies in the palette of the role of ϑ:

$$\begin{array}{lll} \text{inactive: } C_M^4, & \text{horizontal: } \{\mathbf{c} : \mathbf{c}_\mathrm{W} = \mathbf{c}_\mathrm{E}\}, & \text{vertical: } \{\mathbf{c} : \mathbf{c}_\mathrm{S} = \mathbf{c}_\mathrm{N}\}, \\ \text{active: } T_M, & \text{seed: } \{s_M\}. & \end{array}$$

The colour of ϑ on a side X is the pair of the colour of its $\mathbb{A}^\mathrm{r}$-tile on side X and $\mathbf{c}_X$.

So matching tiles of R_M have matching scaffold pairs (Lemma 6.6), equal flows on the shared edge, and equal payload colours on the shared edge. The scaffold $\mathbb{A}$, the quarters, the walls, the flow rules and the roles are fixed; only C_M, T_M and s_M depend on M. The map $M \mapsto R_M$ is total computable: $\mathbb{A}^{\mathrm{r}}$ is a fixed finite set, and R_M is obtained by listing, for each tile of $\mathbb{A}^{\mathrm{r}}$, the payloads in its palette, using Proposition 6.16.

Proposition 6.18 (Nonhalting members tile)**.** *If M does not halt, then R_M tiles.*

Proof. Let (f_m) be a tower with phase $(1/3, 1/3)$ (Lemma 6.2), take the frames with $a_d \equiv 2/3 \pmod{4^{d+1}}$, and let ϑ^{r} be a tiling of $\mathbb{A}^{\mathrm{r}}$ with scaffold pairs f_0^{q}, as in Lemma 6.14(a). By Proposition 6.16(b) there is a tiling Υ of T_M with $\Upsilon(0,0) = s_M$.

Addresses. For $d \geq 1$ and $m \in \mathbb{Z}$ let $\mathrm{ad}_{d,m} \colon \mathcal{F}_d(m) \to \mathbb{Z}$ be the increasing map onto an interval of integers with $\mathrm{ad}_{d,m}(\mathrm{cen}_d(m)) = 0$. For $z \in \mathring{I}_d(m)$ let $\mathrm{ad}^-_{d,m}(z) = \mathrm{ad}_{d,m}(z')$, where z' is the largest element of $\mathcal{F}_d(m)$ with $z' \leq z$, and $\mathrm{ad}^-_{d,m}(z) = \mathrm{ad}_{d,m}(\min \mathcal{F}_d(m)) - 1$ if there is none. Then $\mathrm{ad}^-_{d,m} = \mathrm{ad}_{d,m}$ on $\mathcal{F}_d(m)$; and for $z, z+1 \in \mathring{I}_d(m)$, $\mathrm{ad}^-_{d,m}(z+1) = \mathrm{ad}^-_{d,m}(z)$ if $z+1 \notin \mathcal{F}_d(m)$, while $\mathrm{ad}^-_{d,m}(z+1) = \mathrm{ad}^-_{d,m}(z) + 1$ if $z+1 \in \mathcal{F}_d(m)$.

Payloads. We give each site ξ a payload $\mathrm{pay}(\xi)$ according to the role of $\vartheta^{\mathrm{r}}(\xi)$, described by Lemma 6.14(b),(c). We drop the indices of ad and ad^-, which are determined by the frames containing the coordinates.

- Active or seed, $\xi \in \mathcal{F}_d(m) \times \mathcal{F}_d(m')$: $\mathrm{pay}(\xi) = \Upsilon(\mathrm{ad}(\xi_1), \mathrm{ad}(\xi_2))$.
- Horizontal, with $d = \mathrm{own}(\xi_2)$ and $\xi_1 \in \mathring{I}_d(m)$: if $\xi_2 \in \mathcal{F}_d(m')$ for some m', put $\mathbf{c}_{\mathrm{W}} = \mathbf{c}_{\mathrm{E}} = \Upsilon(\mathrm{ad}^-(\xi_1), \mathrm{ad}(\xi_2))_{\mathrm{E}}$; otherwise put $\mathbf{c}_{\mathrm{W}} = \mathbf{c}_{\mathrm{E}} = \emptyset$.
- Vertical, with $d = \mathrm{own}(\xi_1)$ and $\xi_2 \in \mathring{I}_d(m')$: if $\xi_1 \in \mathcal{F}_d(m)$ for some m, put $\mathbf{c}_{\mathrm{S}} = \mathbf{c}_{\mathrm{N}} = \Upsilon(\mathrm{ad}(\xi_1), \mathrm{ad}^-(\xi_2))_{\mathrm{N}}$; otherwise $\mathbf{c}_{\mathrm{S}} = \mathbf{c}_{\mathrm{N}} = \emptyset$.

These sides are *fixed*: the west and east sides of horizontal carriers, and the south and north sides of vertical carriers. Every other side is *free*. Give a free side the colour of the facing side across its edge if that side is fixed, and $\emptyset$ otherwise. Each free side belongs to one tile and one edge, so this is well defined.

Palettes. Active payloads lie in T_M. A seed tile sits at a pair of centres by Lemma 6.14(c), where $\mathrm{ad} = 0$, so its payload is $\Upsilon(0,0) = s_M$. Horizontal and vertical payloads satisfy $\mathbf{c}_{\mathrm{W}} = \mathbf{c}_{\mathrm{E}}$, respectively $\mathbf{c}_{\mathrm{S}} = \mathbf{c}_{\mathrm{N}}$, and inactive payloads are unconstrained. So $\vartheta(\xi) = (\vartheta^{\mathrm{r}}(\xi), \mathrm{pay}(\xi))$ is a tile of R_M.

Matching. The $\mathbb{A}^{\mathrm{r}}$-parts match because ϑ^{r} is a tiling. Consider a horizontal edge from ξ to $\xi + \mathbf{e}_1$. If at most one of its two sides is fixed, the free side copies the other or both are $\emptyset$. Otherwise both tiles are horizontal carriers. Let $d = \mathrm{own}(\xi_2)$. By Lemma 6.14(b), ξ_1 and $\xi_1 + 1$ lie in interiors of frames of depth d; these interiors are separated by endpoints, so both lie in one $\mathring{I}_d(m)$. If ξ_2 lies in no $\mathcal{F}_d(m')$, it is an endpoint of a frame of depth d, the row carries no active tile by Lemma 6.14(c), and both sides are $\emptyset$. If $\xi_2 \in \mathcal{F}_d(m')$, put $t = \mathrm{ad}(\xi_2)$. A tile on this row inside $\mathring{I}_d(m)$ is active or seed at the free coordinates and horizontal elsewhere. In both cases the east side of ξ is $\Upsilon(\mathrm{ad}^-(\xi_1), t)_{\mathrm{E}}$. If $\xi_1 + 1$ is not free, the west side of $\xi + \mathbf{e}_1$ is $\Upsilon(\mathrm{ad}^-(\xi_1 + 1), t)_{\mathrm{E}} = \Upsilon(\mathrm{ad}^-(\xi_1), t)_{\mathrm{E}}$. If $\xi_1 + 1$ is free, it is $\Upsilon(\mathrm{ad}^-(\xi_1) + 1, t)_{\mathrm{W}}$, which equals $\Upsilon(\mathrm{ad}^-(\xi_1), t)_{\mathrm{E}}$ because Υ is a tiling. Vertical edges are the same with rows and columns exchanged. So ϑ is a tiling of R_M. □

Every product $\mathcal{F}_d(m) \times \mathcal{F}_d(m')$ carries its own copy of Υ, centred at its centre pair. Products without a seed tile receive a copy as well, and nothing links different products: the transverse sides of channels are free.

7 Every halting member has a bounded-width refutation

This section proves the second half of Theorem 7.8.

Theorem 7.1. *Let B be the constant of Theorem 5.18, let $\ell = \binom{276}{2} + 1 = 37951$, and put $j = 10(B + \ell + 100)$. If the machine M halts, then R_M is j-token refutable.*

The proof is by contradiction. Fix a machine M that halts, and let T be its halting time as in Proposition 6.16. Suppose that R_M is not j-token refutable. By Theorem 2.25(iii) there is an E_j family H for R_M. From H we build, in order: a family with one common quarter phase (Section 7.2); a family on the index grid with the same j (Section 7.3); a pin map p on at most $\ell + 3$ sites under which one whole substitution word is forced site by site (Section 7.4); the exact roles on one initialized carrier grid (Section 7.5); and, by a meet argument, a payload assignment on that grid that compresses to a legal T_M-pattern which Proposition 6.16 forbids (Sections 7.7 and 7.8). Section 7.9 checks that j sites suffice at every step. The constants B, ℓ and j do not depend on M, on the depth n chosen below, or on T.

7.1 What we use from Sections 2, 5 and 6

Write $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ in $\mathbb{Z}^2$. Tiles ϑ, ϑ' of a Wang set *match horizontally* if $\vartheta_{\mathrm{E}} = \vartheta'_{\mathrm{W}}$, and *match vertically* if $\vartheta_{\mathrm{N}} = \vartheta'_{\mathrm{S}}$. Integer intervals are written $[a, b] = \{a, a + 1, \dots, b\}$ and $[0, m) = \{0, \dots, m - 1\}$.

Families. We use E_K families for a Wang set as in Definition 2.22: nonempty sets of legal patterns on finite subsets of $\mathbb{Z}^2$, with at most K sites each, closed under restriction and with the extension property for maps on fewer than K sites. If $h \in H$ and h' is supplied by (Extension) for a new site w, then $h'|_{\operatorname{dom} h \cup \{w\}}$ also extends h; so we may and do assume that an extension adds exactly the requested site. Repeating this, a map $h \in H$ extends to a map of H on $\operatorname{dom} h \cup U'$ whenever $|\operatorname{dom} h \cup U'| \le K$ (Remark 2.23). By Theorem 2.25(iii), a Wang set that is not j-token refutable has an E_j family.

The scaffold and the parser. Let $\mathbb{A}$ be the scaffold Wang set of 276 letters and $\sigma\colon \mathbb{A} \to \mathbb{A}^{[0,4)^2}$ its substitution, from Proposition 5.11. For $n \ge 0$ and $\mathfrak{a} \in \mathbb{A}$, $\sigma^n(\mathfrak{a})$ is a legal pattern on $[0, 4^n)^2$. Let $\chi_1, \chi_2\colon \mathbb{A} \to \{0, 1\}$ be the column bit and the row bit of the ruler. For $u \in \mathbb{Z}^2$ put $\mathrm{Anc}_n(u) = \{u,\ u + 4^n\mathbf{e}_1,\ u + 4^n\mathbf{e}_2\}$. A map g into $\mathbb{A}$ defined on $\mathrm{Anc}_n(u)$ is a *good n-anchor at u* if $\chi_1(g(u)) \ne \chi_1(g(u + 4^n\mathbf{e}_1))$ and $\chi_2(g(u)) \ne \chi_2(g(u + 4^n\mathbf{e}_2))$. The port set $\mathrm{Port}_n(u) \subseteq u + [0, 4^n)^2$ has at most ℓ sites, and the readout $\mathfrak{a} \mapsto \big(\sigma^n(\mathfrak{a})(y - u)\big)_{y \in \mathrm{Port}_n(u)}$ is injective on $\mathbb{A}$. We use two consequences of Theorem 5.18, Lemma 5.19, and Proposition 5.21.

(P1) Anchors. For every E_K family G for $\mathbb{A}$ with $K \ge B$ and every $n \ge 0$, there are $u \in \mathbb{Z}^2$ and $g \in G$ with domain $\mathrm{Anc}_n(u)$ that is a good n-anchor at u (Proposition 5.21).

(P2) Type and point certificates. Let G be an E_K family for $\mathbb{A}$ with $K \ge B$, let $n \ge 0$ and $u \in \mathbb{Z}^2$, and let $g \in G$ be defined on $\mathrm{Anc}_n(u) \cup \mathrm{Port}_n(u)$, with $g|_{\mathrm{Anc}_n(u)}$ a good n-anchor at u. Then there is $\mathfrak{a} \in \mathbb{A}$ with $g(y) = \sigma^n(\mathfrak{a})(y - u)$ for every $y \in \mathrm{Port}_n(u)$. If g is also defined at a site $s \in u + [0, 4^n)^2$, then $g(s) = \sigma^n(\mathfrak{a})(s - u)$. (The type and point certificates of Theorem 5.18 have rooted tree decompositions with bags of at most B sites and contain these sites in their boundary; Lemma 5.19 carries their conclusions to every E_K family with $K \ge B$.)

The family R_M. From Propositions 6.11, 6.15 and 6.16 and Definition 6.17 we use the following.

(Q1) Quarters. Every tile ϑ of R_M has a *quadrant* $\mathrm{quad}(\vartheta) \in \{0,1\}^2$ and a *parent* $\mathrm{par}(\vartheta) \in \mathbb{A}$; the pair $(\mathrm{quad}(\vartheta), \mathrm{par}(\vartheta))$ is its *scaffold pair*. Let ϑ, ϑ' match horizontally. Then $\mathrm{quad}(\vartheta') \equiv \mathrm{quad}(\vartheta) + \mathbf{e}_1 \pmod 2$. If the first coordinate of $\mathrm{quad}(\vartheta)$ is 0, the two quarters belong to one parent cell and $\mathrm{par}(\vartheta') = \mathrm{par}(\vartheta)$; if it is 1, then $\mathrm{par}(\vartheta)$ and $\mathrm{par}(\vartheta')$ match horizontally in $\mathbb{A}$. The same holds for vertical matching, with $\mathbf{e}_2$ and the second coordinate.

(Q2) Walls and flows. A scaffold pair may lie on a light vertical wall, as a *left side* or a *right side*, and it may lie on a light horizontal wall, as a *bottom side* or a *top side*; these attributes are functions of the scaffold pair. Each tile ϑ carries flows $\phi_{\mathrm{W}}(\vartheta), \phi_{\mathrm{E}}(\vartheta) \subseteq \{\mathrm{left}, \mathrm{right}\}$ and $\phi_{\mathrm{S}}(\vartheta), \phi_{\mathrm{N}}(\vartheta) \subseteq \{\mathrm{down}, \mathrm{up}\}$. If ϑ, ϑ' match horizontally then $\phi_{\mathrm{E}}(\vartheta) = \phi_{\mathrm{W}}(\vartheta')$, and if they match vertically then $\phi_{\mathrm{N}}(\vartheta) = \phi_{\mathrm{S}}(\vartheta')$. If the scaffold pair of ϑ lies on no light vertical wall, then $\phi_{\mathrm{W}}(\vartheta) = \phi_{\mathrm{E}}(\vartheta)$. On a left side, $\phi_{\mathrm{W}}(\vartheta) \neq \emptyset$ and $\phi_{\mathrm{E}}(\vartheta) \subseteq \{\mathrm{left}\}$. On a right side, $\phi_{\mathrm{E}}(\vartheta) \neq \emptyset$ and $\phi_{\mathrm{W}}(\vartheta) \subseteq \{\mathrm{right}\}$. The vertical rules are the same after rotation: with no light horizontal wall $\phi_{\mathrm{S}}(\vartheta) = \phi_{\mathrm{N}}(\vartheta)$; on a bottom side $\phi_{\mathrm{S}}(\vartheta) \neq \emptyset$ and $\phi_{\mathrm{N}}(\vartheta) \subseteq \{\mathrm{down}\}$; on a top side $\phi_{\mathrm{N}}(\vartheta) \neq \emptyset$ and $\phi_{\mathrm{S}}(\vartheta) \subseteq \{\mathrm{up}\}$.

(Q3) Roles and palettes. A tile ϑ is a *horizontal carrier* if $\phi_{\mathrm{W}}(\vartheta) = \phi_{\mathrm{E}}(\vartheta) = \emptyset$, and a *vertical carrier* if $\phi_{\mathrm{S}}(\vartheta) = \phi_{\mathrm{N}}(\vartheta) = \emptyset$. Its *role* is inactive, horizontal, vertical or active according as it is neither, only a horizontal carrier, only a vertical carrier, or both; an active tile whose scaffold pair is the north-east quarter of an index-0 letter with dark ruler bits has the *seed* role instead. Each tile carries a payload $\mathrm{pay}(\vartheta) \in C_M^4$, a tuple of colours indexed by $\mathrm{W}, \mathrm{E}, \mathrm{S}, \mathrm{N}$. Horizontally matching tiles have $\mathrm{pay}(\vartheta)_{\mathrm{E}} = \mathrm{pay}(\vartheta')_{\mathrm{W}}$, and vertically matching tiles have $\mathrm{pay}(\vartheta)_{\mathrm{N}} = \mathrm{pay}(\vartheta')_{\mathrm{S}}$. The payload lies in the palette of the role: all of C_M^4 for inactive, $\{\mathbf{c} : \mathbf{c}_{\mathrm{W}} = \mathbf{c}_{\mathrm{E}}\}$ for horizontal, $\{\mathbf{c} : \mathbf{c}_{\mathrm{S}} = \mathbf{c}_{\mathrm{N}}\}$ for vertical, T_M for active, and $\{s_M\}$ for seed.

(Q4) The Horn layer. C_M is a finite set with a binary operation $\wedge$ (intersection of packets) that is associative, commutative and idempotent. Write $\wedge$ also coordinatewise on C_M^4. The tile set $T_M \subseteq C_M^4$ is closed under $\wedge$, and $s_M \in T_M$. If M halts at time T, there is no legal T_M-pattern Ξ on $[0, 2T+10)^2$ with $\Xi(0,0) = s_M$ (Proposition 6.16).

(Q5) The initialized board. For a pattern W on $[0, 4^n)^2$ and $\xi \in [0, 2 \cdot 4^n)^2$, the *quarter of W at ξ* is the scaffold pair $(\xi \bmod 2, W(\lfloor \xi/2 \rfloor))$, coordinatewise. Let $n \geq 2$, $\mathfrak{a} \in \mathbb{A}$, $W = \sigma^n(\mathfrak{a})$ and $\Delta_n = 4^{n-1}$, and put $J = [\Delta_n + 1, 3\Delta_n]$ and $J^\circ = [\Delta_n + 2, 3\Delta_n - 1]$. By Proposition 6.15 there is a set $\mathcal{F} \subseteq J^\circ$, the *free coordinates*, such that:

(a) $2\Delta_n + 1 \in \mathcal{F}$, and at least $2^n - 1$ elements of $\mathcal{F}$ are $\geq 2\Delta_n + 1$;

(b) the quarter of W at $(2\Delta_n + 1, 2\Delta_n + 1)$ is the north-east quarter of an index-0 letter with dark ruler bits;

(c) for every $\xi = (\xi_1, \xi_2) \in Z_W = (\mathcal{F} \times J^\circ) \cup (J^\circ \times \mathcal{F})$, the quarter of W at ξ lies on no light wall. In the row of ξ, the nearest quarters of W to the left and to the right of ξ that lie on a light vertical wall both lie in $J \times \{\xi_2\}$, and they are a left side and a right side, in this order, if and only if $\xi_2 \in \mathcal{F}$. In the column of ξ, the nearest quarters below and above ξ that lie on a light horizontal wall both lie in $\{\xi_1\} \times J$, and they are a bottom side and a top side, in this order, if and only if $\xi_1 \in \mathcal{F}$.

Item (c) is the statement that every carrier role on the board is forced by a corridor between two walls inside the board. Since $3\Delta_n < 2 \cdot 4^n$, all these corridors lie in $[0, 2 \cdot 4^n)^2$.

7.2 One adjusted quarter phase

Lemma 7.2. *Let H be an E_K family for R_M with $K \geq 3$. For every $h \in H$ and all $v, w \in \operatorname{dom} h$,*

$$\operatorname{quad}(h(v)) - v \equiv \operatorname{quad}(h(w)) - w \pmod 2.$$

Proof. Choose a path $v = y_0, y_1, \ldots, y_m = w$ of distinct sites with $y_{l+1} - y_l \in \{\pm\mathbf{e}_1, \pm\mathbf{e}_2\}$. If $m \leq 1$, the claim is (Q1) applied to the legal pattern h. Let $m \geq 2$. Put $h_0 = h|_{\{y_0, w\}}$. For $0 \leq l \leq m-2$, extend h_l by the site y_{l+1} and restrict the result to $\{y_{l+1}, w\}$; call it h_{l+1}. Each step extends a map on two sites, which is possible since $K \geq 3$. Put $\vartheta_l = h_l(y_l)$, so that $\vartheta_0 = h(v)$, and note $h_l(w) = h(w)$ for every l. For $l \leq m-2$, the tiles ϑ_l and ϑ_{l+1} are values of one legal pattern at the adjacent sites y_l, y_{l+1}, so (Q1) gives $\operatorname{quad}(\vartheta_{l+1}) - \operatorname{quad}(\vartheta_l) \equiv y_{l+1} - y_l$. Finally ϑ_{m-1} and $h(w)$ are values of h_{m-1} at the adjacent sites y_{m-1}, w. Summing the increments gives the claim. □

For $\gamma \in (\mathbb{Z}/2\mathbb{Z})^2$ let H_γ be the set of $h \in H$ with $\operatorname{quad}(h(v)) - v \equiv \gamma$ for every $v \in \operatorname{dom} h$. We call it the *phase-$\gamma$ component* of H.

Lemma 7.3. *Let H be an E_K family for R_M with $K \geq 3$. Some γ has a map with nonempty domain in H_γ, and for each such γ the set H_γ is an E_K family for R_M.*

Proof. Extending the empty map gives $h \in H$ defined at some site v; with $\gamma = \operatorname{quad}(h(v)) - v$, Lemma 7.2 puts h in H_γ. Now fix such a γ. The set H_γ consists of legal patterns with at most K sites, and it is closed under restriction. Let $h \in H_\gamma$ with $|\operatorname{dom} h| < K$ and let $w \in \mathbb{Z}^2$. If $\operatorname{dom} h \neq \emptyset$, extend h in H by w; by Lemma 7.2 the new value has the phase of the old values, so the extension lies in H_γ. If h is empty, take a site v of some map of H_γ, extend its restriction to $\{v\}$ by w in H (possible since $K \geq 2$), and restrict to $\{w\}$; by Lemma 7.2 the result lies in H_γ. □

Apply this to the E_j family H of R_M; here $j \geq 3$. Fix γ as in Lemma 7.3, and let $\bar\gamma \in \{0,1\}^2$ be its representative. Then $\operatorname{quad}(h(v)) \equiv v - \bar\gamma \pmod 2$ for every $h \in H_\gamma$ and $v \in \operatorname{dom} h$. The phase is common to all maps of H_γ; no site is spent to fix it.

7.3 The index family

For $z \in \mathbb{Z}^2$ put $\iota(z) = \bar\gamma + 2z$. The four sites $\iota(z) + \{0,1\}^2$ form the *cell* of z, and in every map of H_γ the value at $\iota(z) + \varsigma$ has quadrant ς. For $h \in H_\gamma$ with $\operatorname{dom} h \subseteq \iota(\mathbb{Z}^2)$, define $\hat h$ on $\iota^{-1}(\operatorname{dom} h)$ by $\hat h(z) = \operatorname{par}(h(\iota(z)))$, and let $\widehat{H}$ be the set of these maps.

Lemma 7.4. *$\widehat{H}$ is an E_j family for $\mathbb{A}$.*

Proof. Restriction commutes with the hat, and extending h in H_γ by a site $\iota(w)$ extends $\hat h$ by w. It remains to show that each $\hat h$ is a legal pattern. Let $z, z + \mathbf{e}_1 \in \operatorname{dom} \hat h$ and $v = \iota(z)$, so that $\iota(z + \mathbf{e}_1) = v + 2\mathbf{e}_1$. Restrict h to the pair $\{v, v + 2\mathbf{e}_1\}$ and extend the restriction by the site $v + \mathbf{e}_1$; call the result h'. The values $h'(v)$ and $h'(v + \mathbf{e}_1)$ match horizontally and have quadrants $(0,0)$ and $(1,0)$, so (Q1) gives $\operatorname{par}(h'(v)) = \operatorname{par}(h'(v + \mathbf{e}_1))$. The values $h'(v + \mathbf{e}_1)$ and $h'(v + 2\mathbf{e}_1)$ match horizontally, and the first has quadrant $(1,0)$, so their parents match horizontally in $\mathbb{A}$. Since h' agrees with h at v and $v + 2\mathbf{e}_1$, the letters $\hat h(z)$ and $\hat h(z + \mathbf{e}_1)$ match horizontally. The vertical case is the same with $\mathbf{e}_2$. □

The check is binary: we restrict to the pair before adding the middle quarter, so we never extend a map that already has j sites.

7.4 Pins and the forced word

Fix $n \geq 2$ with $2^n - 2 \geq 2T + 10$. By (P1), applied to $\widehat{H}$ with $K = j \geq B$, there are $u \in \mathbb{Z}^2$ and $g_0 \in \widehat{H}$ with domain $\mathrm{Anc}_n(u)$ that is a good n-anchor at u. Extend g_0 in $\widehat{H}$ to a map g_1 on $\mathrm{Anc}_n(u) \cup \mathrm{Port}_n(u)$, a set of at most $\ell + 3 \leq j$ sites. By the first part of (P2) there is $\mathfrak{a} \in \mathbb{A}$ with $g_1(y) = \sigma^n(\mathfrak{a})(y - u)$ for $y \in \mathrm{Port}_n(u)$; it is unique because the readout is injective. Put

$$W = \sigma^n(\mathfrak{a}), \qquad o = \iota(u).$$

By the definition of $\widehat{H}$ there is $p \in H_\gamma$ with $\mathrm{dom}\, p = \iota(\mathrm{Anc}_n(u) \cup \mathrm{Port}_n(u))$ and $\hat{p} = g_1$. These are the *pins*. They occupy at most $\ell + 3$ sites, and every $h \in H_\gamma$ with $p \subseteq h$ projects to an index map that keeps the same good anchor and the same readout of W.

Lemma 7.5 (The forced word)**.** *Let $\xi \in [0, 2 \cdot 4^n)^2$, and let $h \in H_\gamma$ satisfy $p \subseteq h$ and $o + \xi \in \mathrm{dom}\, h$. Then the scaffold pair of $h(o + \xi)$ is the quarter of W at ξ.*

Proof. Write $x = o + \xi$, $\varsigma = \xi \bmod 2$ and $s = u + \lfloor \xi/2 \rfloor$, so that $x = \iota(s) + \varsigma$ lies in the cell of s. The quadrant of $h(x)$ is $x - \bar{\gamma} \equiv \xi \equiv \varsigma$. For the parent, let Y be the path from x to $\iota(s)$ inside the cell of s that has no step if $\varsigma = (0,0)$, one step if $\varsigma \in \{(1,0),(0,1)\}$, and the two steps $x \to \iota(s) + \mathbf{e}_1 \to \iota(s)$ if $\varsigma = (1,1)$. Restrict h to $\mathrm{dom}\, p \cup \{x\}$ and extend it by the at most two other sites of Y; call the result h'. It has at most $\ell + 6$ sites. In each step of Y, the western or southern of the two quarters has coordinate 0 in the direction of the step, so (Q1) gives equal parents along Y. Hence $\mathrm{par}(h(x)) = \mathrm{par}(h'(\iota(s)))$. Let $h'' = h'|_{\mathrm{dom}\, p \cup \{\iota(s)\}}$. Then $\hat{h}'' \in \widehat{H}$ is defined on $\mathrm{Anc}_n(u) \cup \mathrm{Port}_n(u) \cup \{s\}$ and agrees with g_1 on $\mathrm{Anc}_n(u) \cup \mathrm{Port}_n(u)$. So (P2) applies to $\hat{h}''$, and by injectivity of the readout the letter it provides is $\mathfrak{a}$. Since $s \in u + [0, 4^n)^2$, the second part of (P2) gives $\hat{h}''(s) = W(s - u) = W(\lfloor \xi/2 \rfloor)$, and $\hat{h}''(s) = \mathrm{par}(h'(\iota(s)))$. □

The lemma is pointwise. Each query uses the pins and at most three further sites, never all sites of W at once; and it holds for every map of H_γ that keeps the pins, not for one favourable map per query.

7.5 The forced role chart

Let Δ_n, J, J°, $\mathcal{F}$ and Z_W be as in (Q5) for the word W.

Lemma 7.6 (Corridors)**.** *Let $\xi \in Z_W$, and let $h \in H_\gamma$ satisfy $p \subseteq h$ and $v = o + \xi \in \mathrm{dom}\, h$. Then $h(v)$ is a horizontal carrier if and only if $\xi_2 \in \mathcal{F}$, and a vertical carrier if and only if $\xi_1 \in \mathcal{F}$. Consequently the role of $h(v)$ is horizontal if $\xi_1 \notin \mathcal{F}$, vertical if $\xi_2 \notin \mathcal{F}$, active or seed if $\xi \in \mathcal{F}^2$, and seed if $\xi = (2\Delta_n + 1, 2\Delta_n + 1)$.*

Proof. By (Q5)(c), let $\xi + m\mathbf{e}_1$ and $\xi - m'\mathbf{e}_1$, with $m, m' \geq 1$, be the nearest quarters of W to the right and to the left of ξ in its row that lie on a light vertical wall. Both lie in $J \times \{\xi_2\}$.

We walk to the right. Put $h_0 = h$ and $v_l = v + l\mathbf{e}_1$. For $l = 1, \ldots, m$, restrict h_{l-1} to $\mathrm{dom}\, p \cup \{v_{l-1}\}$, a set of at most $\ell + 4$ sites, and extend it by v_l; call the result h_l. Then $p \subseteq h_l$. Put $\vartheta_l = h_l(v_l)$, so that $\vartheta_0 = h(v)$. For $1 \leq l \leq m$, the tiles $\vartheta_{l-1} = h_l(v_{l-1})$ and ϑ_l are values of one legal pattern at horizontally adjacent sites, so $\phi_{\mathrm{E}}(\vartheta_{l-1}) = \phi_{\mathrm{W}}(\vartheta_l)$ by (Q2). By Lemma 7.5, the scaffold pair of ϑ_l is the quarter of W at $\xi + l\mathbf{e}_1$. For $0 \leq l < m$ this quarter lies on no light vertical wall: for $l = 0$ by (Q5)(c), and for $0 < l < m$ by the choice of m. So $\phi_{\mathrm{W}}(\vartheta_l) = \phi_{\mathrm{E}}(\vartheta_l)$ for $0 \leq l < m$. Writing $\psi = \phi_{\mathrm{W}}(\vartheta_0) = \phi_{\mathrm{E}}(\vartheta_0)$, we get $\psi = \phi_{\mathrm{W}}(\vartheta_m)$, where ϑ_m sits at the right wall. Walking to the left in the same way gives a tile ϑ' at the left wall, with the scaffold pair of the quarter of W at $\xi - m'\mathbf{e}_1$, and $\psi = \phi_{\mathrm{E}}(\vartheta')$.

If $\xi_2 \in \mathcal{F}$, the left wall is a left side and the right wall a right side. By (Q2), $\psi = \phi_{\mathrm{E}}(\vartheta') \subseteq \{\text{left}\}$ and $\psi = \phi_{\mathrm{W}}(\vartheta_m) \subseteq \{\text{right}\}$, so $\psi = \emptyset$ and $h(v)$ is a horizontal carrier. If $\xi_2 \notin \mathcal{F}$, then the left wall is not a left side or the right wall is not a right side. In the first case it is a right side and $\psi = \phi_{\mathrm{E}}(\vartheta') \neq \emptyset$; in the second it is a left side and $\psi = \phi_{\mathrm{W}}(\vartheta_m) \neq \emptyset$. Either way $h(v)$ is not a horizontal carrier.

The vertical statement is proved the same way along the column of ξ, with $\mathbf{e}_2$, the flows $\phi_{\mathrm{S}}, \phi_{\mathrm{N}}$, and bottom and top sides. The roles follow from (Q3). At $\xi = (2\Delta_n + 1, 2\Delta_n + 1)$ the tile is active, and by Lemma 7.5 and (Q5)(b) its scaffold pair is the north-east quarter of an index-0 letter with dark ruler bits, so its role is seed. □

The roles do not depend on the nonempty flow values that the family may choose: the walk only uses which flows are empty. An extra active crossing between two consecutive free coordinates would break a wire of the compression in Section 7.8; the lemma excludes it for every map that keeps the pins.

7.6 The selected component

Put $N = 2T + 10$. By (Q5)(a) and $2^n - 1 > N$, let $f_0 < f_1 < \cdots < f_{N-1}$ be the N smallest elements of $\mathcal{F}$ that are at least $2\Delta_n + 1$; thus $f_0 = 2\Delta_n + 1$. Let $F_N = \{f_0, \dots, f_{N-1}\}$ and

$$Z = \big\{\xi : \xi_2 \in F_N,\ f_0 \le \xi_1 \le f_{N-1}\big\} \cup \big\{\xi : \xi_1 \in F_N,\ f_0 \le \xi_2 \le f_{N-1}\big\}.$$

Since $f_0, f_{N-1} \in J^\circ$, we have $Z \subseteq Z_W$. An integer strictly between f_i and f_{i+1} is not in $\mathcal{F}$. The *channel edges* of Z are the pairs $\{\xi, \xi + \mathbf{e}_1\}$ with $\xi_2 \in F_N$ and $f_0 \le \xi_1 < f_{N-1}$, and the pairs $\{\xi, \xi + \mathbf{e}_2\}$ with $\xi_1 \in F_N$ and $f_0 \le \xi_2 < f_{N-1}$. Assign to each $\xi \in Z$ the palette

$$\mathrm{Pal}(\xi) = \begin{cases} \{s_M\} & \text{if } \xi = (f_0, f_0), \\ T_M & \text{if } \xi \in F_N^2 \setminus \{(f_0, f_0)\}, \\ \{\mathbf{c} \in C_M^4 : \mathbf{c}_{\mathrm{W}} = \mathbf{c}_{\mathrm{E}}\} & \text{if } \xi_2 \in F_N,\ \xi_1 \notin F_N, \\ \{\mathbf{c} \in C_M^4 : \mathbf{c}_{\mathrm{S}} = \mathbf{c}_{\mathrm{N}}\} & \text{if } \xi_1 \in F_N,\ \xi_2 \notin F_N. \end{cases}$$

By Lemma 7.6 and (Q3), $\mathrm{pay}(h(o + \xi)) \in \mathrm{Pal}(\xi)$ for every $\xi \in Z$ and every $h \in H_\gamma$ with $p \subseteq h$ and $o + \xi \in \mathrm{dom}\, h$. On $F_N^2 \setminus \{(f_0, f_0)\}$ the role is active or seed, and both palettes lie in T_M because $s_M \in T_M$. Lower-depth initialized components may occur elsewhere in the board; they do not meet Z, and they are not used.

7.7 The projected meet

For $\xi \in Z$ and $v = o + \xi$ put

$$U_v = \big\{\mathrm{pay}(h(v)) : h \in H_\gamma,\ p \subseteq h,\ v \in \mathrm{dom}\, h\big\}, \qquad m_v = \bigwedge U_v.$$

For a channel edge $\{\xi, \xi'\}$ with $\xi' = \xi + \mathbf{e}_1$ or $\xi' = \xi + \mathbf{e}_2$, and $v = o + \xi$, $w = o + \xi'$, put

$$Q_{vw} = \big\{(\mathrm{pay}(h(v)), \mathrm{pay}(h(w))) : h \in H_\gamma,\ p \subseteq h,\ v, w \in \mathrm{dom}\, h\big\}.$$

Lemma 7.7 (Projected meet). *(a) Each U_v is a nonempty finite set, and $m_v \in \mathrm{Pal}(\xi)$.*

(b) The two coordinate projections of Q_{vw} are exactly U_v and U_w.

(c) If $\xi' = \xi + \mathbf{e}_1$ then $(m_v)_{\mathrm{E}} = (m_w)_{\mathrm{W}}$; if $\xi' = \xi + \mathbf{e}_2$ then $(m_v)_{\mathrm{N}} = (m_w)_{\mathrm{S}}$.

Proof. (a) Extending p by v gives an element of U_v, and $U_v \subseteq C_M^4$ is finite. Every palette is closed under $\wedge$: for T_M this is (Q4); for $\{s_M\}$ it is idempotence; and if $\mathbf{c}_\mathrm{W} = \mathbf{c}_\mathrm{E}$ and $\mathbf{c}'_\mathrm{W} = \mathbf{c}'_\mathrm{E}$, then $(\mathbf{c} \wedge \mathbf{c}')_\mathrm{W} = \mathbf{c}_\mathrm{W} \wedge \mathbf{c}'_\mathrm{W} = (\mathbf{c} \wedge \mathbf{c}')_\mathrm{E}$, and likewise for the vertical palette. Since $U_v \subseteq \mathrm{Pal}(\xi)$ by Section 7.6, the meet of its finitely many elements lies in $\mathrm{Pal}(\xi)$.

(b) A pair in Q_{vw} comes from a map that keeps p and is defined at v, so its first coordinate lies in U_v. Conversely, let $\mathbf{c} = \mathrm{pay}(h(v)) \in U_v$. Restrict h to $\mathrm{dom}\, p \cup \{v\}$ and extend by w; the result has at most $\ell + 5$ sites, keeps p, and has the value $h(v)$ at v, so it gives a pair of Q_{vw} with first coordinate $\mathbf{c}$. The second coordinate is symmetric.

(c) Let $\xi' = \xi + \mathbf{e}_1$ and list $Q_{vw} = \{(\mathbf{c}^1, \mathbf{d}^1), \dots, (\mathbf{c}^l, \mathbf{d}^l)\}$. Each pair consists of the payloads of one legal pattern at horizontally adjacent sites, so $\mathbf{c}^i_\mathrm{E} = \mathbf{d}^i_\mathrm{W}$ by (Q3). By (b), $\{\mathbf{c}^1, \dots, \mathbf{c}^l\} = U_v$ and $\{\mathbf{d}^1, \dots, \mathbf{d}^l\} = U_w$. As $\wedge$ acts coordinatewise,

$$(m_v)_\mathrm{E} = \textstyle\bigwedge_i \mathbf{c}^i_\mathrm{E} = \bigwedge_i \mathbf{d}^i_\mathrm{W} = (m_w)_\mathrm{W}.$$

The vertical case is the same. □

So the single assignment $v \mapsto m_v$ satisfies every palette and every channel equality at once. Neither H nor the sets U_v need to be closed under meets; only the palettes and the equality constraints are. This is the classical fact that constraints closed under a semilattice operation are solved by local consistency [14], applied to relations projected from H.

7.8 Compression and the contradiction

Proof of Theorem 7.1. With $N = 2T + 10$, define Ξ on $[0, N)^2$ by $\Xi(i, i') = m_{o+(f_i, f_{i'})}$. By Lemma 7.7(a), $\Xi(i, i') \in T_M$ for all i, i', and $\Xi(0, 0) = s_M$.

Fix i' and $i < N - 1$, and let $v_y = o + (y, f_{i'})$ for $f_i \le y \le f_{i+1}$. These sites lie in $o + Z$, and consecutive ones form channel edges. For $f_i < y < f_{i+1}$ the palette at $(y, f_{i'})$ is horizontal, so $(m_{v_y})_\mathrm{W} = (m_{v_y})_\mathrm{E}$. By Lemma 7.7(c), $(m_{v_y})_\mathrm{E} = (m_{v_{y+1}})_\mathrm{W}$ for $f_i \le y < f_{i+1}$. Chaining these equalities gives $\Xi(i, i')_\mathrm{E} = \Xi(i + 1, i')_\mathrm{W}$. In the same way, along the column f_i through vertical palettes, $\Xi(i, i')_\mathrm{N} = \Xi(i, i' + 1)_\mathrm{S}$.

So Ξ is a legal T_M-pattern on $[0, 2T + 10)^2$ with $\Xi(0, 0) = s_M$. Since M halts at time T, (Q4) says that no such pattern exists. This contradiction shows that R_M has no E_j family. By Theorem 2.25(iii), R_M is j-token refutable. □

The argument excludes every E_j family, so it excludes every possible surviving live symbol at the origin, not only a seeded one. The box radius and the refutation time of R_M depend on M; no bound on them is claimed.

7.9 The token budget

Every map used above has a bounded number of sites, independent of M, n and T:

- Lemmas 7.2 and 7.3: maps on at most 3 sites;
- Lemma 7.4: the binary sibling check uses 3 sites;
- (P1) and (P2) are applied to $\widehat{H}$, which needs $j \ge B$; the maps they are applied to have at most $\ell + 4$ sites, and their proofs use members of $\widehat{H}$ on at most B sites;
- the pins p: at most $\ell + 3$ sites;
- Lemma 7.5: at most $\ell + 6$ sites in H_γ, and at most $\ell + 4$ sites in $\widehat{H}$;

- Lemma 7.6: the walk uses maps on at most $\ell + 5$ sites, and each step calls Lemma 7.5;
- Lemma 7.7: at most $\ell + 5$ sites.

So $j \geq \max(B, \ell + 6)$ suffices, and $j = 10(B + \ell + 100)$ exceeds $B + \ell + 6$. The margin also covers a sibling check through two full parent cells, with 8 quarter sites, in place of the 3-site path of Lemma 7.4. The payload alphabet C_M grows with M; the number of sites does not.

Theorem 7.8 (Theorem C). *Let $j = 10(B + \ell + 100)$ be the integer of Theorem 7.1. The map $M \mapsto R_M$ of Definition 6.17 is total computable, R_M tiles the plane if and only if M does not halt, and R_M is j-token refutable whenever M halts.*

Proof of Theorem 7.8. The map $M \mapsto R_M$ is total computable (Definition 6.17). If M does not halt, R_M tiles (Proposition 6.18). If M halts, R_M is j-token refutable by Theorem 7.1, so it does not tile by Theorem 2.25(iv). Hence R_M tiles if and only if M does not halt, and every halting member is j-token refutable for the one integer $j = 10(B + \ell + 100)$. □

8 Proof of the main theorem and corollaries

Throughout this section $j = 10(B + \ell + 100)$ is the integer of Theorem 7.1, with $\ell = \binom{276}{2} + 1 = 37951$, and

$$k = \max(2, 2j) = 2j, \qquad q = k + 1, \qquad a = 2q^2 + 1, \qquad D_0 = 50q, \qquad \varepsilon_0 = \varepsilon_k = (4D_0^2)^{-a}.$$

The dimension is $d = 2 + k$. The family $M \mapsto R_M$ is the one of Theorem 7.8. We write $\delta_O(t)$ as in Definition 2.11, adding a superscript M when the Wang set is R_M.

Lemma 8.1 (One threshold). *Let $0 < \varepsilon \leq \varepsilon_0$. Then $\beta = 2D_0^2 \varepsilon^{1/a} \leq \frac{1}{2}$, the hypothesis of Theorem 4.2 holds, and $\varepsilon < 10^{-9}$. Both conditions on the rate have the form $\varepsilon \leq \varepsilon_0$, so they hold at every rate in $(0, \varepsilon_0]$.*

Proof. The map $\varepsilon \mapsto \beta$ is increasing, and at ε_0 it equals $2D_0^2(4D_0^2)^{-1} = \frac{1}{2}$. Theorem 4.2 asks for $0 < \varepsilon \leq (4D_0^2)^{-a} = \varepsilon_0$. Since $q \geq 3$, $D_0 \geq 150$ and $a \geq 19$, we get $\varepsilon_0 \leq (90000)^{-19} < 10^{-9}$. □

Theorem 8.2 (Theorem A). *For every $\varepsilon \in (0, \varepsilon_0]$ and every Turing machine M, the noisy quorum rule $F_{R_M}^{[k]}$ on $\mathbb{Z}^{2+k}$ with uniform replacement at rate ε is a positive-rate probabilistic cellular automaton, and it is ergodic if and only if M halts. The map $M \mapsto F_{R_M}^{[k]}$ is total computable.*

Proof of Theorem 8.2. Fix $\varepsilon \in (0, \varepsilon_0]$ and use uniform replacement noise at rate ε. The map $M \mapsto R_M$ is total computable by Theorem 7.8, and T_R, $\mathcal{A}_R$ and the local rule of $F_R^{[k]}$ on $\mathcal{N} = \Lambda \times \{c_0, \ldots, c_k\}$ are computed from R by finite enumeration. So $M \mapsto F_{R_M}^{[k]}$ is total computable.

Suppose M halts. By Theorem 7.8, R_M is j-token refutable; let $\tau = \tau(R_M)$ be its refutation time. The hypotheses of Theorem 3.1 hold: $j \geq 1$, $k \geq \max(2, 2j)$, $0 < \varepsilon < 1$, and $\beta \leq \frac{1}{2} < 1$ by Lemma 8.1. So for every window O and every $t \geq 0$,

$$\delta_O^M(t) \leq 2|O|\, \varepsilon^{-\tau/a}\, \frac{\beta^t}{1 - \beta} \leq 4|O|\, \varepsilon^{-\tau/a}\, 2^{-t}, \tag{13}$$

which tends to 0. The process forgets, and it is ergodic by Lemma 2.12.

Suppose M does not halt. By Theorem 7.8, R_M tiles; let ξ be the cloned tiling configuration of a tiling. By Theorem 4.2 and Lemma 8.1, for every site v and every $t \geq 0$,

$$\mathbb{P}_{\xi}\big(X_t(v) \neq \xi(v)\big) < 1.01\,\varepsilon, \qquad \mathbb{P}_{\perp}\big(X_t(v) \text{ is live}\big) < 1.01\,\varepsilon. \tag{14}$$

Take the window $\{v\}$ and the event that v carries the live symbol $\xi(v)$. Then $\mathbb{P}_{\xi}(X_t(v) = \xi(v)) > 1 - 1.01\varepsilon$ and $\mathbb{P}_{\perp}(X_t(v) = \xi(v)) < 1.01\varepsilon$, so the process has memory with $\alpha = 1 - 2.02\varepsilon > 0$. By Lemma 2.13 it has two different invariant measures, and it is not ergodic.

Both cases hold for every $\varepsilon \in (0, \varepsilon_0]$, and j, k and ε_0 do not depend on M. □

Remark 8.3 (Scope of the constants). Since $B \geq 1$, we have $j \geq 10(1 + 37951 + 100) = 380520$, so $d = 2 + 2j$ is a fixed integer, at least 761042 in this construction. Its exact value depends on the constant B of Theorem 5.18, a fixed integer, shown to exist, that we do not evaluate. The rate ε_0 depends only on k, and β only on k and ε; the Wang set enters the forgetting bound only through $\tau(R_M)$. The alphabet $\mathcal{A}_{R_M}$ varies with M, so the least transition probability $\varepsilon/|\mathcal{A}_{R_M}|$ is not bounded below over the family. A fixed noise rate, as in the title, means a fixed replacement rate ε with uniform replacement on the alphabet of each member.

8.1 Corollaries

Corollary 8.4 (Every small rate)**.** *For every machine M the following are equivalent:*

(i) *M halts;*

(ii) *$F_{R_M}^{[k]}$ with uniform replacement noise is ergodic at some rate $\varepsilon \in (0, \varepsilon_0]$;*

(iii) *it is ergodic at every rate $\varepsilon \in (0, \varepsilon_0]$.*

Consequently, for each fixed $\varepsilon \in (0, \varepsilon_0]$, no algorithm decides, given a finite Wang set R, whether $F_R^{[k]}$ with uniform replacement noise at rate ε is ergodic.

Proof. By Lemma 8.1 the hypotheses of Theorems 3.1 and 4.2 used in the proof of Theorem 8.2 hold at every rate in $(0, \varepsilon_0]$. So (i) implies (iii), (iii) implies (ii), and if M does not halt the process is not ergodic at any rate in $(0, \varepsilon_0]$, which is (ii) implies (i). For the last claim, a decision procedure at rate ε, composed with the computable map $M \mapsto R_M$, would decide halting. □

Corollary 8.5 (Every Wang set at once)**.** *There is a total computable map $R \mapsto G_R$ from finite Wang sets to positive-rate probabilistic cellular automata on $\mathbb{Z}^{2+k}$, each with uniform replacement noise, such that for every $\varepsilon \in (0, \varepsilon_0]$ and every finite Wang set R, the automaton G_R at rate ε is ergodic if and only if R does not tile the plane.*

Proof. Let M_R be the machine that, for $n = 1, 2, \ldots$ in turn, checks whether R has a legal pattern on the $n \times n$ square, and halts at the first n for which it has none. The map $R \mapsto M_R$ is computable, and each check is finite. If R tiles, restrictions of a tiling give legal squares of every size, so M_R does not halt. If M_R does not halt, R has legal squares of every size; by compactness of $R^{\mathbb{Z}^2}$ (König's lemma on the finitely branching tree of legal centred squares) it tiles. So M_R halts if and only if R does not tile. Put $G_R = F_{R_{M_R}}^{[k]}$ with uniform replacement noise, and apply Theorem 8.2 to M_R. □

The automaton G_R is built from the recoded set R_{M_R}, not from R itself. So Corollary 8.5 gives one fixed dimension for every Wang set, whatever its token width, through the family of Theorem 7.8; it says nothing new about the quorum rule of R itself.

The next corollary concerns other noise. For a probability law ν on $\mathcal{A}_R$, *ν-replacement noise* at rate ε is defined as in Definition 2.8, with the values $\zeta_t(v)$ independent with law ν in place of the uniform law. Uniform replacement is the case where ν is uniform. Lemmas 2.10, 2.12 and 2.13 and their proofs hold verbatim for ν-replacement noise: they use only the mixture formula and the fact that $X_1|_O$ depends on finitely many coordinates of X_0 and on independent noise.

Corollary 8.6 (Other replacement laws)**.** *For each machine M let ν_M be a probability law on $\mathcal{A}_{R_M}$, and let $\varepsilon \in (0, \varepsilon_0]$.*

(i) $F_{R_M}^{[k]}$ with ν_M-replacement noise at rate ε is ergodic if and only if M halts. Every transition probability of this process is at least $\varepsilon \min_s \nu_M(s)$, which is positive when ν_M has full support.

(ii) Suppose M does not halt. Let the replacement events be as in Definition 2.8, and let the values written at events be arbitrary random variables, which may depend on the event indicators, on the past of the run and on further randomness. Then the bounds (14) hold for the run from every cloned tiling configuration ξ and for the run from $\perp$, at every site and time.

Proof. (ii) The proof of Theorem 4.2 has a deterministic part and a counting part. The deterministic part (Lemma 4.5) holds for every realization of the events and of the values: a bad point that is not an event has two bad points one step earlier, in two different provider planes, and no point of time 0 is bad. The counting part uses the noise only through the bound $\mathbb{P}(\text{every point of } S \text{ is an event}) \le \varepsilon^{|S|}$ for finite sets S of points of positive time (Remark 4.9). Both parts hold under the stated hypotheses, so the bounds hold.

(i) Suppose M does not halt. Values drawn independently from ν_M are a special case of (ii), so (14) holds, the process has memory as in the proof of Theorem 8.2, and Lemma 2.13 shows that it is not ergodic.

Suppose M halts. In the proof of Theorem 3.1, the runs from x and from $\perp$ use the same indicators η and the same values ζ. The covering field E is a function of η alone (Definition 3.5). The transport and cover lemmas and Corollary 3.7 hold for every realization of η and ζ. The discrepancy path of Lemma 3.9 uses only that both runs take the common value at an event. The final count uses only the independence of the indicators and $\mathbb{P}(\eta_t(v) = 1) = \varepsilon$. None of these steps uses the law of ζ. So (3) and (4) hold for ν_M-replacement noise with the same constants, the process forgets, and Lemma 2.12 shows that it is ergodic.

Finally, given X_t, the value $X_{t+1}(v)$ equals s with probability $(1-\varepsilon)\mathbf{1}[F_{R_M}^{[k]}(X_t)(v) = s] + \varepsilon\,\nu_M(s) \ge \varepsilon\,\nu_M(s)$. □

The forgetting half needs the values to be drawn without looking at the configuration, because its coupling gives both runs the same value at each event. The memory half needs nothing about the values.

Corollary 8.7 (Mixing times are not computably bounded)**.** *Let $\varepsilon \in (0, \varepsilon_0]$ be rational, and use uniform replacement noise at rate ε.*

(i) For every machine M, window O and time t, the supremum in $\delta_O^M(t)$ is a maximum over finitely many restrictions of the start, and $\delta_O^M(t)$ is a rational number computable from M, O and t.

(ii) Let $t_{\text{mix}}(M)$ be the least $t_0 \ge 0$ such that $\delta_{\{0\}}^M(t) < \frac{1}{2}$ for every $t \ge t_0$, and $t_{\text{mix}}(M) = \infty$ if there is none. Then $t_{\text{mix}}(M) < \infty$ if and only if M halts.

(iii) No computable function b satisfies $t_{\text{mix}}(M) \le b(M)$ for every halting M. No computable function b satisfies $\tau(R_M) \le b(M)$ for every halting M.

Proof. (i) Let $\mathcal{N} = \Lambda \times \{c_0, \dots, c_k\}$ and $\mathcal{N}_t$ be as in Section 2, and write $\mathcal{A} = \mathcal{A}_{R_M}$ and $F = F^{[k]}_{R_M}$. Given X_s, the values $X_{s+1}(v)$ are independent, and $X_{s+1}(v) = s'$ with probability $(1-\varepsilon)\mathbf{1}[F(X_s)(v) = s'] + \varepsilon/|\mathcal{A}|$, where $F(X_s)(v)$ depends on X_s only on $v + \mathcal{N}$. Hence, for a window S and $w \in \mathcal{A}^S$,

$$\mathbb{P}_x\big(X_{s+1}|_S = w\big) = \sum_{y \in \mathcal{A}^{S+\mathcal{N}}} \mathbb{P}_x\big(X_s|_{S+\mathcal{N}} = y\big) \prod_{v \in S} \Big((1-\varepsilon)\mathbf{1}\big[F(y)(v) = w(v)\big] + \frac{\varepsilon}{|\mathcal{A}|} \Big).$$

where $F(y)(v)$ is the local rule applied to y on $v + \mathcal{N}$. Apply this t times, with $S = O + \mathcal{N}_{t-s-1}$ at step s, from $X_0|_{O+\mathcal{N}_t} = x|_{O+\mathcal{N}_t}$ to $X_t|_O$. Each step is a finite sum of products of rationals computable from M. So $\mathrm{Law}_x(X_t|_O)$ is a rational vector, computable from M, O, t and $x|_{O+\mathcal{N}_t}$, and it depends on x only through this restriction. The supremum over x is a maximum over the finite set $\mathcal{A}^{O+\mathcal{N}_t}$, and the total variation distance of two rational vectors is rational.

(ii) If M halts, (13) with $O = \{0\}$ gives $\delta^M_{\{0\}}(t) \to 0$, so $t_{\mathrm{mix}}(M) < \infty$. If M does not halt, let ξ be a cloned tiling configuration. By (14), for every t,

$$\delta^M_{\{0\}}(t) \geq \mathbb{P}_\xi\big(X_t(0) = \xi(0)\big) - \mathbb{P}_\perp\big(X_t(0) = \xi(0)\big) > 1 - 2.02\,\varepsilon > \tfrac{1}{2},$$

so $t_{\mathrm{mix}}(M) = \infty$.

(iii) Suppose b is computable and $t_{\mathrm{mix}}(M) \leq b(M)$ for every halting M. Given M, compute $\delta^M_{\{0\}}(b(M))$ by (i) and answer "halts" exactly when it is below $\frac{1}{2}$. If M halts, then $b(M) \geq t_{\mathrm{mix}}(M)$, so the value is below $\frac{1}{2}$. If M does not halt, the value exceeds $\frac{1}{2}$ by the proof of (ii). This decides the halting problem, which is impossible.

For the refutation time, let M halt and $\tau = \tau(R_M)$. By (13) with $|O| = 1$, $\delta^M_{\{0\}}(t) < \frac{1}{2}$ whenever $2^t > 8\varepsilon^{-\tau/a}$, that is, whenever $t > 3 + (\tau/a)\log_2(1/\varepsilon)$. So $t_{\mathrm{mix}}(M) \leq 4 + \tau L$ with $L = \lceil \log_2(1/\varepsilon) \rceil$, since $a \geq 1$. A computable b with $\tau(R_M) \leq b(M)$ for every halting M would give the computable bound $4 + b(M)L$ on $t_{\mathrm{mix}}(M)$, which we have excluded. □

The process and the all-bottom start are invariant under translations of $\mathbb{Z}^{2+k}$, so $\delta^M_{\{v\}} = \delta^M_{\{0\}}$ for every site v. Restricting to a smaller window does not increase total variation, so $\delta^M_O \geq \delta^M_{\{0\}}$ for every nonempty window O. The conclusions of Corollary 8.7 therefore hold for the mixing time of every nonempty window. Forgetting happens for every halting member, but its onset is not computably bounded. The bound (13) reads $\delta^M_O(t) \leq 2^{-(t-t_1)}$ with $t_1 = (\tau(R_M)/a)\log_2(1/\varepsilon) + \log_2(4|O|)$: a delay set by the refutation time, followed by decay at a rate that does not depend on M.

Ergodicity also has an equilibrium reading. A PCA on $\mathbb{Z}^d$ with a strictly positive transition table defines a Gibbs specification on $\mathbb{Z}^{d+1}$, its space-time interaction, in which time is one more lattice direction [7]. For $F^{[k]}_{R_M}$ with uniform replacement noise at rate ε, the table is

$$p_M(s \mid y) = (1-\varepsilon)\,\mathbf{1}\big[s = f_M(y)\big] + \frac{\varepsilon}{|\mathcal{A}_{R_M}|}, \qquad s \in \mathcal{A}_{R_M},\ y \in \mathcal{A}_{R_M}^{\mathcal{N}},$$

where f_M is the local rule of $F^{[k]}_{R_M}$ on $\mathcal{N}$. Each entry is at least $\varepsilon/|\mathcal{A}_{R_M}| > 0$. Write a space-time point as $(v,t) \in \mathbb{Z}^{2+k} \times \mathbb{Z}$; its parents are the points $(v + w, t-1)$, $w \in \mathcal{N}$. The *space-time interaction* Ψ_M assigns to each point (v,t) the energy $-\log p_M\big(\omega(v,t) \mid \omega \text{ at the parents of } (v,t)\big)$ of a configuration $\omega \in \mathcal{A}_{R_M}^{\mathbb{Z}^{3+k}}$. It has finite range and is invariant under all translations of $\mathbb{Z}^{3+k}$. A *Gibbs measure* for Ψ_M is a DLR state of its specification, as defined in [7].

Corollary 8.8 (Space-time Gibbs measures). *Let $\varepsilon \in (0, \varepsilon_0]$. If M halts, Ψ_M has exactly one translation-invariant Gibbs measure. If M does not halt, it has at least two. For rational ε, the table p_M is computable from M; so it is undecidable whether a member of this computable family of finite-range, translation-invariant interactions on $\mathbb{Z}^{3+k}$ has a unique translation-invariant Gibbs measure.*

Proof. We use three results of [7], stated there for every spatial dimension, every finite alphabet with at least two symbols, every finite neighbourhood containing the origin, and every strictly positive table. Our process fits: the alphabet is finite, $\mathcal{N}$ is finite and contains $0 = (0, c_0)$, and every entry of p_M is positive. The three results are: the lemma that PCA histories satisfy the DLR equations, so that every two-sided stationary trajectory law is a Gibbs measure; the corollary that periodic states are stationary trajectories, which with that lemma identifies the translation-invariant Gibbs measures with the trajectory laws of translation-invariant stationary laws, the correspondence of Goldstein, Kuik, Lebowitz and Maes [12]; and item (i) of the theorem on the Gibbs simplex of an ergodic PCA: an ergodic strictly positive PCA has exactly one periodic, in particular one translation-invariant, Gibbs measure. The notion of ergodicity in [7], a stationary law that attracts every initial law, is the one of Definition 2.9.

If $\mathcal{A}_{R_M} = \{\perp\}$, then R_M has no legal 3×3 block, so it does not tile and M halts by Theorem 7.8. The configuration space is then one point, which carries exactly one measure, and the claim holds. Otherwise $\mathcal{A}_{R_M}$ has at least three symbols.

Suppose M halts. By Theorem 8.2 the process is ergodic, with invariant measure π. The process commutes with the translations of $\mathbb{Z}^{2+k}$, so every translate of π is invariant, and π is translation invariant by uniqueness. Its two-sided stationary trajectory law is invariant under all translations of $\mathbb{Z}^{3+k}$ and is a Gibbs measure. By the theorem on the Gibbs simplex, item (i), it is the only translation-invariant one.

Suppose M does not halt. By Theorem 7.8, R_M tiles, so the set $\mathcal{T}_M \subseteq R_M^{\mathbb{Z}^2}$ of tilings is nonempty. It is closed in the compact product space and invariant under the shifts of $\mathbb{Z}^2$. Fix $z_0 \in \mathcal{T}_M$ and let m_n be the average of the point masses at the translates $z_0(\cdot + w)$, $w \in [-n, n]^2$. A weak limit point m of (m_n) is a probability measure on $\mathcal{T}_M$, and it is shift invariant: for a unit vector e of $\mathbb{Z}^2$, the measures m_n and their translates by e differ by at most $2(2n+1)/(2n+1)^2$ in total variation. For a tiling z let ξ_z be its cloned tiling configuration. The map $z \mapsto \xi_z$ is continuous, since $\xi_z(u, P)$ depends on z only on $u + \Lambda$. It satisfies $\xi_{z(\cdot+w)} = \xi_z(\cdot + (w, 0))$, and ξ_z is invariant under translations of the transverse coordinate. So the image κ of m is a probability measure on Ω that is invariant under all translations of $\mathbb{Z}^{2+k}$.

Fix a site v. The function $x \mapsto \mathbb{P}_x(X_t(v) \text{ is live})$ depends on finitely many coordinates of x, and by (14) it exceeds $1 - 1.01\varepsilon$ at every ξ_z. By the mixture formula of Section 2, $\mathbb{P}_\kappa(X_t(v) \text{ is live}) \geq 1 - 1.01\varepsilon$ for every t. The laws $\kappa\Phi^t$ are translation invariant, because κ is and the process commutes with translations. By Lemma 2.10, the averages $\frac{1}{n}\sum_{t<n} \kappa\Phi^t$ have a weak limit point π^+, which is invariant. It is translation invariant, as a weak limit of translation-invariant measures, and $\pi^+(\omega(v) \text{ is live}) \geq 1 - 1.01\varepsilon$, because the cylinder is open and closed. In the same way the averages from the all-bottom configuration, which is translation invariant, have an invariant, translation-invariant limit point π^-, with $\pi^-(\omega(v) \text{ is live}) \leq 1.01\varepsilon$ by (14). As $1.01\varepsilon < 1 - 1.01\varepsilon$, the measures π^+ and π^- differ.

For an invariant π, its two-sided stationary trajectory law is the measure on $\mathcal{A}_{R_M}^{\mathbb{Z}^{3+k}}$ whose restriction to the times $t_0, \ldots, t_0 + n$ is the law of $(X_0, \ldots, X_n)$ under Law_π; these restrictions are consistent because π is invariant, and Kolmogorov's theorem gives the measure. For $\pi = \pi^\pm$ it is invariant under time translations, and under space translations because $\pi^\pm$ is and the process commutes with them. So both trajectory laws are translation-invariant Gibbs measures. Their marginals at time 0 are $\pi^+ \neq \pi^-$, so they differ.

The last claim follows: an algorithm deciding uniqueness of the translation-invariant Gibbs measure for the tables p_M would decide halting. □

Remark 8.9 (Arithmetical placement). For a PCA with a strictly positive rational table, ergodicity is a Π_2^0 property of the table [7]. The reason is short. By Halawi's theorem [13], an ergodic positive PCA is uniformly ergodic. So ergodicity says: for every window O and every $m \geq 1$ there is t with $\Delta_O(t) < 1/m$, where $\Delta_O(t) = \sup_{x,y} d_{\mathrm{TV}}\big(\mathrm{Law}_x(X_t|_O), \mathrm{Law}_y(X_t|_O)\big)$. The argument of Corollary 8.7(i) computes $\Delta_O(t)$ exactly, and $\Delta_O(t)$ does not increase with t [7]. Theorem 8.2 gives a lower bound at the first level. For each rational $\varepsilon \in (0, \varepsilon_0]$, the computable map $M \mapsto F_{R_M}^{[k]}$ reduces the halting problem to ergodicity at rate ε. So ergodicity of positive-rate PCA on $\mathbb{Z}^{2+k}$ at the fixed rate ε, with uniform replacement, is Σ_1^0-hard, and non-ergodicity is Π_1^0-hard; in particular the non-ergodic members do not form a computably enumerable set. Whether ergodicity at a fixed rate is Π_2^0-complete is not settled by Theorem 8.2.

9 Open problems

The dimension. Theorem A holds on $\mathbb{Z}^{2+k}$ with $k = \max(2, 2j)$, and the j of Theorem C is fixed but very large. Its size comes from the parser of Section 5, whose readout uses $\binom{276}{2} + 1$ observation points; part (ii) of Theorem B needs far fewer, and the computation layer adds only a few to those observation points. Two routes to a small dimension are open. One is a family with small token width: width 2 would place Theorem A on $\mathbb{Z}^6$. The other is a forgetting mechanism whose number of extra dimensions does not grow with j.

Open problem 9.1. What is the least dimension d for which ergodicity of positive-rate cellular automata on $\mathbb{Z}^d$, at one fixed noise rate, is undecidable? What is the least token width of a computable family of Wang sets with undecidable tiling problem?

The exact degree. For rational positive rates, ergodicity is a Π_2^0 property of the table [7, Remark on the Π_2^0 placement]. Theorem A shows it is Σ_1^0-hard at one fixed rate.

Open problem 9.2. Is ergodicity of positive-rate cellular automata at one fixed noise rate Π_2^0-complete?

Natural rules. The rules of Theorem A are built to order. The quorum rule with $k = 2$ on $\mathbb{Z}^4$ is the smallest member of its kind, and part (i) of Theorem B holds for it.

Open problem 9.3. Is there a fixed rule shape in low dimension, applied to arbitrary Wang sets, whose ergodicity at a fixed noise rate is undecidable?

How this work was done

The authors chose the question, directed the work and judged every result. AI models from two developers explored the problem, derived and drafted the arguments, and challenged each other's results: OpenAI's GPT-6.1 Sol and Anthropic's Claude Fable 5.2 and Claude Opus 5.5. The forgetting theorem and the construction of Theorem C were each derived by one model and then re-derived refutation-first, step by step, by a model of the other developer, which did not write them. The memory theorem rests on the contour lemma of the forgetting proof, which was cross-checked in this way; its remaining steps were written and checked by

models of one developer, the check in a separate run that did not write them. These reviews found one gap in the parser of Section 5 (a missing transport of the phase from the anchor to a queried block in the point certificate) and one constant in the memory bound that rested on an unproved property; both were repaired and re-checked. The reconstruction of the manuscript from the proofs was itself read refutation-first by a reviewer of the other developer. Models are not listed as authors; the authors take responsibility for every claim.

Verification status

- **Proved in the text.** Theorems A, B and C and every lemma and corollary of Sections 2 to 8, given the cited inputs below.
- **Imported with stated scope.** For Corollary 8.8, the correspondence between translation-invariant Gibbs states of the space–time interaction and stationary trajectories, and the uniqueness of the periodic Gibbs state of an ergodic positive-rate automaton [7, 12]; for the remark on the arithmetical degree, the Π^0_2 placement of [7].
- **Existence only.** The parser constant B of Theorem 5.18, and hence j, k and ε_0, are shown to exist; their values were not computed. The map $M \mapsto R_M$ does not use them.
- **Checked by machine.** No computation is used in any proof.
- **Not formalized.** Nothing is verified in Lean.
- **Internal only.** Every review is internal; none of it is journal peer review.

A Tables for the scaffold

This appendix lists the finite data of Section 5: Ollinger's substitution [25], in the transcription of [18], and its ruler decoration. Every entry below was derived again from the definitions in that section; the counts are obtained by listing, not by computer search. Vertical lane atoms are $\mathsf{r}_0, \mathsf{r}_2, \mathsf{g}_0, \mathsf{g}_2$ and horizontal lane atoms are $\mathsf{r}_1, \mathsf{r}_3, \mathsf{g}_1, \mathsf{g}_3$; the letter r means red and g means green.

	∂_{N}	∂_{S}	∂_{E}	∂_{W}
thin a	r_0	g_2	r_1	g_3
thin b	g_0	r_2	g_1	r_3
thin c	r_2	g_0	g_3	r_1
thin d	g_2	r_0	r_3	g_1
corner $\angle$a	g_2	r_0	r_3	g_1
corner $\angle$b	r_0	g_2	r_1	g_3
corner $\angle$c	r_2	g_0	g_3	r_1
corner $\angle$d	g_0	r_2	g_1	r_3
line $\langle \alpha, \gamma \rangle$	α	α	γ	γ

	∂_{N}	∂_{S}	∂_{E}	∂_{W}
black A	0	1	0	1
black B	1	0	0	0
black C	1	0	1	1
black D	1	1	1	0
black E	0	0	1	0

Table 1: Endpoints of the thin and thick components, and bits of the black component. A line $\langle \alpha, \gamma \rangle$ has a vertical atom α and a horizontal atom γ; all sixteen lines occur. The edge colour of a tile (θ, κ, ω) on a side is the triple of the three entries on that side.

The left table gives the following consequences, used in Lemma 5.4.

- Each corner has exactly the four endpoints of one thin type: $F(\mathsf{a}) = \angle\mathsf{b}$, $F(\mathsf{b}) = \angle\mathsf{d}$, $F(\mathsf{c}) = \angle\mathsf{c}$, $F(\mathsf{d}) = \angle\mathsf{a}$.
- Thin types match horizontally exactly in the pairs (a,c), (c,a), (b,d), (d,b) (west tile first), and vertically exactly in the pairs (a,d), (d,a), (b,c), (c,b) (south tile first).
- ∂_{S} of the thin types $\mathsf{a},\mathsf{b},\mathsf{c},\mathsf{d}$ is $\mathsf{g}_2,\mathsf{r}_2,\mathsf{g}_0,\mathsf{r}_0$; ∂_{N} is $\mathsf{r}_0,\mathsf{g}_0,\mathsf{r}_2,\mathsf{g}_2$; ∂_{W} is $\mathsf{g}_3,\mathsf{r}_3,\mathsf{r}_1,\mathsf{g}_1$; ∂_{E} is $\mathsf{r}_1,\mathsf{g}_1,\mathsf{g}_3,\mathsf{r}_3$.

The black maps of the substitution are

ω	A	B	C	D	E
$\mathrm{se}(\omega)$	B	B	C	C	C
$\mathrm{nw}(\omega)$	E	D	D	D	E

They are characterized by $\partial_{\mathrm{W}}\,\mathrm{se}(\omega) = \partial_{\mathrm{E}}\omega$ with $\mathrm{se}(\omega) \in \{\mathsf{B},\mathsf{C}\}$, and $\partial_{\mathrm{S}}\,\mathrm{nw}(\omega) = \partial_{\mathrm{N}}\omega$ with $\mathrm{nw}(\omega) \in \{\mathsf{D},\mathsf{E}\}$. They also satisfy $\partial_{\mathrm{E}}\,\mathrm{se}(\omega) = \partial_{\mathrm{E}}\omega$, $\partial_{\mathrm{N}}\,\mathrm{nw}(\omega) = \partial_{\mathrm{N}}\omega$, $\partial_{\mathrm{S}}\,\mathrm{se}(\omega) = 0$, $\partial_{\mathrm{W}}\,\mathrm{nw}(\omega) = 0$, $\partial_{\mathrm{N}}\,\mathrm{se}(\omega) = 1$ and $\partial_{\mathrm{E}}\,\mathrm{nw}(\omega) = 1$ for every ω.

thin	thick	black	condition	tiles
a	corner	A	none	4
a	line $\langle\alpha,\gamma\rangle$	B	$\alpha \neq \mathsf{r}_0$	12
a	line $\langle\alpha,\gamma\rangle$	C	$\alpha \neq \mathsf{r}_2$	12
a	line $\langle\alpha,\gamma\rangle$	D	$\gamma \neq \mathsf{r}_3$	12
a	line $\langle\alpha,\gamma\rangle$	E	$\gamma \neq \mathsf{r}_1$	12
b	corner	A	none	4
c	line $\langle\alpha,\gamma\rangle$	B	$\alpha \neq \mathsf{r}_0$	12
c	line $\langle\alpha,\gamma\rangle$	C	$\alpha \neq \mathsf{r}_2$	12
d	line $\langle\alpha,\gamma\rangle$	D	$\gamma \neq \mathsf{r}_3$	12
d	line $\langle\alpha,\gamma\rangle$	E	$\gamma \neq \mathsf{r}_1$	12
			total	104

Table 2: The 104 tiles of $\mathbb{O}$. A line is excluded in a black class by one of four red atoms.

child	thin	thick	black
$(0,0)$	a	κ	ω
$(1,0)$	c	$\langle\partial_{\mathrm{S}}\theta,\ \partial_{\mathrm{E}}\kappa\rangle$	$\mathrm{se}(\omega)$
$(0,1)$	d	$\langle\partial_{\mathrm{N}}\kappa,\ \partial_{\mathrm{W}}\theta\rangle$	$\mathrm{nw}(\omega)$
$(1,1)$	b	$F(\theta)$	A

child	χ_1	χ_2
$(0,0)$	$1-b_1$	$1-b_2$
$(1,0)$	0	$1-b_2$
$(0,1)$	$1-b_1$	0
$(1,1)$	0	0

Table 3: The substitution φ of a tile (θ,κ,ω) (left), and the bits of the decorated substitution φ' of a letter with bits (b_1,b_2) (right). Child $(1,0)$ is south-east and child $(0,1)$ is north-west.

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θ	blacks allowed by children $(1,0)$, $(0,1)$	formal parents	in $\mathbb{O}$	excluded
a	all	52	52	0
b	A	4	4	0
c	B, C, D	36	24	12: black D, line with $\gamma \neq r_3$
d	C, D, E	36	24	12: black C, line with $\alpha \neq r_2$
total		128	104	24

Table 4: Formal parents of an aligned legal 2×2 block (Lemma 5.5). The count of formal parents uses the 52 pairs (κ, ω) allowed at child $(0,0)$.

thin	base tiles	bit rule	variants each	letters
a	4 corners	$b_1 = b_2$	2	8
a	8 red crossings	$b_1 \neq b_2$	2	16
a	40 other lines	none	4	160
b	4 corners	$b_1 = b_2 = 0$	1	4
c	4 red crossings	$b_1 = 0, b_2 = 1$	1	4
c	20 other lines	$b_1 = 0$	2	40
d	4 red crossings	$b_1 = 1, b_2 = 0$	1	4
d	20 other lines	$b_2 = 0$	2	40
			total	276

Table 5: The 276 letters of $\mathbb{A}$. A red crossing is a line whose two atoms are red. Each black class B, C, D, E admits exactly two of the four red crossings (Table 2), which gives 8 red crossings with thin type a and 4 each with thin types c and d.

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position	thin	thick	red crossing exactly when
$(2,1)$	d	$\langle \partial_{\mathrm{S}}\theta,\ \mathsf{r}_1 \rangle$	$\theta \in \{\mathsf{b},\mathsf{d}\}$
$(2,3)$	d	$\langle \partial_{\mathrm{N}}\theta,\ \mathsf{r}_3 \rangle$	$\theta \in \{\mathsf{a},\mathsf{c}\}$
$(1,2)$	c	$\langle \mathsf{r}_0,\ \partial_{\mathrm{W}}\theta \rangle$	$\theta \in \{\mathsf{b},\mathsf{c}\}$
$(3,2)$	c	$\langle \mathsf{r}_2,\ \partial_{\mathrm{E}}\theta \rangle$	$\theta \in \{\mathsf{a},\mathsf{d}\}$

Table 6: Four cells of the depth-two block $\varphi^2(\theta,\kappa,\omega)$, for every base tile. Position $(2,1)$ is child $(0,1)$ of child $(1,0)$; $(2,3)$ is child $(0,1)$ of child $(1,1)$; $(1,2)$ is child $(1,0)$ of child $(0,1)$; $(3,2)$ is child $(1,0)$ of child $(1,1)$.

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